1.a

In [2]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.io import loadmat
from sklearn.decomposition import PCA
import pandas as pd


plt.style.use('seaborn-v0_8-darkgrid')
sns.set_palette("husl")


# 1. LOAD THE DATASET

print("="*70)
print("LOADING PAVIA UNIVERSITY DATASET")
print("="*70)

# Load the hyperspectral image and ground truth
data = loadmat('PaviaU.mat')
gt = loadmat('PaviaU_gt.mat')

# Extract the actual arrays
X = data['paviaU']  # Hyperspectral image
y = gt['paviaU_gt']  # Ground truth labels

print(f"Hyperspectral Image Shape: {X.shape}")
print(f"Ground Truth Shape: {y.shape}")
print(f"Data type: {X.dtype}")
print(f"Ground Truth data type: {y.dtype}")


# 2. BASIC DATASET STATISTICS

print("\n" + "="*70)
print("BASIC DATASET STATISTICS")
print("="*70)

height, width, bands = X.shape
total_pixels = height * width
print(f"Image Dimensions: {height} x {width} pixels")
print(f"Number of Spectral Bands: {bands}")
print(f"Total Pixels: {total_pixels:,}")
print(f"Spectral Range: 430-960 nm (5 nm spacing)")
print(f"\nData Range:")
print(f"  Min value: {X.min():.4f}")
print(f"  Max value: {X.max():.4f}")
print(f"  Mean value: {X.mean():.4f}")
print(f"  Std deviation: {X.std():.4f}")


# 3. CLASS DISTRIBUTION ANALYSIS
```

```
print("\n" + "="*70)
print("CLASS DISTRIBUTION ANALYSIS")
print("="*70)

# Class names for Pavia University
class_names = {
    0: 'Unlabeled',
    1: 'Asphalt',
    2: 'Meadows',
    3: 'Gravel',
    4: 'Trees',
    5: 'Painted metal sheets',
    6: 'Bare Soil',
    7: 'Bitumen',
    8: 'Self-Blocking Bricks',
    9: 'Shadows'
}

# Count samples per class
unique_classes, counts = np.unique(y, return_counts=True)
class_distribution = dict(zip(unique_classes, counts))

print("\nClass Distribution:")
print("-" * 70)
labeled_samples = 0
for class_id in sorted(class_distribution.keys()):
    count = class_distribution[class_id]
    percentage = (count / total_pixels) * 100
    if class_id == 0:
        print(f"Class {class_id} ({class_names[class_id]}): {count:,} ({percentage:
    else:
        labeled_samples += count
        print(f"Class {class_id} ({class_names[class_id]}): {count:,} ({percentage:

print(f"\nTotal Labeled Samples: {labeled_samples:,}")
print(f"Total Unlabeled Samples: {class_distribution[0]:,}")
print(f"Percentage Labeled: {(labeled_samples/total_pixels)*100:.2f}%")

# Calculate class imbalance ratio
labeled_counts = [counts[i] for i in range(1, len(counts))]
imbalance_ratio = max(labeled_counts) / min(labeled_counts)
print(f"\nClass Imbalance Ratio (max/min): {imbalance_ratio:.2f}:1")


# 4. VISUALIZATION: CLASS DISTRIBUTION

fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# Bar plot of class distribution (excluding unlabeled)
labeled_classes = [c for c in unique_classes if c != 0]
labeled_class_counts = [class_distribution[c] for c in labeled_classes]
labeled_class_names = [class_names[c] for c in labeled_classes]

axes[0].bar(range(len(labeled_classes)), labeled_class_counts, color='steelblue')
axes[0].set_xlabel('Class', fontsize=12, fontweight='bold')
```

```
axes[0].set_ylabel('Number of Samples', fontsize=12, fontweight='bold')
axes[0].set_title('Class Distribution (Labeled Classes Only)', fontsize=14, fontwei
axes[0].set_xticks(range(len(labeled_classes)))
axes[0].set_xticklabels(labeled_class_names, rotation=45, ha='right')
axes[0].grid(axis='y', alpha=0.3)

# Add value labels on bars
for i, v in enumerate(labeled_class_counts):
    axes[0].text(i, v + 500, str(v), ha='center', va='bottom', fontweight='bold')

# Pie chart showing labeled vs unlabeled
pie_data = [class_distribution[0], labeled_samples]
pie_labels = ['Unlabeled', 'Labeled']
colors = ['lightgray', 'steelblue']
axes[1].pie(pie_data, labels=pie_labels, autopct='%1.1f%%', colors=colors,
            startangle=90, textprops={'fontsize': 12, 'fontweight': 'bold'})
axes[1].set_title('Labeled vs Unlabeled Pixels', fontsize=14, fontweight='bold')

plt.tight_layout()
plt.show()


# 5. GROUND TRUTH VISUALIZATION

print("\n" + "="*70)
print("VISUALIZING GROUND TRUTH MAP")
print("="*70)

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(y, cmap='tab10')
ax.set_title('Ground Truth Labels - Pavia University', fontsize=14, fontweight='bol
ax.set_xlabel('Width (pixels)', fontsize=12)
ax.set_ylabel('Height (pixels)', fontsize=12)

# Create custom colorbar
cbar = plt.colorbar(im, ax=ax, ticks=range(10))
cbar.set_label('Class Labels', fontsize=12, fontweight='bold')
cbar.ax.set_yticklabels([class_names[i] for i in range(10)], fontsize=9)

plt.tight_layout()
plt.show()


# 6. SPECTRAL BAND ANALYSIS

print("\n" + "="*70)
print("SPECTRAL BAND ANALYSIS")
print("="*70)

# Calculate statistics for each band
band_means = np.mean(X, axis=(0, 1))
band_stds = np.std(X, axis=(0, 1))
band_mins = np.min(X, axis=(0, 1))
band_maxs = np.max(X, axis=(0, 1))

# Calculate wavelengths (430-960 nm in 5nm spacing)
```

```
wavelengths = np.linspace(430, 960, bands)

print(f"Band Statistics Summary:")
print(f"  Average band mean: {band_means.mean():.4f}")
print(f"  Average band std: {band_stds.mean():.4f}")
print(f"  Band with highest mean: {np.argmax(band_means)} (wavelength: {wavelengths
print(f"  Band with lowest mean: {np.argmin(band_means)} (wavelength: {wavelengths[

# Visualize spectral statistics
fig, axes = plt.subplots(2, 2, figsize=(15, 10))

# Mean reflectance per band
axes[0, 0].plot(wavelengths, band_means, linewidth=2, color='darkblue')
axes[0, 0].fill_between(wavelengths, band_means - band_stds, band_means + band_stds
                        alpha=0.3, color='lightblue')
axes[0, 0].set_xlabel('Wavelength (nm)', fontsize=11, fontweight='bold')
axes[0, 0].set_ylabel('Mean Reflectance', fontsize=11, fontweight='bold')
axes[0, 0].set_title('Mean Spectral Signature (±1 std)', fontsize=12, fontweight='b
axes[0, 0].grid(alpha=0.3)

# Standard deviation per band
axes[0, 1].plot(wavelengths, band_stds, linewidth=2, color='darkred')
axes[0, 1].set_xlabel('Wavelength (nm)', fontsize=11, fontweight='bold')
axes[0, 1].set_ylabel('Standard Deviation', fontsize=11, fontweight='bold')
axes[0, 1].set_title('Spectral Variability Across Bands', fontsize=12, fontweight='
axes[0, 1].grid(alpha=0.3)

# Min-Max range per band
axes[1, 0].fill_between(wavelengths, band_mins, band_maxs, alpha=0.5, color='green'
axes[1, 0].plot(wavelengths, band_means, linewidth=2, color='darkgreen', label='Mea
axes[1, 0].set_xlabel('Wavelength (nm)', fontsize=11, fontweight='bold')
axes[1, 0].set_ylabel('Reflectance Value', fontsize=11, fontweight='bold')
axes[1, 0].set_title('Min-Max Range Across Bands', fontsize=12, fontweight='bold')
axes[1, 0].legend()
axes[1, 0].grid(alpha=0.3)

# Heatmap of correlation between first 20 bands
sample_bands = 20
band_subset = X[:, :, :sample_bands].reshape(-1, sample_bands)
# Sample for computational efficiency
sample_indices = np.random.choice(band_subset.shape[0], min(5000, band_subset.shape
correlation_matrix = np.corrcoef(band_subset[sample_indices].T)

im = axes[1, 1].imshow(correlation_matrix, cmap='coolwarm', aspect='auto', vmin=-1,
axes[1, 1].set_xlabel('Band Index', fontsize=11, fontweight='bold')
axes[1, 1].set_ylabel('Band Index', fontsize=11, fontweight='bold')
axes[1, 1].set_title(f'Band Correlation Matrix (First {sample_bands} Bands)', fonts
plt.colorbar(im, ax=axes[1, 1])

plt.tight_layout()
plt.show()


# 7. CLASS-SPECIFIC SPECTRAL SIGNATURES

print("\n" + "="*70)
```

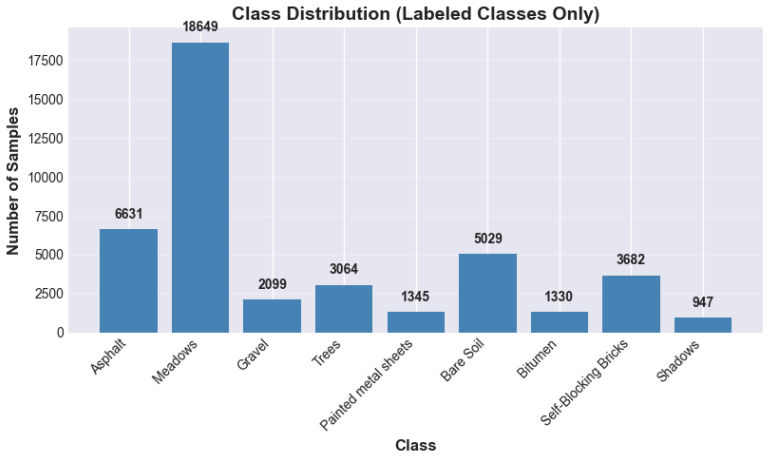
```
print("CLASS-SPECIFIC SPECTRAL SIGNATURES")
print("="*70)

# Extract spectral signatures for each class
fig, ax = plt.subplots(figsize=(14, 8))

for class_id in range(1, 10):
    # Get all pixels belonging to this class
    mask = (y == class_id)
    class_pixels = X[mask]

    if len(class_pixels) > 0:
        # Calculate mean spectral signature
        mean_signature = np.mean(class_pixels, axis=0)
        std_signature = np.std(class_pixels, axis=0)

        # Plot with confidence interval
        ax.plot(wavelengths, mean_signature, linewidth=2, label=class_names[class_i
        ax.fill_between(wavelengths, mean_signature - std_signature/2,
                        mean_signature + std_signature/2, alpha=0.1)

ax.set_xlabel('Wavelength (nm)', fontsize=12, fontweight='bold')
ax.set_ylabel('Mean Reflectance', fontsize=12, fontweight='bold')
ax.set_title('Mean Spectral Signatures by Class', fontsize=14, fontweight='bold')
ax.legend(bbox_to_anchor=(1.05, 1), loc='upper left', fontsize=10)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()


# 8. RGB VISUALIZATION

print("\n" + "="*70)
print("RGB COMPOSITE VISUALIZATION")
print("="*70)

# Create false-color RGB composite using bands (approximate RGB wavelengths)
# Red: ~640nm (band 54), Green: ~550nm (band 24), Blue: ~470nm (band 8)
red_band = 54
green_band = 24
blue_band = 8

rgb_image = np.stack([
    X[:, :, red_band],
    X[:, :, green_band],
    X[:, :, blue_band]
], axis=2)

# Normalize to 0-1 range for display
rgb_normalized = (rgb_image - rgb_image.min()) / (rgb_image.max() - rgb_image.min()

fig, axes = plt.subplots(1, 2, figsize=(15, 6))

axes[0].imshow(rgb_normalized)
axes[0].set_title('False Color RGB Composite', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Width (pixels)', fontsize=12)
```

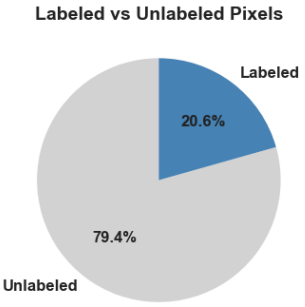
```
axes[0].set_ylabel('Height (pixels)', fontsize=12)
axes[0].axis('on')

# Ground truth overlay
axes[1].imshow(rgb_normalized)
masked_gt = np.ma.masked_where(y == 0, y)
axes[1].imshow(masked_gt, cmap='tab10', alpha=0.5)
axes[1].set_title('RGB with Ground Truth Overlay', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Width (pixels)', fontsize=12)
axes[1].set_ylabel('Height (pixels)', fontsize=12)

plt.tight_layout()
plt.show()


# 9. DIMENSIONALITY ANALYSIS WITH PCA

print("\n" + "="*70)
print("DIMENSIONALITY REDUCTION ANALYSIS (PCA)")
print("="*70)

# Reshape data for PCA
X_reshaped = X.reshape(-1, bands)

# Sample for computational efficiency
sample_size = min(10000, X_reshaped.shape[0])
sample_indices = np.random.choice(X_reshaped.shape[0], sample_size, replace=False)
X_sample = X_reshaped[sample_indices]

# Apply PCA
pca = PCA()
pca.fit(X_sample)

# Calculate cumulative variance
cumulative_variance = np.cumsum(pca.explained_variance_ratio_)

# Find number of components for 95% and 99% variance
n_95 = np.argmax(cumulative_variance >= 0.95) + 1
n_99 = np.argmax(cumulative_variance >= 0.99) + 1

print(f"Number of components for 95% variance: {n_95}")
print(f"Number of components for 99% variance: {n_99}")
print(f"Variance explained by first 3 components: {cumulative_variance[2]:.2%}")
print(f"Variance explained by first 10 components: {cumulative_variance[9]:.2%}")

# Visualize PCA results
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# Scree plot
axes[0].plot(range(1, 31), pca.explained_variance_ratio_[:30], 'bo-', linewidth=2)
axes[0].set_xlabel('Principal Component', fontsize=12, fontweight='bold')
axes[0].set_ylabel('Explained Variance Ratio', fontsize=12, fontweight='bold')
axes[0].set_title('Scree Plot (First 30 Components)', fontsize=14, fontweight='bold
axes[0].grid(alpha=0.3)

# Cumulative variance
```

```
axes[1].plot(range(1, len(cumulative_variance) + 1), cumulative_variance, 'r-', lin
axes[1].axhline(y=0.95, color='g', linestyle='--', label='95% variance')
axes[1].axhline(y=0.99, color='b', linestyle='--', label='99% variance')
axes[1].axvline(x=n_95, color='g', linestyle=':', alpha=0.5)
axes[1].axvline(x=n_99, color='b', linestyle=':', alpha=0.5)
axes[1].set_xlabel('Number of Components', fontsize=12, fontweight='bold')
axes[1].set_ylabel('Cumulative Explained Variance', fontsize=12, fontweight='bold')
axes[1].set_title('Cumulative Explained Variance', fontsize=14, fontweight='bold')
axes[1].legend()
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.show()


# 10. DATA QUALITY CHECKS

print("\n" + "="*70)
print("DATA QUALITY CHECKS")
print("="*70)

# Check for missing values
nan_count = np.isnan(X).sum()
inf_count = np.isinf(X).sum()
print(f"NaN values: {nan_count}")
print(f"Inf values: {inf_count}")

# Check for negative values
negative_count = (X < 0).sum()
print(f"Negative values: {negative_count} ({(negative_count/X.size)*100:.4f}%)")

# Check data range consistency
print(f"\nData range check:")
print(f"  Expected range: [0, ~1] or [0, ~10000] depending on scaling")
print(f"  Actual range: [{X.min():.4f}, {X.max():.4f}]")

# Spatial continuity check
from scipy.ndimage import binary_dilation

isolated_pixels = 0
for class_id in range(1, 10):
    mask = (y == class_id)
    dilated = binary_dilation(mask, iterations=1)
    isolated = mask & ~dilated
    isolated_pixels += isolated.sum()

print(f"\nSpatial coherence:")
print(f"  Isolated labeled pixels: {isolated_pixels}")

print("\n" + "="*70)
print("EDA COMPLETE")
print("="*70)
print("\nKey Findings Summary:")
print(f"1. Dataset has {bands} spectral bands covering {wavelengths[0]:.0f}-{wavele
print(f"2. {len(labeled_classes)} classes with imbalance ratio of {imbalance_ratio:
print(f"3. Only {(labeled_samples/total_pixels)*100:.2f}% of pixels are labeled")
```

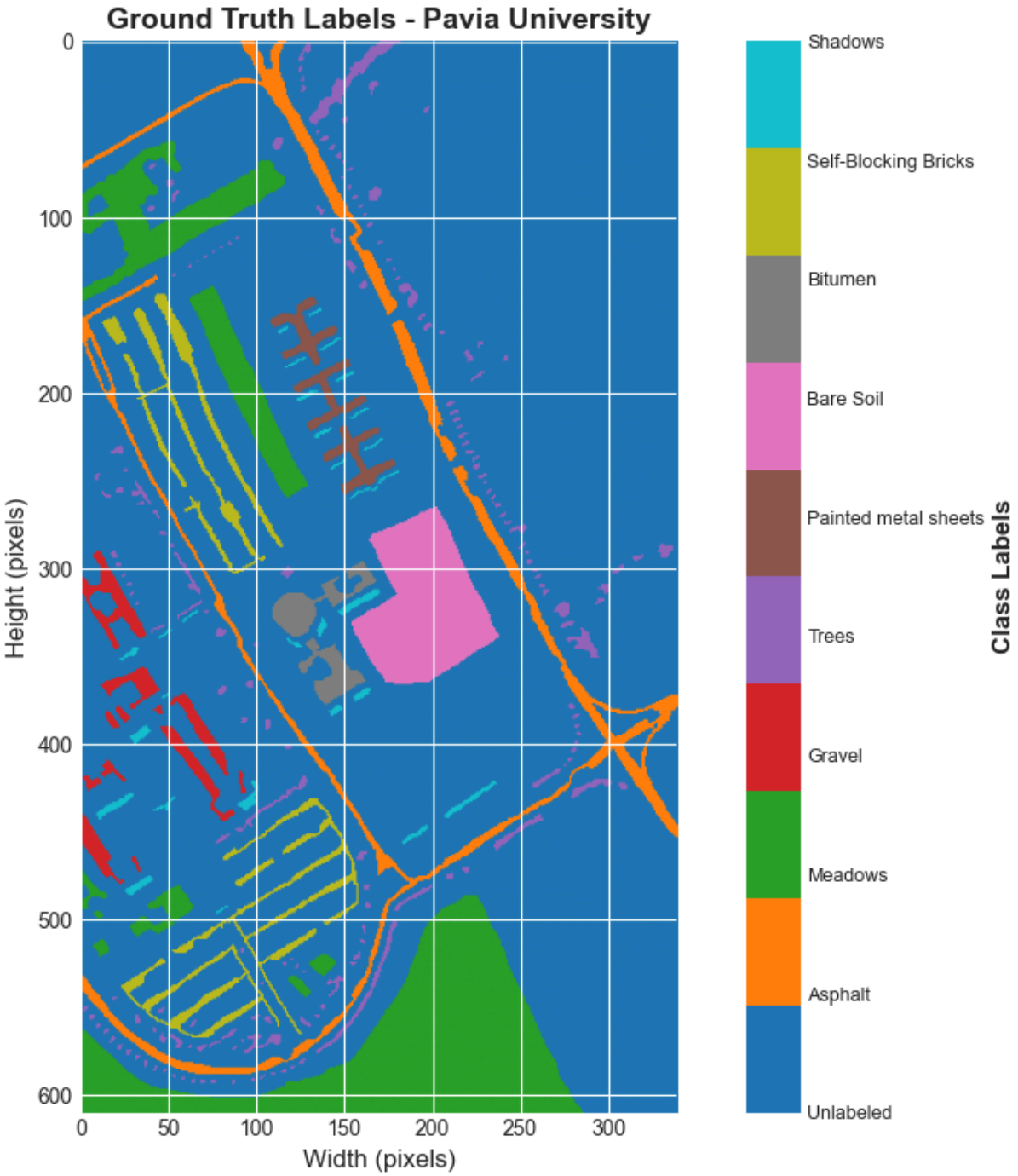
```
print(f"4. {n_95} components capture 95% of spectral variance")
print(f"5. Data quality: {'GOOD' if nan_count == 0 and inf_count == 0 else 'NEEDS A
```

```
================================================================================
LOADING PAVIA UNIVERSITY DATASET
================================================================================
Hyperspectral Image Shape: (610, 340, 103)
Ground Truth Shape: (610, 340)
Data type: uint16
Ground Truth data type: uint8

================================================================================
BASIC DATASET STATISTICS
================================================================================
Image Dimensions: 610 x 340 pixels
Number of Spectral Bands: 103
Total Pixels: 207,400
Spectral Range: 430-960 nm (5 nm spacing)

Data Range:
  Min value: 0.0000
  Max value: 8000.0000
  Mean value: 1389.1253
  Std deviation: 897.6575

================================================================================
CLASS DISTRIBUTION ANALYSIS
================================================================================

Class Distribution:
--------------------------------------------------------------------------------
Class 0 (Unlabeled): 164,624 (79.38%)
Class 1 (Asphalt): 6,631 (3.20%)
Class 2 (Meadows): 18,649 (8.99%)
Class 3 (Gravel): 2,099 (1.01%)
Class 4 (Trees): 3,064 (1.48%)
Class 5 (Painted metal sheets): 1,345 (0.65%)
Class 6 (Bare Soil): 5,029 (2.42%)
Class 7 (Bitumen): 1,330 (0.64%)
Class 8 (Self-Blocking Bricks): 3,682 (1.78%)
Class 9 (Shadows): 947 (0.46%)

Total Labeled Samples: 42,776
Total Unlabeled Samples: 164,624
Percentage Labeled: 20.62%

Class Imbalance Ratio (max/min): 19.69:1
```

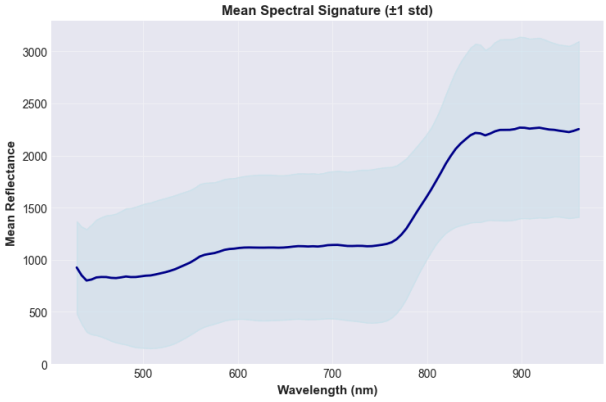


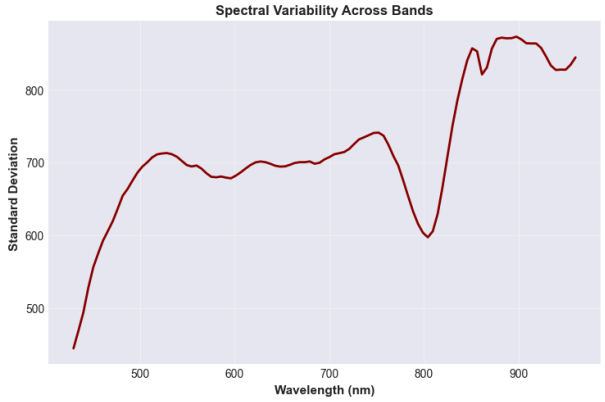


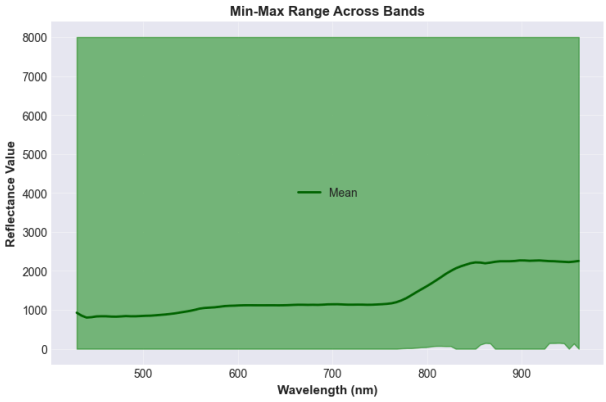

```
======================================================================
VISUALIZING GROUND TRUTH MAP
======================================================================
```

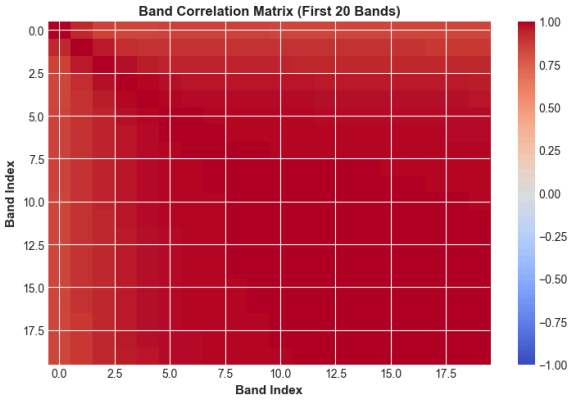



```
======================================================================
SPECTRAL BAND ANALYSIS
======================================================================
Band Statistics Summary:
  Average band mean: 1389.1253
  Average band std: 716.4461
  Band with highest mean: 90 (wavelength: 897.6 nm)
  Band with lowest mean: 2 (wavelength: 440.4 nm)
```





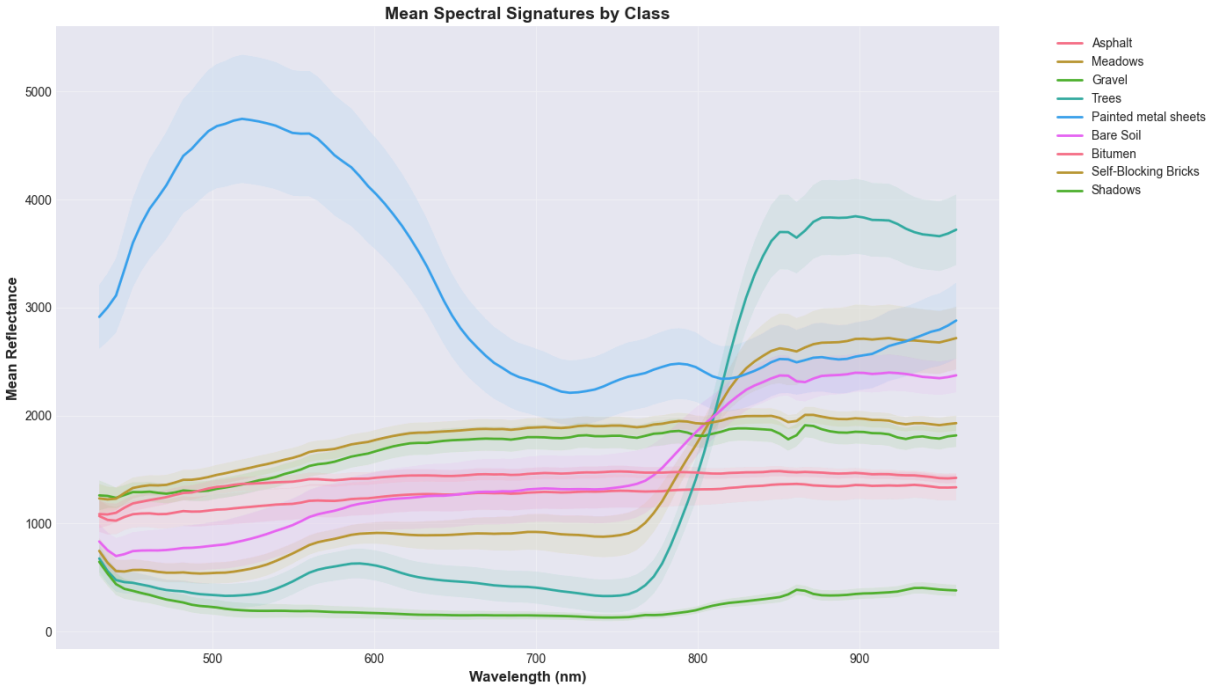




==========

CLASS-SPECIFIC SPECTRAL SIGNATURES

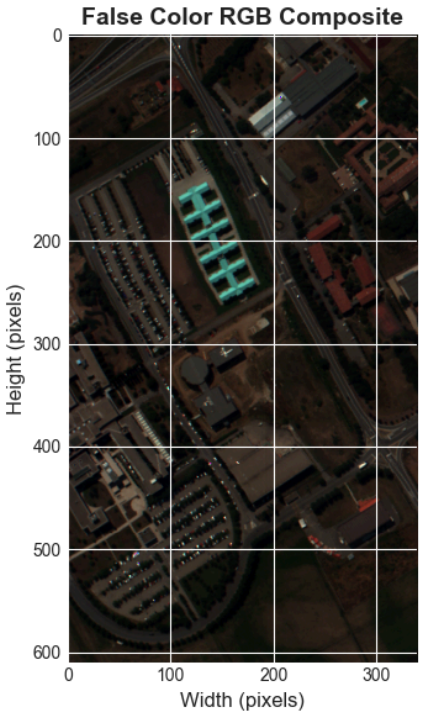
==========

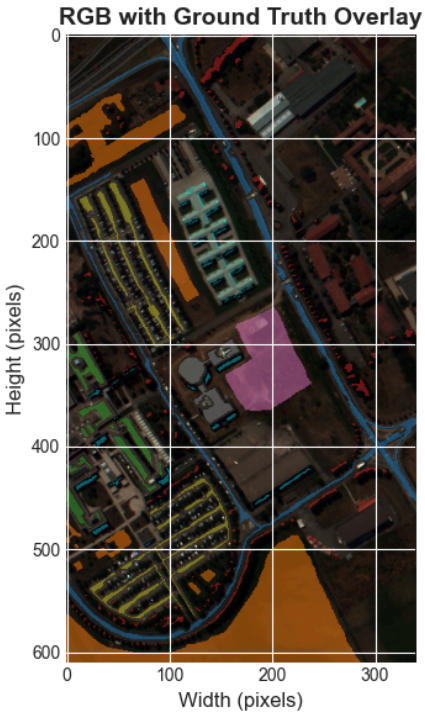


==========

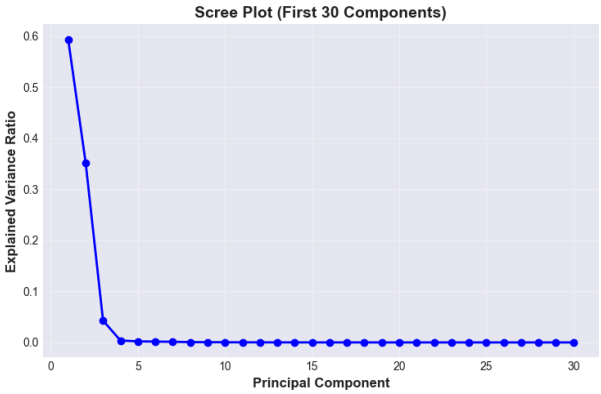
RGB COMPOSITE VISUALIZATION

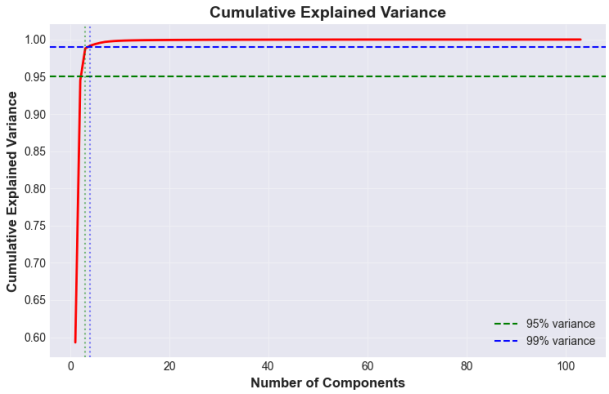
==========






```
======================================================================
DIMENSIONALITY REDUCTION ANALYSIS (PCA)
======================================================================
Number of components for 95% variance: 3
Number of components for 99% variance: 4
Variance explained by first 3 components: 98.80%
Variance explained by first 10 components: 99.84%
```





```
======================================================================
DATA QUALITY CHECKS
======================================================================
NaN values: 0
Inf values: 0
Negative values: 0 (0.0000%)

Data range check:
  Expected range: [0, ~1] or [0, ~10000] depending on scaling
  Actual range: [0.0000, 8000.0000]

Spatial coherence:
  Isolated labeled pixels: 0

======================================================================
EDA COMPLETE
======================================================================

Key Findings Summary:
1. Dataset has 103 spectral bands covering 430-960 nm
2. 9 classes with imbalance ratio of 19.69:1
3. Only 20.62% of pixels are labeled
4. 3 components capture 95% of spectral variance
5. Data quality: GOOD
```

Explanation: Performance in categorization will be impacted by several significant aspects of the PaviaU dataset. Because of the stark class imbalance (19.69:1 ratio), with Meadows dominating at 8.99% and Shadows being the minority class at only 0.46%, classifiers are likely to bias toward majority classes unless proper balancing procedures are used. Since almost 80% of pixels are unlabeled, semi-supervised learning approaches may be used. The spectrum analysis reveals important insights: The reflectance patterns of materials and vegetation that are frequently found in urban settings are consistent with the higher mean reflectance at 897.6 nm (near-infrared) as opposed to 440.4 nm (blue visible).

1.b

In [5]:
```python
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.io import loadmat
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (classification_report, confusion_matrix,
                             accuracy_score, precision_recall_fscore_support,
                             roc_curve, auc, roc_auc_score)
from sklearn.model_selection import train_test_split
import pandas as pd

# Set random seed for reproducibility
np.random.seed(42)


# 1. LOAD AND PREPARE DATA
```

```
print("="*70)
print("BINARY CLASSIFICATION: VEGETATION vs NON-VEGETATION")
print("="*70)

# Load data
data = loadmat('PaviaU.mat')
gt = loadmat('PaviaU_gt.mat')

X = data['paviaU']
y = gt['paviaU_gt']

height, width, bands = X.shape
print(f"\nDataset Shape: {height} x {width} x {bands}")


# 2. CREATE BINARY LABELS

print("\n" + "="*70)
print("CREATING BINARY LABELS")
print("="*70)

# Class mapping
class_names = {
    0: 'Unlabeled',
    1: 'Asphalt',
    2: 'Meadows',          # VEGETATION
    3: 'Gravel',
    4: 'Trees',            # VEGETATION
    5: 'Painted metal sheets',
    6: 'Bare Soil',
    7: 'Bitumen',
    8: 'Self-Blocking Bricks',
    9: 'Shadows'
}

# Define vegetation classes
vegetation_classes = [2, 4]  # Meadows and Trees
non_vegetation_classes = [1, 3, 5, 6, 7, 8, 9]  # All others except unlabeled

print("Vegetation Classes:")
for c in vegetation_classes:
    print(f"  Class {c}: {class_names[c]}")

print("\nNon-Vegetation Classes:")
for c in non_vegetation_classes:
    print(f"  Class {c}: {class_names[c]}")



# 3. ANALYZE ORIGINAL CLASS DISTRIBUTION IN BINARY PROBLEM

print("\n" + "="*70)
print("ORIGINAL CLASS DISTRIBUTION IN BINARY PROBLEM")
print("="*70)
```

```
# Count samples per original class in each binary category
print("\nVEGETATION Group:")
print("-" * 50)
veg_total = 0
for class_id in vegetation_classes:
    count = np.sum(y == class_id)
    veg_total += count
    print(f"  {class_names[class_id]}: {count:,} samples")
print(f"Total Vegetation: {veg_total:,}")

print("\nNON-VEGETATION Group:")
print("-" * 50)
non_veg_total = 0
class_counts_non_veg = {}
for class_id in non_vegetation_classes:
    count = np.sum(y == class_id)
    non_veg_total += count
    class_counts_non_veg[class_id] = count
    print(f"  {class_names[class_id]}: {count:,} samples")
print(f"Total Non-Vegetation: {non_veg_total:,}")

print(f"\nBinary Class Balance:")
print(f"  Vegetation: {veg_total:,} ({veg_total/(veg_total+non_veg_total)*100:.2f}%
print(f"  Non-Vegetation: {non_veg_total:,} ({non_veg_total/(veg_total+non_veg_tota
print(f"  Imbalance Ratio: {max(veg_total, non_veg_total)/min(veg_total, non_veg_to


# 4. STRATIFIED SAMPLING STRATEGY

print("\n" + "="*70)
print("STRATIFIED SAMPLING STRATEGY")
print("="*70)



# 5. PREPARE DATA FOR CLASSIFICATION

print("\n" + "="*70)
print("PREPARING DATA")
print("="*70)

# Reshape spectral data
X_reshaped = X.reshape(-1, bands)
y_flat = y.ravel()

# CRITICAL: Get only labeled pixels
labeled_mask = y_flat > 0
X_labeled = X_reshaped[labeled_mask]
y_original = y_flat[labeled_mask]

print(f"Total pixels in image: {len(y_flat):,}")
print(f"Unlabeled pixels (class 0): {np.sum(y_flat == 0):,}")
print(f"Total LABELED samples (excluding class 0): {len(y_original):,}")
```

```python
# Create binary labels from the labeled data only
# Vegetation (Trees=4, Meadows=2) = 1
# Non-Vegetation (all others: 1,3,5,6,7,8,9) = 0
y_binary_final = np.zeros(len(y_original), dtype=int)
y_binary_final[np.isin(y_original, vegetation_classes)] = 1

print(f"\nBinary Label Distribution:")
print(f"  Vegetation (1): {np.sum(y_binary_final == 1):,}")
print(f"  Non-Vegetation (0): {np.sum(y_binary_final == 0):,}")
print(f"  Total: {len(y_binary_final):,}")


# 6. STRATIFIED TRAIN-TEST SPLIT

print("\n" + "="*70)
print("PERFORMING STRATIFIED TRAIN-TEST SPLIT")
print("="*70)

# Use original class labels for stratification to ensure equal representation
test_size = 0.30  # 70/30 split
print(f"Split Ratio: {(1-test_size)*100:.0f}% Train / {test_size*100:.0f}% Test")

# Stratified split based on ORIGINAL 9-class labels
X_train, X_test, y_train_binary, y_test_binary, y_train_original, y_test_original =
    X_labeled,
    y_binary_final,
    y_original,
    test_size=test_size,
    stratify=y_original,  # Stratify by original classes
    random_state=42
)

print(f"\nTraining set size: {len(X_train):,}")
print(f"Testing set size: {len(X_test):,}")


# 7. VERIFY STRATIFICATION

print("\n" + "="*70)
print("VERIFYING STRATIFICATION")
print("="*70)

print("\nOriginal Class Distribution in Train/Test Sets:")
print("-" * 70)
print(f"{'Class':<25} {'Original':>10} {'Train':>10} {'Test':>10} {'Train %':>10} {
print("-" * 70)

for class_id in range(1, 10):
    original_count = np.sum(y_original == class_id)
    train_count = np.sum(y_train_original == class_id)
    test_count = np.sum(y_test_original == class_id)
    train_pct = (train_count / original_count) * 100 if original_count > 0 else 0
    test_pct = (test_count / original_count) * 100 if original_count > 0 else 0

    print(f"{class_names[class_id]:<25} {original_count:>10,} {train_count:>10,} "
```

```
          f"{test_count:>10,} {train_pct:>9.1f}% {test_pct:>9.1f}%")

print("\nBinary Class Distribution:")
print("-" * 70)
print(f"{'Category':<25} {'Train Count':>15} {'Train %':>10} {'Test Count':>15} {'T
print("-" * 70)

veg_train = np.sum(y_train_binary == 1)
non_veg_train = np.sum(y_train_binary == 0)
veg_test = np.sum(y_test_binary == 1)
non_veg_test = np.sum(y_test_binary == 0)

print(f"{'Vegetation':<25} {veg_train:>15,} {veg_train/len(y_train_binary)*100:>9.1
      f"{veg_test:>15,} {veg_test/len(y_test_binary)*100:>9.1f}%")
print(f"{'Non-Vegetation':<25} {non_veg_train:>15,} {non_veg_train/len(y_train_bina
      f"{non_veg_test:>15,} {non_veg_test/len(y_test_binary)*100:>9.1f}%")


# 8. SELECT SPECIFIC SPECTRAL BANDS

print("\n" + "="*70)
print("SELECTING SPECIFIC SPECTRAL BANDS")
print("="*70)

# Calculate wavelengths
wavelengths = np.linspace(430, 960, bands)
print(f"Spectral range: {wavelengths[0]:.1f} - {wavelengths[-1]:.1f} nm")
print(f"Spectral resolution: {(wavelengths[1] - wavelengths[0]):.2f} nm/band")

#target wavelengths for each spectral region
target_wavelengths = {
    'Blue': 470,        # Blue region: ~450-495 nm
    'Green': 550,       # Green region: ~495-570 nm
    'Red': 670,         # Red region: ~620-750 nm
    'Red-edge': 720,    # Red-edge region: ~690-730 nm (vegetation transition)
    'NIR': 850          # Near-Infrared: ~750-900 nm
}

# Find closest band index for each target wavelength
selected_bands = {}
selected_band_indices = []

print("\nSelected Bands:")
print("-" * 70)
print(f"{'Region':<15} {'Target (nm)':<15} {'Actual (nm)':<15} {'Band Index':<15}")
print("-" * 70)

for region, target_wl in target_wavelengths.items():
    # Find closest wavelength
    idx = np.argmin(np.abs(wavelengths - target_wl))
    actual_wl = wavelengths[idx]
    selected_bands[region] = idx
    selected_band_indices.append(idx)
    print(f"{region:<15} {target_wl:<15.1f} {actual_wl:<15.2f} {idx:<15}")

print(f"\nTotal selected bands: {len(selected_band_indices)}")
```

```
print(f"Using bands: {selected_band_indices}")

# Extract only the selected bands from training and testing data
X_train_selected = X_train[:, selected_band_indices]
X_test_selected = X_test[:, selected_band_indices]

print(f"\nReduced feature dimensions:")
print(f"  Original: {X_train.shape[1]} bands")
print(f"  Selected: {X_train_selected.shape[1]} bands")
print(f"  Reduction: {(1 - X_train_selected.shape[1]/X_train.shape[1])*100:.1f}%")


# 9. FEATURE SCALING

print("\n" + "="*70)
print("FEATURE SCALING")
print("="*70)

# Standardize features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_selected)
X_test_scaled = scaler.transform(X_test_selected)

print("Applied StandardScaler (zero mean, unit variance)")
print(f"Training data shape: {X_train_scaled.shape}")
print(f"  Mean: {X_train_scaled.mean():.6f}")
print(f"  Std: {X_train_scaled.std():.6f}")

# Visualize selected bands on spectral signature
print("\nVisualizing selected bands on mean spectral signature...")
fig, ax = plt.subplots(figsize=(14, 6))

# Plot mean spectral signature of all training data
mean_signature = np.mean(X_train, axis=0)
ax.plot(wavelengths, mean_signature, linewidth=2, color='gray', alpha=0.7,
        label='All 103 bands')

# Highlight selected bands
colors = {'Blue': 'blue', 'Green': 'green', 'Red': 'red',
          'Red-edge': 'orange', 'NIR': 'darkred'}

for region, idx in selected_bands.items():
    ax.scatter(wavelengths[idx], mean_signature[idx],
               s=200, c=colors[region], marker='o',
               edgecolors='black', linewidths=2, zorder=5,
               label=f'{region} ({wavelengths[idx]:.1f} nm)')
    ax.axvline(x=wavelengths[idx], color=colors[region],
               linestyle='--', alpha=0.3)

ax.set_xlabel('Wavelength (nm)', fontsize=12, fontweight='bold')
ax.set_ylabel('Mean Reflectance', fontsize=12, fontweight='bold')
ax.set_title('Selected Spectral Bands for Classification', fontsize=14, fontweight=
ax.legend(loc='best', fontsize=10)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

```
# 9. TRAIN LOGISTIC REGRESSION MODEL

print("\n" + "="*70)
print("TRAINING LOGISTIC REGRESSION MODEL")
print("="*70)

# Train model with balanced class weights to handle imbalance
lr_model = LogisticRegression(
    random_state=42,
    max_iter=1000,
    class_weight='balanced',  # Handle class imbalance
    solver='lbfgs',
    verbose=1
)

print(f"Training model with {X_train_scaled.shape[1]} selected spectral bands...")
lr_model.fit(X_train_scaled, y_train_binary)
print("Training complete!")

print(f"\nModel Parameters:")
print(f"  Number of features: {X_train_scaled.shape[1]}")
print(f"  Number of coefficients: {len(lr_model.coef_[0])}")
print(f"  Intercept: {lr_model.intercept_[0]:.6f}")


# 10. MAKE PREDICTIONS

print("\n" + "="*70)
print("MAKING PREDICTIONS")
print("="*70)

# Predictions
y_train_pred = lr_model.predict(X_train_scaled)
y_test_pred = lr_model.predict(X_test_scaled)

# Prediction probabilities
y_train_proba = lr_model.predict_proba(X_train_scaled)[:, 1]
y_test_proba = lr_model.predict_proba(X_test_scaled)[:, 1]

print("Predictions complete!")


# 11. EVALUATE MODEL PERFORMANCE

print("\n" + "="*70)
print("MODEL PERFORMANCE EVALUATION")
print("="*70)

# Training accuracy
train_accuracy = accuracy_score(y_train_binary, y_train_pred)
print(f"\nTraining Accuracy: {train_accuracy*100:.2f}%")

# Testing accuracy
test_accuracy = accuracy_score(y_test_binary, y_test_pred)
```

```
print(f"Testing Accuracy: {test_accuracy*100:.2f}%")

# Detailed metrics
print("\nClassification Report (Test Set):")
print("-" * 70)
target_names = ['Non-Vegetation', 'Vegetation']
print(classification_report(y_test_binary, y_test_pred, target_names=target_names,

# Per-class metrics
precision, recall, f1, support = precision_recall_fscore_support(
    y_test_binary, y_test_pred, average=None
)



# ROC-AUC Score
roc_auc = roc_auc_score(y_test_binary, y_test_proba)
print(f"\nROC-AUC Score: {roc_auc:.4f}")


# 12. CONFUSION MATRIX

print("\n" + "="*70)
print("CONFUSION MATRIX")
print("="*70)

cm = confusion_matrix(y_test_binary, y_test_pred)
print("\nConfusion Matrix:")
print(cm)

# Visualize confusion matrix
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Raw counts
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=target_names, yticklabels=target_names,
            ax=axes[0], cbar_kws={'label': 'Count'})
axes[0].set_title('Confusion Matrix (Counts)', fontsize=14, fontweight='bold')
axes[0].set_ylabel('True Label', fontsize=12, fontweight='bold')
axes[0].set_xlabel('Predicted Label', fontsize=12, fontweight='bold')

# Normalized
cm_normalized = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
sns.heatmap(cm_normalized, annot=True, fmt='.2%', cmap='Blues',
            xticklabels=target_names, yticklabels=target_names,
            ax=axes[1], cbar_kws={'label': 'Percentage'})
axes[1].set_title('Confusion Matrix (Normalized)', fontsize=14, fontweight='bold')
axes[1].set_ylabel('True Label', fontsize=12, fontweight='bold')
axes[1].set_xlabel('Predicted Label', fontsize=12, fontweight='bold')

plt.tight_layout()
plt.show()


# 13. ROC CURVE
```

```
print("\n" + "="*70)
print("ROC CURVE ANALYSIS")
print("="*70)

# Calculate ROC curve
fpr, tpr, thresholds = roc_curve(y_test_binary, y_test_proba)

# Plot ROC curve
fig, ax = plt.subplots(figsize=(10, 8))
ax.plot(fpr, tpr, linewidth=3, label=f'Logistic Regression (AUC = {roc_auc:.4f})',
ax.plot([0, 1], [0, 1], 'k--', linewidth=2, label='Random Classifier (AUC = 0.50)')
ax.set_xlabel('False Positive Rate', fontsize=12, fontweight='bold')
ax.set_ylabel('True Positive Rate', fontsize=12, fontweight='bold')
ax.set_title('ROC Curve - Vegetation vs Non-Vegetation', fontsize=14, fontweight='b
ax.legend(fontsize=11, loc='lower right')
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()


# 14. PREDICTION DISTRIBUTION

print("\n" + "="*70)
print("PREDICTION PROBABILITY DISTRIBUTION")
print("="*70)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Training set
axes[0].hist(y_train_proba[y_train_binary == 0], bins=50, alpha=0.6,
             label='Non-Vegetation', color='red', edgecolor='black')
axes[0].hist(y_train_proba[y_train_binary == 1], bins=50, alpha=0.6,
             label='Vegetation', color='green', edgecolor='black')
axes[0].set_xlabel('Predicted Probability (Vegetation)', fontsize=12, fontweight='b
axes[0].set_ylabel('Frequency', fontsize=12, fontweight='bold')
axes[0].set_title('Training Set - Prediction Probabilities', fontsize=13, fontweigh
axes[0].legend()
axes[0].grid(alpha=0.3)

# Testing set
axes[1].hist(y_test_proba[y_test_binary == 0], bins=50, alpha=0.6,
             label='Non-Vegetation', color='red', edgecolor='black')
axes[1].hist(y_test_proba[y_test_binary == 1], bins=50, alpha=0.6,
             label='Vegetation', color='green', edgecolor='black')
axes[1].set_xlabel('Predicted Probability (Vegetation)', fontsize=12, fontweight='b
axes[1].set_ylabel('Frequency', fontsize=12, fontweight='bold')
axes[1].set_title('Testing Set - Prediction Probabilities', fontsize=13, fontweight
axes[1].legend()
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.show()


# 15. FEATURE IMPORTANCE (COEFFICIENTS)
```

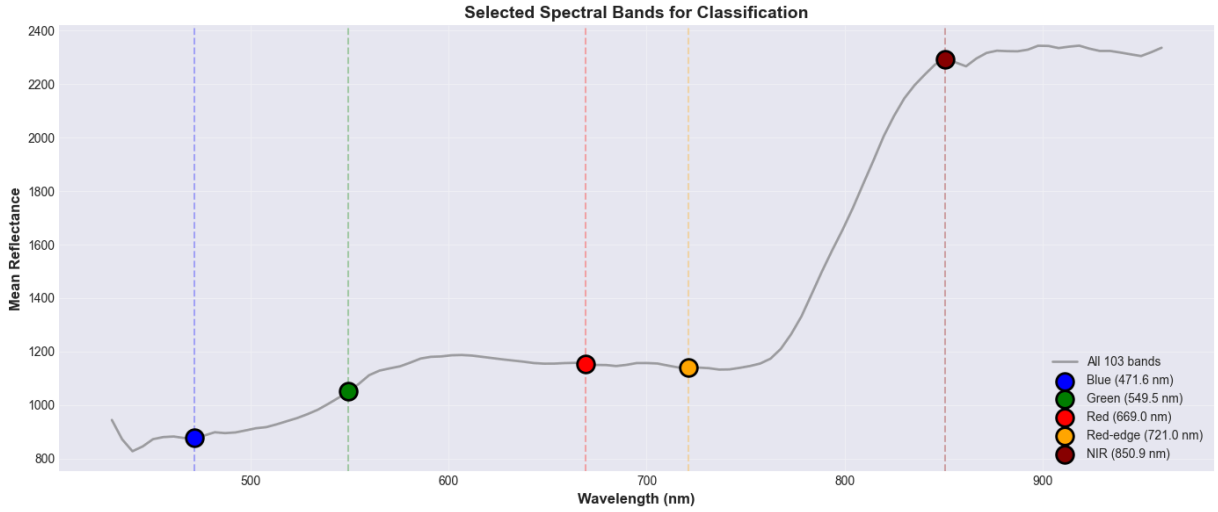
```
print("\n" + "="*70)
print("FEATURE IMPORTANCE ANALYSIS")
print("="*70)

# Get coefficients for the 5 selected bands
coefficients = lr_model.coef_[0]

print("\nLogistic Regression Coefficients for Selected Bands:")
print("-" * 70)
print(f"{'Region':<15} {'Wavelength (nm)':<18} {'Band Index':<15} {'Coefficient':<1
print("-" * 70)

regions_list = ['Blue', 'Green', 'Red', 'Red-edge', 'NIR']
for i, region in enumerate(regions_list):
    idx = selected_bands[region]
    wl = wavelengths[idx]
    coef = coefficients[i]
    print(f"{region:<15} {wl:<18.2f} {idx:<15} {coef:<15.6f}")

print("\nInterpretation:")
print("-" * 70)
for i, region in enumerate(regions_list):
    coef = coefficients[i]
    if coef > 0:
        print(f"  {region}: Positive coefficient ({coef:.4f}) → Higher reflectance
    else:
        print(f"  {region}: Negative coefficient ({coef:.4f}) → Higher reflectance

# Visualize coefficients
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# Bar plot of coefficients
colors_bar = [colors[region] for region in regions_list]
axes[0].bar(regions_list, coefficients, color=colors_bar, edgecolor='black', linewi
axes[0].axhline(y=0, color='black', linestyle='-', linewidth=1.5)
axes[0].set_xlabel('Spectral Region', fontsize=12, fontweight='bold')
axes[0].set_ylabel('Coefficient Value', fontsize=12, fontweight='bold')
axes[0].set_title('Logistic Regression Coefficients by Spectral Region',
                  fontsize=13, fontweight='bold')
axes[0].grid(axis='y', alpha=0.3)

# Add value labels on bars
for i, (region, coef) in enumerate(zip(regions_list, coefficients)):
    axes[0].text(i, coef + (0.1 if coef > 0 else -0.1), f'{coef:.3f}',
                 ha='center', va='bottom' if coef > 0 else 'top',
                 fontweight='bold', fontsize=10)

# Coefficient importance (absolute values)
abs_coef = np.abs(coefficients)
axes[1].bar(regions_list, abs_coef, color=colors_bar, edgecolor='black', linewidth=
axes[1].set_xlabel('Spectral Region', fontsize=12, fontweight='bold')
axes[1].set_ylabel('Absolute Coefficient Value', fontsize=12, fontweight='bold')
axes[1].set_title('Feature Importance (Absolute Coefficients)',
                  fontsize=13, fontweight='bold')
axes[1].grid(axis='y', alpha=0.3)
```

```python
# Add value labels
for i, (region, abs_c) in enumerate(zip(regions_list, abs_coef)):
    axes[1].text(i, abs_c + 0.05, f'{abs_c:.3f}',
                 ha='center', va='bottom', fontweight='bold', fontsize=10)

plt.tight_layout()
plt.show()

# Find most important feature
most_important_idx = np.argmax(abs_coef)
most_important_region = regions_list[most_important_idx]
print(f"\nMost important spectral region: {most_important_region} "
      f"(coefficient = {coefficients[most_important_idx]:.4f})")


# 16. SUMMARY

print("\n" + "="*70)
print("SUMMARY")
print("="*70)
print(f"""
Binary Classification Results:
-------------------------------
Training Accuracy:     {train_accuracy*100:.2f}%
Testing Accuracy:      {test_accuracy*100:.2f}%
ROC-AUC Score:         {roc_auc:.4f}

Precision (Veg):       {precision[1]:.4f}
Recall (Veg):          {recall[1]:.4f}
F1-Score (Veg):        {f1[1]:.4f}

Precision (Non-Veg):   {precision[0]:.4f}
Recall (Non-Veg):      {recall[0]:.4f}
F1-Score (Non-Veg):    {f1[0]:.4f}

Data Split:
-----------
Training samples:      {len(X_train):,}
Testing samples:       {len(X_test):,}
Split ratio:           70/30

Stratification:        ✓ By original 9 classes
Class weighting:       ✓ Balanced
Feature scaling:       ✓ StandardScaler
""")

print("="*70)
print("CLASSIFICATION COMPLETE")
print("="*70)
```

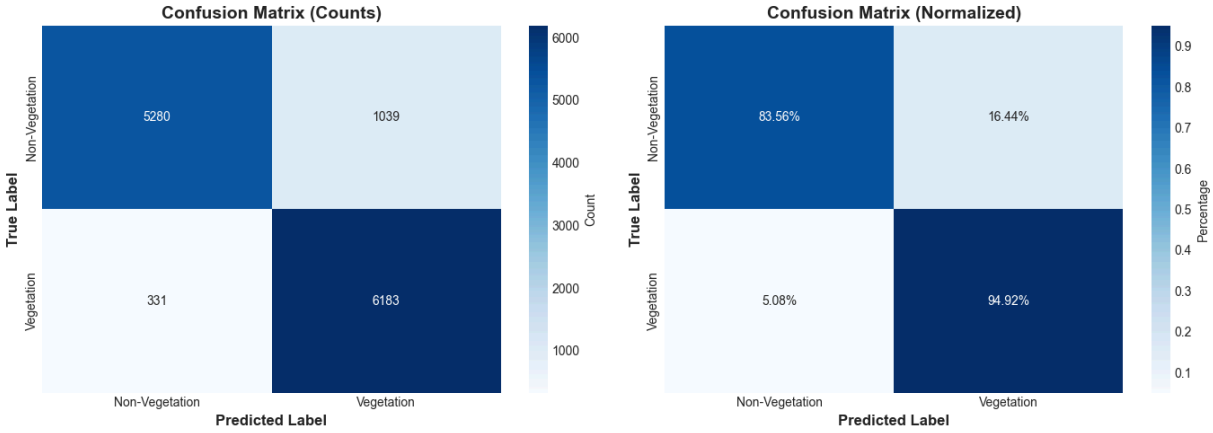
```
================================================================================
BINARY CLASSIFICATION: VEGETATION vs NON-VEGETATION
================================================================================

Dataset Shape: 610 x 340 x 103

================================================================================
CREATING BINARY LABELS
================================================================================
Vegetation Classes:
  Class 2: Meadows
  Class 4: Trees

Non-Vegetation Classes:
  Class 1: Asphalt
  Class 3: Gravel
  Class 5: Painted metal sheets
  Class 6: Bare Soil
  Class 7: Bitumen
  Class 8: Self-Blocking Bricks
  Class 9: Shadows

================================================================================
ORIGINAL CLASS DISTRIBUTION IN BINARY PROBLEM
================================================================================

VEGETATION Group:
--------------------------------------------------
  Meadows: 18,649 samples
  Trees: 3,064 samples
Total Vegetation: 21,713

NON-VEGETATION Group:
--------------------------------------------------
  Asphalt: 6,631 samples
  Gravel: 2,099 samples
  Painted metal sheets: 1,345 samples
  Bare Soil: 5,029 samples
  Bitumen: 1,330 samples
  Self-Blocking Bricks: 3,682 samples
  Shadows: 947 samples
Total Non-Vegetation: 21,063

Binary Class Balance:
  Vegetation: 21,713 (50.76%)
  Non-Vegetation: 21,063 (49.24%)
  Imbalance Ratio: 1.03:1

================================================================================
STRATIFIED SAMPLING STRATEGY
================================================================================

================================================================================
PREPARING DATA
================================================================================
Total pixels in image: 207,400
```

```
Unlabeled pixels (class 0): 164,624
Total LABELED samples (excluding class 0): 42,776

Binary Label Distribution:
  Vegetation (1): 21,713
  Non-Vegetation (0): 21,063
  Total: 42,776

======================================================================
PERFORMING STRATIFIED TRAIN-TEST SPLIT
======================================================================
Split Ratio: 70% Train / 30% Test

Training set size: 29,943
Testing set size: 12,833

======================================================================
VERIFYING STRATIFICATION
======================================================================

Original Class Distribution in Train/Test Sets:
----------------------------------------------------------------------
Class                       Original      Train       Test    Train %     Test %
----------------------------------------------------------------------
Asphalt                        6,631      4,642      1,989      70.0%      30.0%
Meadows                       18,649     13,054      5,595      70.0%      30.0%
Gravel                         2,099      1,469        630      70.0%      30.0%
Trees                          3,064      2,145        919      70.0%      30.0%
Painted metal sheets           1,345        942        403      70.0%      30.0%
Bare Soil                      5,029      3,520      1,509      70.0%      30.0%
Bitumen                        1,330        931        399      70.0%      30.0%
Self-Blocking Bricks           3,682      2,577      1,105      70.0%      30.0%
Shadows                          947        663        284      70.0%      30.0%

Binary Class Distribution:
----------------------------------------------------------------------
Category                    Train Count    Train %      Test Count     Test %
----------------------------------------------------------------------
Vegetation                       15,199      50.8%           6,514      50.8%
Non-Vegetation                   14,744      49.2%           6,319      49.2%

======================================================================
SELECTING SPECIFIC SPECTRAL BANDS
======================================================================
Spectral range: 430.0 - 960.0 nm
Spectral resolution: 5.20 nm/band

Selected Bands:
----------------------------------------------------------------------
Region          Target (nm)     Actual (nm)     Band Index
----------------------------------------------------------------------
Blue            470.0           471.57          8
Green           550.0           549.51          23
Red             670.0           669.02          46
Red-edge        720.0           720.98          56
NIR             850.0           850.88          81
```

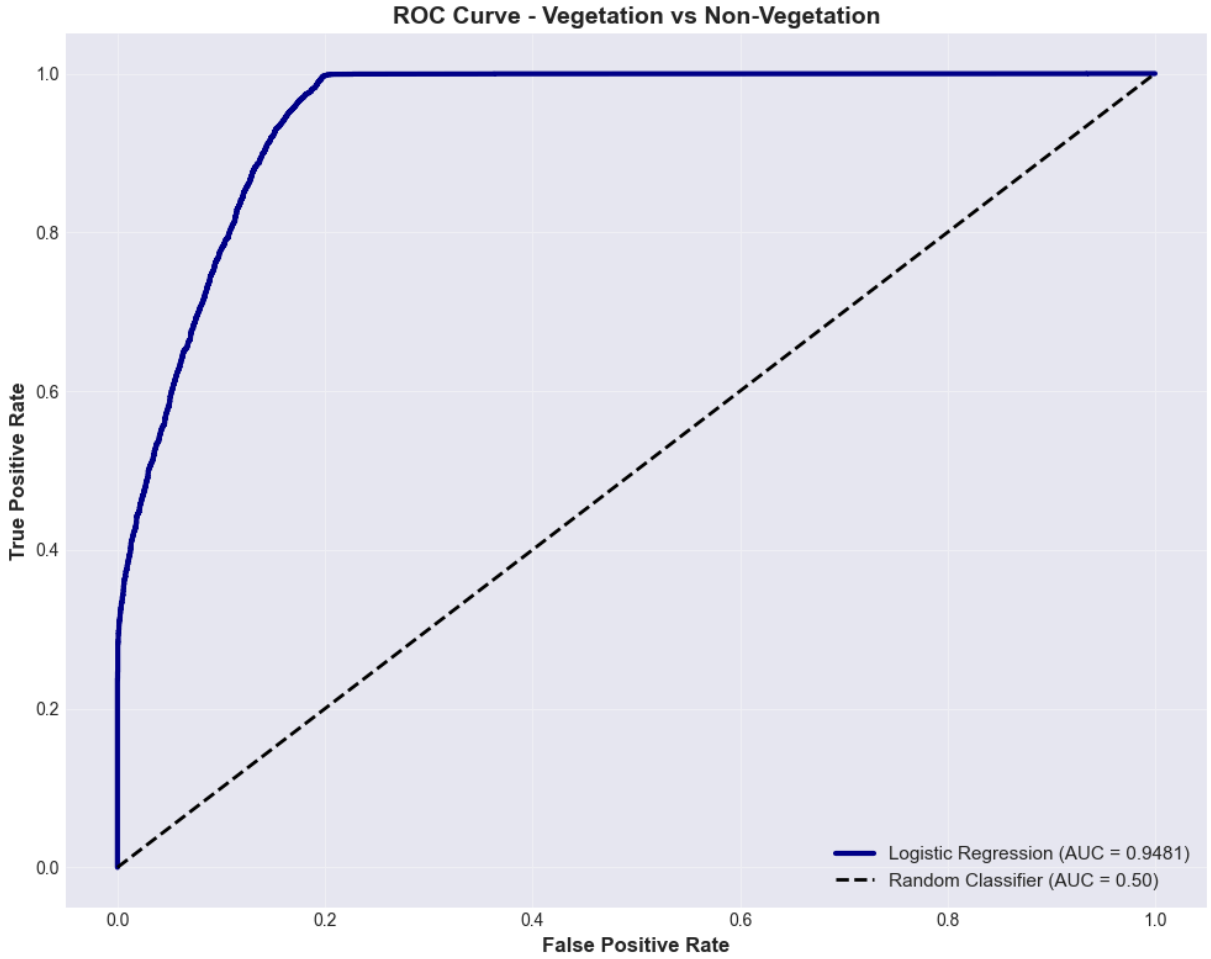
```
Total selected bands: 5
Using bands: [np.int64(8), np.int64(23), np.int64(46), np.int64(56), np.int64(81)]

Reduced feature dimensions:
  Original: 103 bands
  Selected: 5 bands
  Reduction: 95.1%

======================================================================
FEATURE SCALING
======================================================================
Applied StandardScaler (zero mean, unit variance)
Training data shape: (29943, 5)
  Mean: 0.000000
  Std: 1.000000

Visualizing selected bands on mean spectral signature...
```



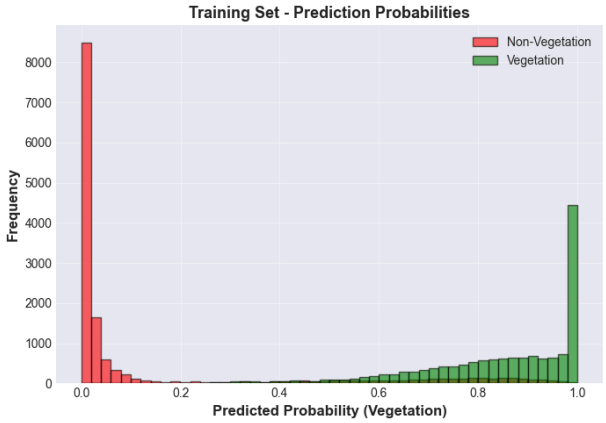
```
======================================================================
TRAINING LOGISTIC REGRESSION MODEL
======================================================================
Training model with 5 selected spectral bands...
Training complete!

Model Parameters:
  Number of features: 5
  Number of coefficients: 5
  Intercept: -0.427524

======================================================================
MAKING PREDICTIONS
======================================================================
Predictions complete!

======================================================================
MODEL PERFORMANCE EVALUATION
======================================================================

Training Accuracy: 89.53%
Testing Accuracy: 89.32%

Classification Report (Test Set):
----------------------------------------------------------------------
                precision    recall  f1-score   support

Non-Vegetation     0.9410    0.8356    0.8852      6319
    Vegetation     0.8561    0.9492    0.9003      6514

      accuracy                         0.8932     12833
     macro avg     0.8986    0.8924    0.8927     12833
  weighted avg     0.8979    0.8932    0.8928     12833


ROC-AUC Score: 0.9481

======================================================================
CONFUSION MATRIX
======================================================================

Confusion Matrix:
[[5280 1039]
 [ 331 6183]]
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:    0.0s finished
```

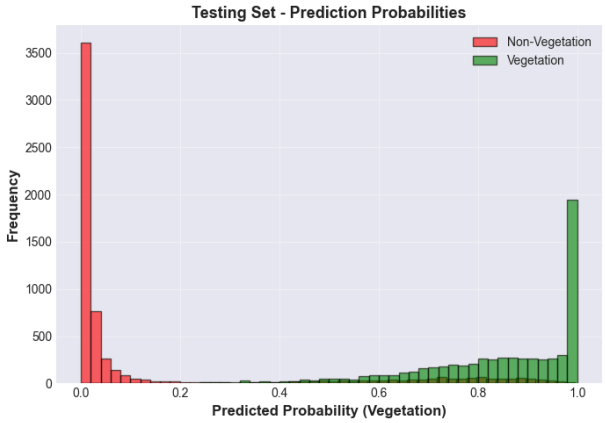



```
======================================================================
ROC CURVE ANALYSIS
======================================================================
```

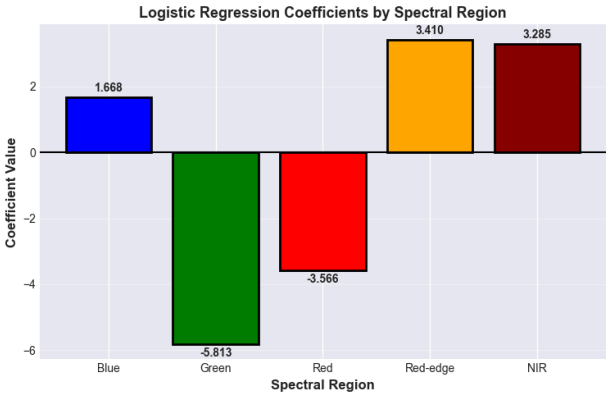


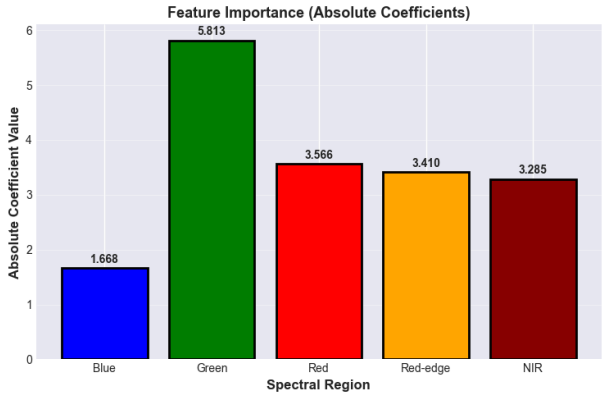

```
======================================================================
PREDICTION PROBABILITY DISTRIBUTION
======================================================================
```






```
======================================================================
FEATURE IMPORTANCE ANALYSIS
======================================================================

Logistic Regression Coefficients for Selected Bands:
----------------------------------------------------------------------
Region          Wavelength (nm)    Band Index      Coefficient
----------------------------------------------------------------------
Blue            471.57             8               1.667714
Green           549.51             23              -5.812935
Red             669.02             46              -3.565698
Red-edge        720.98             56              3.410441
NIR             850.88             81              3.285229

Interpretation:
----------------------------------------------------------------------
  Blue: Positive coefficient (1.6677) → Higher reflectance favors VEGETATION
  Green: Negative coefficient (-5.8129) → Higher reflectance favors NON-VEGETATION
  Red: Negative coefficient (-3.5657) → Higher reflectance favors NON-VEGETATION
  Red-edge: Positive coefficient (3.4104) → Higher reflectance favors VEGETATION
  NIR: Positive coefficient (3.2852) → Higher reflectance favors VEGETATION
```





```
Most important spectral region: Green (coefficient = -5.8129)

======================================================================
SUMMARY
======================================================================

Binary Classification Results:
------------------------------
Training Accuracy:     89.53%
Testing Accuracy:      89.32%
ROC-AUC Score:         0.9481

Precision (Veg):       0.8561
Recall (Veg):          0.9492
F1-Score (Veg):        0.9003

Precision (Non-Veg):   0.9410
Recall (Non-Veg):      0.8356
F1-Score (Non-Veg):    0.8852

Data Split:
-----------
Training samples:      29,943
Testing samples:       12,833
Split ratio:           70/30

Stratification:        ✓ By original 9 classes
Class weighting:       ✓ Balanced
Feature scaling:       ✓ StandardScaler

======================================================================
CLASSIFICATION COMPLETE
======================================================================
```

In [ ]:

```python
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import (accuracy_score, precision_recall_fscore_support,
                             confusion_matrix, roc_curve, auc, roc_auc_score)


# COMPREHENSIVE MODEL EVALUATION AND INTERPRETATION


print("="*80)
print("COMPREHENSIVE MODEL EVALUATION REPORT")
print("Binary Classification: Vegetation vs Non-Vegetation")
print("="*80)



# 1. ACCURACY METRICS

print("\n" + "="*80)
print("1. ACCURACY METRICS")
```

```
print("="*80)

# Training Accuracy
train_accuracy = accuracy_score(y_train_binary, y_train_pred)
print(f"\nTraining Set:")
print(f"  Overall Accuracy: {train_accuracy*100:.2f}%")

# Per-class accuracy for training
train_precision, train_recall, train_f1, train_support = precision_recall_fscore_su
    y_train_binary, y_train_pred, average=None
)
train_per_class_acc = []
for class_idx in [0, 1]:
    mask = (y_train_binary == class_idx)
    class_correct = np.sum((y_train_pred[mask] == class_idx))
    class_total = np.sum(mask)
    class_acc = class_correct / class_total
    train_per_class_acc.append(class_acc)

train_mean_per_class_acc = np.mean(train_per_class_acc)
print(f"  Per-Class Accuracy (Non-Vegetation): {train_per_class_acc[0]*100:.2f}%")
print(f"  Per-Class Accuracy (Vegetation): {train_per_class_acc[1]*100:.2f}%")
print(f"  Mean Per-Class Accuracy: {train_mean_per_class_acc*100:.2f}%")

# Testing Accuracy
test_accuracy = accuracy_score(y_test_binary, y_test_pred)
print(f"\nTesting Set:")
print(f"  Overall Accuracy: {test_accuracy*100:.2f}%")

# Per-class accuracy for testing
test_precision, test_recall, test_f1, test_support = precision_recall_fscore_suppor
    y_test_binary, y_test_pred, average=None
)
test_per_class_acc = []
for class_idx in [0, 1]:
    mask = (y_test_binary == class_idx)
    class_correct = np.sum((y_test_pred[mask] == class_idx))
    class_total = np.sum(mask)
    class_acc = class_correct / class_total
    test_per_class_acc.append(class_acc)

test_mean_per_class_acc = np.mean(test_per_class_acc)
print(f"  Per-Class Accuracy (Non-Vegetation): {test_per_class_acc[0]*100:.2f}%")
print(f"  Per-Class Accuracy (Vegetation): {test_per_class_acc[1]*100:.2f}%")
print(f"  Mean Per-Class Accuracy: {test_mean_per_class_acc*100:.2f}%")


# 2. PRECISION, RECALL, F1-SCORE - DETAILED

print("\n" + "="*80)
print("2. PRECISION, RECALL, AND F1-SCORE")
print("="*80)

class_names = ['Non-Vegetation', 'Vegetation']

# Training Set Metrics
```

```
print("\nTRAINING SET:")
print("-" * 80)
print(f"{'Class':<20} {'Precision':<12} {'Recall':<12} {'F1-Score':<12} {'Support':
print("-" * 80)
for i, name in enumerate(class_names):
    print(f"{name:<20} {train_precision[i]:<12.4f} {train_recall[i]:<12.4f} "
          f"{train_f1[i]:<12.4f} {train_support[i]:<12,}")

train_macro_precision = np.mean(train_precision)
train_macro_recall = np.mean(train_recall)
train_macro_f1 = np.mean(train_f1)

print("-" * 80)
print(f"{'Macro Average':<20} {train_macro_precision:<12.4f} {train_macro_recall:<1
      f"{train_macro_f1:<12.4f}")

# Testing Set Metrics
print("\nTESTING SET:")
print("-" * 80)
print(f"{'Class':<20} {'Precision':<12} {'Recall':<12} {'F1-Score':<12} {'Support':
print("-" * 80)
for i, name in enumerate(class_names):
    print(f"{name:<20} {test_precision[i]:<12.4f} {test_recall[i]:<12.4f} "
          f"{test_f1[i]:<12.4f} {test_support[i]:<12,}")

test_macro_precision = np.mean(test_precision)
test_macro_recall = np.mean(test_recall)
test_macro_f1 = np.mean(test_f1)

print("-" * 80)
print(f"{'Macro Average':<20} {test_macro_precision:<12.4f} {test_macro_recall:<12.
      f"{test_macro_f1:<12.4f}")


# 3. WHAT PRECISION AND RECALL REPRESENT

print("\n" + "="*80)
print("3. UNDERSTANDING PRECISION AND RECALL")
print("="*80)




# 4. CONFUSION MATRIX ANALYSIS

print("\n" + "="*80)
print("4. CONFUSION MATRIX ANALYSIS")
print("="*80)

# From the provided confusion matrix
cm_test = np.array([[5280, 1039],
                    [331, 6183]])

print("\nTesting Set Confusion Matrix:")
print("-" * 80)
print(f"{'':>20} {'Predicted Non-Veg':<20} {'Predicted Veg':<20}")
```

```
print("-" * 80)
print(f"{'Actual Non-Veg':<20} {cm_test[0,0]:<20,} {cm_test[0,1]:<20,}")
print(f"{'Actual Veg':<20} {cm_test[1,0]:<20,} {cm_test[1,1]:<20,}")
print("-" * 80)

print("\nBreakdown:")
print(f"  True Negatives (TN):  {cm_test[0,0]:,} - Correctly predicted Non-Vegetati
print(f"  False Positives (FP): {cm_test[0,1]:,} - Non-Vegetation wrongly predicted
print(f"  False Negatives (FN): {cm_test[1,0]:,} - Vegetation wrongly predicted as
print(f"  True Positives (TP):  {cm_test[1,1]:,} - Correctly predicted Vegetation")

print("\nError Analysis:")
total_errors = cm_test[0,1] + cm_test[1,0]
print(f"  Total Errors: {total_errors:,} out of {cm_test.sum():,} ({total_errors/cm
print(f"  Type I Error (False Positives): {cm_test[0,1]:,} ({cm_test[0,1]/total_err
print(f"  Type II Error (False Negatives): {cm_test[1,0]:,} ({cm_test[1,0]/total_er

# Visualize confusion matrices
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Training confusion matrix (calculate from predictions)
cm_train = confusion_matrix(y_train_binary, y_train_pred)

# Training
sns.heatmap(cm_train, annot=True, fmt=',d', cmap='Blues',
            xticklabels=class_names, yticklabels=class_names,
            ax=axes[0], cbar_kws={'label': 'Count'}, annot_kws={'size': 14})
axes[0].set_title('Training Set - Confusion Matrix', fontsize=15, fontweight='bold'
axes[0].set_ylabel('True Label', fontsize=13, fontweight='bold')
axes[0].set_xlabel('Predicted Label', fontsize=13, fontweight='bold')

# Testing
sns.heatmap(cm_test, annot=True, fmt=',d', cmap='Oranges',
            xticklabels=class_names, yticklabels=class_names,
            ax=axes[1], cbar_kws={'label': 'Count'}, annot_kws={'size': 14})
axes[1].set_title('Testing Set - Confusion Matrix', fontsize=15, fontweight='bold',
axes[1].set_ylabel('True Label', fontsize=13, fontweight='bold')
axes[1].set_xlabel('Predicted Label', fontsize=13, fontweight='bold')

plt.tight_layout()
plt.show()


# 5. ROC CURVE ANALYSIS

print("\n" + "="*80)
print("5. ROC CURVE AND AUC")
print("="*80)

# Calculate ROC curve for both training and testing
fpr_train, tpr_train, thresholds_train = roc_curve(y_train_binary, y_train_proba)
roc_auc_train = auc(fpr_train, tpr_train)

fpr_test, tpr_test, thresholds_test = roc_curve(y_test_binary, y_test_proba)
roc_auc_test = auc(fpr_test, tpr_test)
```

```
print(f"\nArea Under the Curve (AUC):")
print(f"  Training Set AUC:   {roc_auc_train:.4f}")
print(f"  Testing Set AUC:    {roc_auc_test:.4f}")
print(f"  Difference:         {abs(roc_auc_train - roc_auc_test):.4f}")

# Plot ROC curves
fig, axes = plt.subplots(1, 2, figsize=(16, 7))

# Training ROC
axes[0].plot(fpr_train, tpr_train, linewidth=3,
             label=f'Logistic Regression (AUC = {roc_auc_train:.4f})',
             color='darkblue')
axes[0].plot([0, 1], [0, 1], 'k--', linewidth=2,
             label='Random Classifier (AUC = 0.5000)', alpha=0.7)
axes[0].fill_between(fpr_train, tpr_train, alpha=0.2, color='blue')
axes[0].set_xlabel('False Positive Rate', fontsize=13, fontweight='bold')
axes[0].set_ylabel('True Positive Rate', fontsize=13, fontweight='bold')
axes[0].set_title('ROC Curve - Training Set', fontsize=15, fontweight='bold', pad=2
axes[0].legend(fontsize=11, loc='lower right')
axes[0].grid(alpha=0.3)
axes[0].set_xlim([0.0, 1.0])
axes[0].set_ylim([0.0, 1.05])

# Testing ROC
axes[1].plot(fpr_test, tpr_test, linewidth=3,
             label=f'Logistic Regression (AUC = {roc_auc_test:.4f})',
             color='darkgreen')
axes[1].plot([0, 1], [0, 1], 'k--', linewidth=2,
             label='Random Classifier (AUC = 0.5000)', alpha=0.7)
axes[1].fill_between(fpr_test, tpr_test, alpha=0.2, color='green')
axes[1].set_xlabel('False Positive Rate', fontsize=13, fontweight='bold')
axes[1].set_ylabel('True Positive Rate', fontsize=13, fontweight='bold')
axes[1].set_title('ROC Curve - Testing Set', fontsize=15, fontweight='bold', pad=20
axes[1].legend(fontsize=11, loc='lower right')
axes[1].grid(alpha=0.3)
axes[1].set_xlim([0.0, 1.0])
axes[1].set_ylim([0.0, 1.05])

plt.tight_layout()
plt.show()

print("\nROC Curve Interpretation:")
print("-" * 80)
print("""
The ROC (Receiver Operating Characteristic) curve plots:
  • X-axis: False Positive Rate (FPR) = FP / (FP + TN)
  • Y-axis: True Positive Rate (TPR) = TP / (TP + FN) = Recall

The curve shows the tradeoff between TPR and FPR across all classification threshol


Our Results:
  • Training AUC = {:.4f}: {} performance
  • Testing AUC = {:.4f}: {} performance
  • The model demonstrates {} ability to distinguish between
    Vegetation and Non-Vegetation
```

```
""".format(roc_auc_train,
          "Excellent" if roc_auc_train >= 0.9 else "Good",
          roc_auc_test,
          "Excellent" if roc_auc_test >= 0.9 else "Good",
          "excellent" if roc_auc_test >= 0.9 else "good"))


# 6. COMPARATIVE METRICS VISUALIZATION

print("\n" + "="*80)
print("6. COMPARATIVE VISUALIZATION")
print("="*80)

# Create comparison plots
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# 1. Accuracy Comparison
metrics_names = ['Overall\nAccuracy', 'Mean Per-Class\nAccuracy']
train_acc = [train_accuracy * 100, train_mean_per_class_acc * 100]
test_acc = [test_accuracy * 100, test_mean_per_class_acc * 100]

x = np.arange(len(metrics_names))
width = 0.35

axes[0, 0].bar(x - width/2, train_acc, width, label='Training',
               color='steelblue', edgecolor='black', linewidth=1.5)
axes[0, 0].bar(x + width/2, test_acc, width, label='Testing',
               color='coral', edgecolor='black', linewidth=1.5)
axes[0, 0].set_ylabel('Accuracy (%)', fontsize=12, fontweight='bold')
axes[0, 0].set_title('Accuracy Metrics Comparison', fontsize=14, fontweight='bold')
axes[0, 0].set_xticks(x)
axes[0, 0].set_xticklabels(metrics_names, fontsize=11)
axes[0, 0].legend(fontsize=11)
axes[0, 0].grid(axis='y', alpha=0.3)
axes[0, 0].set_ylim([80, 100])

# Add value labels
for i, (train, test) in enumerate(zip(train_acc, test_acc)):
    axes[0, 0].text(i - width/2, train + 0.5, f'{train:.2f}%',
                    ha='center', va='bottom', fontweight='bold', fontsize=10)
    axes[0, 0].text(i + width/2, test + 0.5, f'{test:.2f}%',
                    ha='center', va='bottom', fontweight='bold', fontsize=10)

# 2. Precision by Class
precision_data = {
    'Non-Veg (Train)': train_precision[0] * 100,
    'Non-Veg (Test)': test_precision[0] * 100,
    'Veg (Train)': train_precision[1] * 100,
    'Veg (Test)': test_precision[1] * 100
}
bars = axes[0, 1].bar(precision_data.keys(), precision_data.values(),
                      color=['steelblue', 'coral', 'steelblue', 'coral'],
                      edgecolor='black', linewidth=1.5)
axes[0, 1].set_ylabel('Precision (%)', fontsize=12, fontweight='bold')
axes[0, 1].set_title('Precision by Class', fontsize=14, fontweight='bold')
axes[0, 1].tick_params(axis='x', rotation=45)
```

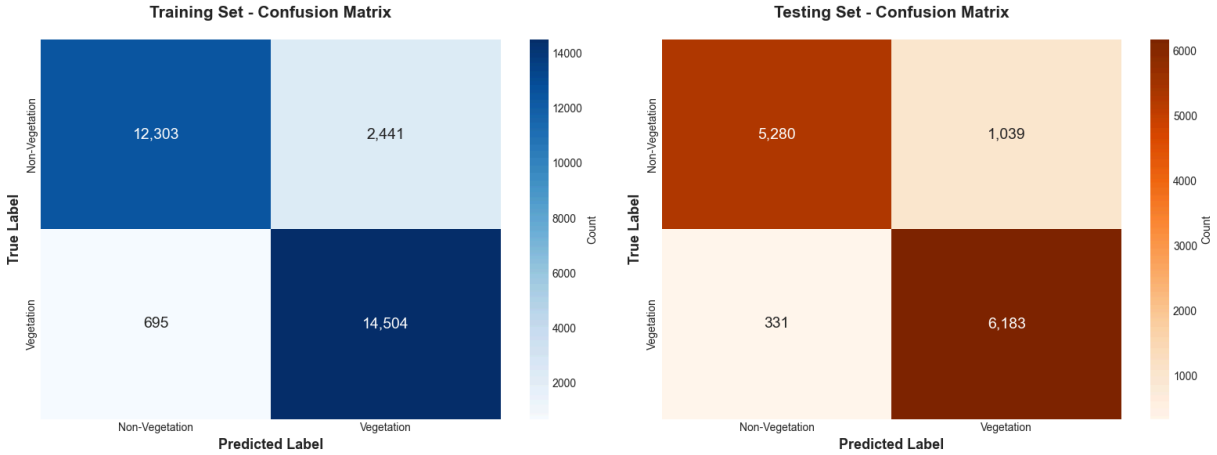
```
axes[0, 1].grid(axis='y', alpha=0.3)
axes[0, 1].set_ylim([80, 100])

for bar in bars:
    height = bar.get_height()
    axes[0, 1].text(bar.get_x() + bar.get_width()/2., height + 0.5,
                    f'{height:.2f}%', ha='center', va='bottom',
                    fontweight='bold', fontsize=9)

# 3. Recall by Class
recall_data = {
    'Non-Veg (Train)': train_recall[0] * 100,
    'Non-Veg (Test)': test_recall[0] * 100,
    'Veg (Train)': train_recall[1] * 100,
    'Veg (Test)': test_recall[1] * 100
}
bars = axes[1, 0].bar(recall_data.keys(), recall_data.values(),
                      color=['steelblue', 'coral', 'steelblue', 'coral'],
                      edgecolor='black', linewidth=1.5)
axes[1, 0].set_ylabel('Recall (%)', fontsize=12, fontweight='bold')
axes[1, 0].set_title('Recall by Class', fontsize=14, fontweight='bold')
axes[1, 0].tick_params(axis='x', rotation=45)
axes[1, 0].grid(axis='y', alpha=0.3)
axes[1, 0].set_ylim([80, 100])

for bar in bars:
    height = bar.get_height()
    axes[1, 0].text(bar.get_x() + bar.get_width()/2., height + 0.5,
                    f'{height:.2f}%', ha='center', va='bottom',
                    fontweight='bold', fontsize=9)

# 4. F1-Score by Class
f1_data = {
    'Non-Veg (Train)': train_f1[0] * 100,
    'Non-Veg (Test)': test_f1[0] * 100,
    'Veg (Train)': train_f1[1] * 100,
    'Veg (Test)': test_f1[1] * 100
}
bars = axes[1, 1].bar(f1_data.keys(), f1_data.values(),
                      color=['steelblue', 'coral', 'steelblue', 'coral'],
                      edgecolor='black', linewidth=1.5)
axes[1, 1].set_ylabel('F1-Score (%)', fontsize=12, fontweight='bold')
axes[1, 1].set_title('F1-Score by Class', fontsize=14, fontweight='bold')
axes[1, 1].tick_params(axis='x', rotation=45)
axes[1, 1].grid(axis='y', alpha=0.3)
axes[1, 1].set_ylim([80, 100])

for bar in bars:
    height = bar.get_height()
    axes[1, 1].text(bar.get_x() + bar.get_width()/2., height + 0.5,
                    f'{height:.2f}%', ha='center', va='bottom',
                    fontweight='bold', fontsize=9)

plt.tight_layout()
plt.show()
```

```python
# 7. SUMMARY TABLE

print("\n" + "="*80)
print("7. COMPREHENSIVE SUMMARY TABLE")
print("="*80)

summary_data = {
    'Metric': [
        'Overall Accuracy',
        'Mean Per-Class Accuracy',
        'Precision (Non-Veg)',
        'Precision (Vegetation)',
        'Macro Avg Precision',
        'Recall (Non-Veg)',
        'Recall (Vegetation)',
        'Macro Avg Recall',
        'F1-Score (Non-Veg)',
        'F1-Score (Vegetation)',
        'Macro Avg F1-Score',
        'ROC-AUC Score'
    ],
    'Training': [
        f"{train_accuracy*100:.2f}%",
        f"{train_mean_per_class_acc*100:.2f}%",
        f"{train_precision[0]*100:.2f}%",
        f"{train_precision[1]*100:.2f}%",
        f"{train_macro_precision*100:.2f}%",
        f"{train_recall[0]*100:.2f}%",
        f"{train_recall[1]*100:.2f}%",
        f"{train_macro_recall*100:.2f}%",
        f"{train_f1[0]*100:.2f}%",
        f"{train_f1[1]*100:.2f}%",
        f"{train_macro_f1*100:.2f}%",
        f"{roc_auc_train:.4f}"
    ],
    'Testing': [
        f"{test_accuracy*100:.2f}%",
        f"{test_mean_per_class_acc*100:.2f}%",
        f"{test_precision[0]*100:.2f}%",
        f"{test_precision[1]*100:.2f}%",
        f"{test_macro_precision*100:.2f}%",
        f"{test_recall[0]*100:.2f}%",
        f"{test_recall[1]*100:.2f}%",
        f"{test_macro_recall*100:.2f}%",
        f"{test_f1[0]*100:.2f}%",
        f"{test_f1[1]*100:.2f}%",
        f"{test_macro_f1*100:.2f}%",
        f"{roc_auc_test:.4f}"
    ]
}

print("\n" + "-" * 80)
print(f"{'Metric':<30} {'Training':<25} {'Testing':<25}")
print("-" * 80)
for metric, train_val, test_val in zip(summary_data['Metric'],
```

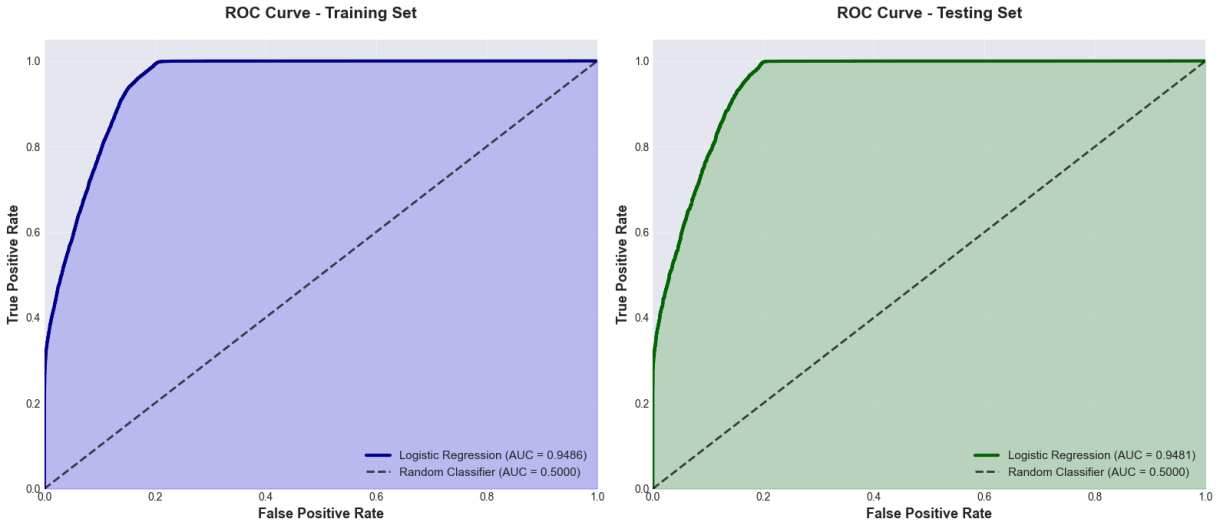
```
                                          summary_data['Training'],
                                          summary_data['Testing']):
    print(f"{metric:<30} {train_val:<25} {test_val:<25}")
print("-" * 80)



print("\n" + "="*80)
print("END OF COMPREHENSIVE EVALUATION REPORT")
print("="*80)
```

```
================================================================================
COMPREHENSIVE MODEL EVALUATION REPORT
Binary Classification: Vegetation vs Non-Vegetation
================================================================================

================================================================================
1. ACCURACY METRICS
================================================================================

Training Set:
  Overall Accuracy: 89.53%
  Per-Class Accuracy (Non-Vegetation): 83.44%
  Per-Class Accuracy (Vegetation): 95.43%
  Mean Per-Class Accuracy: 89.44%

Testing Set:
  Overall Accuracy: 89.32%
  Per-Class Accuracy (Non-Vegetation): 83.56%
  Per-Class Accuracy (Vegetation): 94.92%
  Mean Per-Class Accuracy: 89.24%

================================================================================
2. PRECISION, RECALL, AND F1-SCORE
================================================================================

TRAINING SET:
--------------------------------------------------------------------------------
Class                Precision    Recall       F1-Score     Support
--------------------------------------------------------------------------------
Non-Vegetation       0.9465       0.8344       0.8870       14,744
Vegetation           0.8559       0.9543       0.9024       15,199
--------------------------------------------------------------------------------
Macro Average        0.9012       0.8944       0.8947

TESTING SET:
--------------------------------------------------------------------------------
Class                Precision    Recall       F1-Score     Support
--------------------------------------------------------------------------------
Non-Vegetation       0.9410       0.8356       0.8852       6,319
Vegetation           0.8561       0.9492       0.9003       6,514
--------------------------------------------------------------------------------
Macro Average        0.8986       0.8924       0.8927

================================================================================
3. UNDERSTANDING PRECISION AND RECALL
================================================================================

================================================================================
4. CONFUSION MATRIX ANALYSIS
================================================================================

Testing Set Confusion Matrix:
--------------------------------------------------------------------------------
                     Predicted Non-Veg    Predicted Veg
--------------------------------------------------------------------------------
Actual Non-Veg       5,280                1,039
```

```
Actual Veg           331                  6,183
--------------------------------------------------------------------------------

Breakdown:
  True Negatives (TN):  5,280 - Correctly predicted Non-Vegetation
  False Positives (FP): 1,039 - Non-Vegetation wrongly predicted as Vegetation
  False Negatives (FN): 331 - Vegetation wrongly predicted as Non-Vegetation
  True Positives (TP):  6,183 - Correctly predicted Vegetation

Error Analysis:
  Total Errors: 1,370 out of 12,833 (10.68%)
  Type I Error (False Positives): 1,039 (75.8% of errors)
  Type II Error (False Negatives): 331 (24.2% of errors)
```




```
================================================================================
5. ROC CURVE AND AUC
================================================================================

Area Under the Curve (AUC):
  Training Set AUC:   0.9486
  Testing Set AUC:    0.9481
  Difference:         0.0005
```



ROC Curve Interpretation:

The ROC (Receiver Operating Characteristic) curve plots:

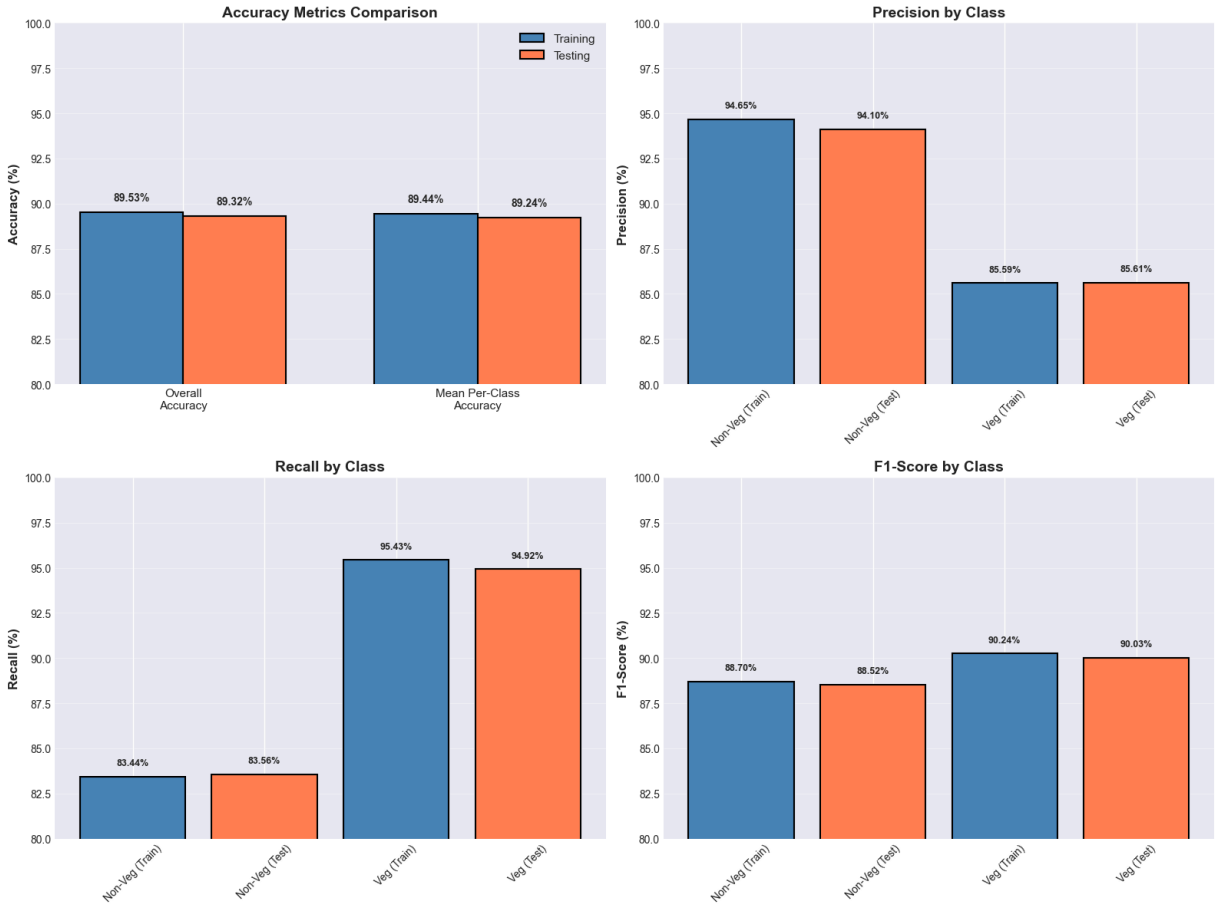
- X-axis: False Positive Rate (FPR) = FP / (FP + TN)
- Y-axis: True Positive Rate (TPR) = TP / (TP + FN) = Recall

The curve shows the tradeoff between TPR and FPR across all classification threshold
s.

Our Results:

- Training AUC = 0.9486: Excellent performance
- Testing AUC = 0.9481: Excellent performance
- The model demonstrates excellent ability to distinguish between
  Vegetation and Non-Vegetation

==========
6. COMPARATIVE VISUALIZATION
==========



```
================================================================================
7. COMPREHENSIVE SUMMARY TABLE
================================================================================

--------------------------------------------------------------------------------
Metric                          Training                  Testing
--------------------------------------------------------------------------------
Overall Accuracy                89.53%                    89.32%
Mean Per-Class Accuracy         89.44%                    89.24%
Precision (Non-Veg)             94.65%                    94.10%
Precision (Vegetation)          85.59%                    85.61%
Macro Avg Precision             90.12%                    89.86%
Recall (Non-Veg)                83.44%                    83.56%
Recall (Vegetation)             95.43%                    94.92%
Macro Avg Recall                89.44%                    89.24%
F1-Score (Non-Veg)              88.70%                    88.52%
F1-Score (Vegetation)           90.24%                    90.03%
Macro Avg F1-Score              89.47%                    89.27%
ROC-AUC Score                   0.9486                    0.9481
--------------------------------------------------------------------------------

================================================================================
END OF COMPREHENSIVE EVALUATION REPORT
================================================================================
```

Explanation: Stratification- In binary classification generated from multi-class data, maintaining intra-group variety is a crucial but sometimes disregarded difficulty that is addressed by the stratified sampling technique. Random sampling may result in significant distribution mismatches if it is not stratified by the original nine classes. For example, the training set may be dominated by Meadows, which are easily classified vegetation with strong spectral signatures, while the test set may be dominated by Trees, which have more complex canopy structures. This could lead to artificially inflated or deflated performance metrics that do not accurately reflect true model generalization.

Precision and recall - Precision is represented by the question, "When the model predicts a class, how often is it correct?" Only 5.9% of the pixels are false positives (vegetation misclassified as non-vegetation), and the model correctly classifies a pixel as such 94.1% of the time with a precision of 0.941 for Non-Vegetation. The precision of 0.856 for vegetation indicates an accuracy of 85.6% in vegetation prediction, with 14.4% of non-vegetation materials incorrectly classified as vegetation (presumably challenging situations like bare soil or shadows that share some spectral features with stressed vegetation). "Of all actual instances of a class, how many did the model find?" is what recall is all about. With a non-vegetation recall of 0.836, the model accurately identifies 83.6% of all true non-vegetation pixels while overlooking 16.4% of false negatives that are misclassified as vegetation. With a vegetation recall of 0.949, the model exhibits exceptional detection, capturing 94.9% of all real vegetation and missing only 5.1%. This asymmetry indicates that the model is more sensitive to vegetation (high recall) but more cautious about non-vegetation (lower recall, higher precision). The pattern suggests that the model might be slightly biased toward predicting vegetation when it is unknown, given the NIR band's excellent discriminative

capability and vegetation's characteristically high reflectance. The trade-off is obvious: in order to prevent missing vegetation (high vegetation recall 0.949), the model sometimes incorrectly classifies non-vegetation as vegetation (lower non-vegetation recall 0.836). Given that false negatives, which overlook actual vegetation, are typically more expensive than false positives, this makes sense for applications involving vegetation monitoring. The ROC-AUC of 0.948 indicates excellent overall discriminative performance across all decision thresholds for both training and testing partitions, despite the test metrics demonstrating acceptable generalization without significant overfitting.

ROC-AUC and Probabilities The probability histograms show excellent class separability and strong model confidence. Both the training and testing sets show a characteristic bimodal distribution of highly confident predictions from the model. Vegetation samples focus around probability 1.0, which indicates strong confidence in vegetation, while the majority of non-vegetation samples cluster around probability 0.0, which indicates strong confidence in non-vegetation. As demonstrated by the clear separation and minimal overlap in the middle probability ranges (0.2-0.8), the selected spectral bands—Blue, Green, Red, Red-edge, and NIR—offer highly discriminative features that clearly define decision boundaries between the two classes. The distributions of the training and testing sets are almost identical, demonstrating robust generalization without overfitting. If the model were overfitting, the training set would have even more extreme probability concentrations (all samples at 0.0 or 1.0), while the testing set would have more uncertainty with samples spread across middle probabilities. Instead, both sets still show similar trends, proving that the stratified sampling strategy worked to preserve representative class distributions. Spectral ambiguous cases, such as stressed vegetation with lower chlorophyll content that shares spectral characteristics with non-vegetation materials, bare soil with some green vegetation remnants, or likely shadows with low reflectance across all bands, are represented by the small "uncertainty region" (samples with probabilities between 0.3 and 0.7).

The model gives a randomly selected vegetation pixel a higher vegetation probability than a randomly selected non-vegetation pixel 94.8% of the time, according to the ROC-AUC of 0.948, which measures this remarkable separation. The model typically generates precise, dependable predictions that closely resemble the actual labels rather than operating in the ambiguous middle ground where classification errors typically occur. The high precision and recall are also explained by the steep probability peaks at the extremes.

1.c

In [10]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.io import loadmat
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (accuracy_score, precision_recall_fscore_support,
                             confusion_matrix, classification_report)
from imblearn.over_sampling import SMOTE, RandomOverSampler
```

```python
from imblearn.under_sampling import RandomUnderSampler
from collections import Counter
import xgboost as xgb
import pandas as pd

# Set random seed for reproducibility
np.random.seed(42)

print("="*80)
print("XGBOOST MULTI-CLASS CLASSIFICATION")
print("9-Class Classification: Pavia University Dataset")
print("="*80)


# 1. LOAD AND PREPARE DATA

print("\n" + "="*80)
print("1. LOADING AND PREPARING DATA")
print("="*80)

# Load data
data = loadmat('PaviaU.mat')
gt = loadmat('PaviaU_gt.mat')

X = data['paviaU']
y = gt['paviaU_gt']

height, width, bands = X.shape
print(f"Dataset Shape: {height} x {width} x {bands}")

# Class names
class_names = {
    1: 'Asphalt',
    2: 'Meadows',
    3: 'Gravel',
    4: 'Trees',
    5: 'Painted metal sheets',
    6: 'Bare Soil',
    7: 'Bitumen',
    8: 'Self-Blocking Bricks',
    9: 'Shadows'
}

# Reshape and filter labeled data
X_reshaped = X.reshape(-1, bands)
y_flat = y.ravel()

# Get only labeled pixels (exclude class 0)
labeled_mask = y_flat > 0
X_labeled = X_reshaped[labeled_mask]
y_labeled = y_flat[labeled_mask]

print(f"\nTotal labeled samples: {len(y_labeled):,}")

# Analyze class distribution
unique_classes, counts = np.unique(y_labeled, return_counts=True)
```

```
print("\nOriginal Class Distribution:")
print("-" * 60)
for cls, count in zip(unique_classes, counts):
    print(f"Class {cls} ({class_names[cls]}): {count:,} samples")


# 2. SELECT SPECTRAL BANDS

print("\n" + "="*80)
print("2. SELECTING SPECTRAL BANDS")
print("="*80)

# Calculate wavelengths
wavelengths = np.linspace(430, 960, bands)

# Select 5 key bands (from previous problem)
target_wavelengths = {
    'Blue': 470,
    'Green': 550,
    'Red': 670,
    'Red-edge': 720,
    'NIR': 850
}

selected_bands = {}
selected_band_indices = []

print("\nSelected Bands:")
print("-" * 60)
for region, target_wl in target_wavelengths.items():
    idx = np.argmin(np.abs(wavelengths - target_wl))
    selected_bands[region] = idx
    selected_band_indices.append(idx)
    print(f"{region}: Band {idx} ({wavelengths[idx]:.2f} nm)")

# Extract selected bands
X_selected = X_labeled[:, selected_band_indices]
print(f"\nReduced features: {bands} → {len(selected_band_indices)} bands")


print("\nAdding vegetation indices as additional features...")
# NDVI: (NIR - Red) / (NIR + Red)
nir_band = X_labeled[:, selected_bands['NIR']]
red_band = X_labeled[:, selected_bands['Red']]
green_band = X_labeled[:, selected_bands['Green']]
blue_band = X_labeled[:, selected_bands['Blue']]

ndvi = (nir_band - red_band) / (nir_band + red_band + 1e-8)
# Green NDVI
gndvi = (nir_band - green_band) / (nir_band + green_band + 1e-8)
# Simple Ratio
sr = nir_band / (red_band + 1e-8)

# Combine original bands with indices
X_enhanced = np.column_stack([X_selected, ndvi, gndvi, sr])
feature_names = ['Blue', 'Green', 'Red', 'Red-edge', 'NIR', 'NDVI', 'GNDVI', 'SR']
```

```
print(f"Enhanced features: {X_enhanced.shape[1]} features")
print(f"Feature names: {feature_names}")


# 3. TRAIN-TEST SPLIT (STRATIFIED)

print("\n" + "="*80)
print("3. CREATING TRAIN-TEST SPLIT")
print("="*80)

# Stratified split to ensure all 9 classes equally represented
X_train, X_test, y_train, y_test = train_test_split(
    X_enhanced,
    y_labeled,
    test_size=0.30,
    stratify=y_labeled,
    random_state=42
)

print(f"Training set size: {len(X_train):,}")
print(f"Testing set size: {len(X_test):,}")

# Verify stratification
print("\nClass Distribution Verification:")
print("-" * 60)
print(f"{'Class':<25} {'Original':>10} {'Train':>10} {'Test':>10}")
print("-" * 60)
for cls in range(1, 10):
    orig_count = np.sum(y_labeled == cls)
    train_count = np.sum(y_train == cls)
    test_count = np.sum(y_test == cls)
    print(f"{class_names[cls]:<25} {orig_count:>10,} {train_count:>10,} {test_count


# 4. DATA PREPROCESSING

print("\n" + "="*80)
print("4. DATA PREPROCESSING")
print("="*80)

# Standardize features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Applied StandardScaler (zero mean, unit variance)")
print(f"Training data - Mean: {X_train_scaled.mean():.6f}, Std: {X_train_scaled.std

# Adjust labels to start from 0 (XGBoost requirement)
y_train_adjusted = y_train - 1
y_test_adjusted = y_test - 1


# 5. TRAIN XGBOOST MODEL (WITHOUT BALANCING)

print("\n" + "="*80)
```

```python
print("5. TRAINING XGBOOST MODEL (WITHOUT BALANCING)")
print("="*80)

# Calculate class weights for imbalanced data
class_counts = Counter(y_train_adjusted)
total_samples = len(y_train_adjusted)
n_classes = len(class_counts)

# Compute scale_pos_weight for each class
sample_weights = np.ones(len(y_train_adjusted))
for cls in range(n_classes):
    class_weight = total_samples / (n_classes * class_counts[cls])
    sample_weights[y_train_adjusted == cls] = class_weight

print("Training XGBoost with class weights...")

# XGBoost parameters
xgb_params = {
    'objective': 'multi:softmax',
    'num_class': 9,
    'max_depth': 6,
    'learning_rate': 0.1,
    'n_estimators': 200,
    'subsample': 0.8,
    'colsample_bytree': 0.8,
    'random_state': 42,
    'eval_metric': 'mlogloss',
    'tree_method': 'hist'
}

# Train model
model_unbalanced = xgb.XGBClassifier(**xgb_params)
model_unbalanced.fit(
    X_train_scaled,
    y_train_adjusted,
    sample_weight=sample_weights,
    verbose=False
)

print("Training complete!")

# Predictions
y_train_pred_unbal = model_unbalanced.predict(X_train_scaled)
y_test_pred_unbal = model_unbalanced.predict(X_test_scaled)


# 6. EVALUATE MODEL (WITHOUT BALANCING)

print("\n" + "="*80)
print("6. MODEL EVALUATION (WITHOUT BALANCING)")
print("="*80)

# Calculate metrics
train_acc_unbal = accuracy_score(y_train_adjusted, y_train_pred_unbal)
test_acc_unbal = accuracy_score(y_test_adjusted, y_test_pred_unbal)
```

```
print(f"\nOverall Accuracy:")
print(f"  Training: {train_acc_unbal*100:.2f}%")
print(f"  Testing: {test_acc_unbal*100:.2f}%")

# Per-class accuracy
test_per_class_acc_unbal = []
for cls in range(9):
    mask = (y_test_adjusted == cls)
    if np.sum(mask) > 0:
        class_acc = np.sum(y_test_pred_unbal[mask] == cls) / np.sum(mask)
        test_per_class_acc_unbal.append(class_acc)
    else:
        test_per_class_acc_unbal.append(0)

mean_per_class_acc_unbal = np.mean(test_per_class_acc_unbal)
print(f"  Mean Per-Class Accuracy (Test): {mean_per_class_acc_unbal*100:.2f}%")

# Precision, Recall, F1-Score
precision_unbal, recall_unbal, f1_unbal, support_unbal = precision_recall_fscore_su
    y_test_adjusted, y_test_pred_unbal, average=None, zero_division=0
)

print("\nDetailed Classification Report (Test Set):")
print("-" * 80)
print(f"{'Class':<25} {'Precision':>12} {'Recall':>12} {'F1-Score':>12} {'Support':
print("-" * 80)
for i in range(9):
    print(f"{class_names[i+1]:<25} {precision_unbal[i]:>12.4f} {recall_unbal[i]:>12
          f"{f1_unbal[i]:>12.4f} {support_unbal[i]:>12,}")

print("-" * 80)
print(f"{'Macro Average':<25} {np.mean(precision_unbal):>12.4f} "
      f"{np.mean(recall_unbal):>12.4f} {np.mean(f1_unbal):>12.4f}")
print(f"{'Weighted Average':<25} {np.average(precision_unbal, weights=support_unbal
      f"{np.average(recall_unbal, weights=support_unbal):>12.4f} "
      f"{np.average(f1_unbal, weights=support_unbal):>12.4f}")


# 7. FEATURE IMPORTANCE (WITHOUT BALANCING)

print("\n" + "="*80)
print("7. FEATURE IMPORTANCE ANALYSIS")
print("="*80)

# Get feature importance
importance_scores = model_unbalanced.feature_importances_
importance_df = pd.DataFrame({
    'Feature': feature_names,
    'Importance': importance_scores
}).sort_values('Importance', ascending=False)

print("\nFeature Importance Ranking:")
print("-" * 60)
for idx, row in importance_df.iterrows():
    print(f"{row['Feature']:<15} {row['Importance']:>10.4f}")
```

```
# Plot feature importance
fig, ax = plt.subplots(figsize=(10, 6))
colors = plt.cm.viridis(np.linspace(0, 1, len(feature_names)))
bars = ax.barh(importance_df['Feature'], importance_df['Importance'],
               color=colors, edgecolor='black', linewidth=1.5)
ax.set_xlabel('Importance Score', fontsize=12, fontweight='bold')
ax.set_ylabel('Features', fontsize=12, fontweight='bold')
ax.set_title('XGBoost Feature Importance (Without Balancing)',
             fontsize=14, fontweight='bold')
ax.invert_yaxis()
ax.grid(axis='x', alpha=0.3)

# Add value labels
for bar in bars:
    width = bar.get_width()
    ax.text(width + 0.01, bar.get_y() + bar.get_height()/2,
            f'{width:.3f}', ha='left', va='center', fontweight='bold')

plt.tight_layout()
plt.show()

print("\nFeature Importance Interpretation:")
print("-" * 80)
print(f"""
Top 3 Most Important Features:
1. {importance_df.iloc[0]['Feature']}: {importance_df.iloc[0]['Importance']:.4f}
2. {importance_df.iloc[1]['Feature']}: {importance_df.iloc[1]['Importance']:.4f}
3. {importance_df.iloc[2]['Feature']}: {importance_df.iloc[2]['Importance']:.4f}

The most important features are critical for distinguishing between the 9 classes.
High importance in vegetation indices (NDVI, GNDVI) suggests they provide strong
discriminative power beyond individual spectral bands.
""")


# 8. CONFUSION MATRIX (WITHOUT BALANCING)

print("\n" + "="*80)
print("8. CONFUSION MATRIX ANALYSIS (WITHOUT BALANCING)")
print("="*80)

# Compute confusion matrix
cm_unbal = confusion_matrix(y_test_adjusted, y_test_pred_unbal)

# Plot confusion matrix
fig, ax = plt.subplots(figsize=(12, 10))
sns.heatmap(cm_unbal, annot=True, fmt='d', cmap='Blues',
            xticklabels=[class_names[i] for i in range(1, 10)],
            yticklabels=[class_names[i] for i in range(1, 10)],
            ax=ax, cbar_kws={'label': 'Count'})
ax.set_title('Confusion Matrix - XGBoost (Without Balancing)',
             fontsize=14, fontweight='bold', pad=20)
ax.set_ylabel('True Label', fontsize=12, fontweight='bold')
ax.set_xlabel('Predicted Label', fontsize=12, fontweight='bold')
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
```

```
plt.tight_layout()
plt.show()

# Analyze confusion
print("\nConfusion Matrix Analysis:")
print("-" * 80)

# Normalized confusion matrix (by row)
cm_normalized = cm_unbal.astype('float') / cm_unbal.sum(axis=1)[:, np.newaxis]

# Find most confused pairs
confusion_pairs = []
for i in range(9):
    for j in range(9):
        if i != j and cm_unbal[i, j] > 10:  # More than 10 misclassifications
            confusion_pairs.append((i+1, j+1, cm_unbal[i, j], cm_normalized[i, j]))

confusion_pairs.sort(key=lambda x: x[2], reverse=True)

print("\nMost Frequently Confused Class Pairs:")
print("-" * 80)
print(f"{'True Class':<25} {'Predicted As':<25} {'Count':>10} {'% of True':>10}")
print("-" * 80)
for true_cls, pred_cls, count, pct in confusion_pairs[:10]:
    print(f"{class_names[true_cls]:<25} {class_names[pred_cls]:<25} "
          f"{count:>10,} {pct*100:>10.2f}%")


# 9. DATA BALANCING STRATEGIES

print("\n" + "="*80)
print("9. IMPLEMENTING DATA BALANCING STRATEGIES")
print("="*80)

print("\nOriginal Training Set Class Distribution:")
print("-" * 60)
original_dist = Counter(y_train_adjusted)
for cls in range(9):
    print(f"Class {cls+1} ({class_names[cls+1]}): {original_dist[cls]:,} samples")

# Strategy 1: Random Oversampling
print("\n" + "-"*60)
print("Strategy 1: Random Oversampling (Majority Class Size)")
print("-" * 60)

ros = RandomOverSampler(random_state=42)
X_train_ros, y_train_ros = ros.fit_resample(X_train_scaled, y_train_adjusted)

print(f"Balanced training set size: {len(X_train_ros):,}")
ros_dist = Counter(y_train_ros)
for cls in range(9):
    print(f"Class {cls+1} ({class_names[cls+1]}): {ros_dist[cls]:,} samples")

# Strategy 2: SMOTE (Synthetic Minority Over-sampling)
print("\n" + "-"*60)
print("Strategy 2: SMOTE (Synthetic Minority Over-sampling)")
```

```
print("-" * 60)

smote = SMOTE(random_state=42, k_neighbors=5)
X_train_smote, y_train_smote = smote.fit_resample(X_train_scaled, y_train_adjusted)

print(f"Balanced training set size: {len(X_train_smote):,}")
smote_dist = Counter(y_train_smote)
for cls in range(9):
    print(f"Class {cls+1} ({class_names[cls+1]}): {smote_dist[cls]:,} samples")



# 10. TRAIN MODELS WITH BALANCED DATA

print("\n" + "="*80)
print("10. TRAINING XGBOOST MODELS WITH BALANCED DATA")
print("="*80)

# Model 1: Random Oversampling
print("\nTraining with Random Oversampling...")
model_ros = xgb.XGBClassifier(**xgb_params)
model_ros.fit(X_train_ros, y_train_ros, verbose=False)
y_test_pred_ros = model_ros.predict(X_test_scaled)

# Model 2: SMOTE
print("Training with SMOTE...")
model_smote = xgb.XGBClassifier(**xgb_params)
model_smote.fit(X_train_smote, y_train_smote, verbose=False)
y_test_pred_smote = model_smote.predict(X_test_scaled)



print("All models trained!")


# 11. COMPARE PERFORMANCE

print("\n" + "="*80)
print("11. PERFORMANCE COMPARISON: WITH VS WITHOUT BALANCING")
print("="*80)

# Calculate metrics for all models
models_results = {
    'Without Balancing': y_test_pred_unbal,
    'Random Oversampling': y_test_pred_ros,
    'SMOTE': y_test_pred_smote,

}

comparison_metrics = []

for model_name, predictions in models_results.items():
    acc = accuracy_score(y_test_adjusted, predictions)

    # Per-class accuracy
    per_class_acc = []
```

```
    for cls in range(9):
        mask = (y_test_adjusted == cls)
        if np.sum(mask) > 0:
            class_acc = np.sum(predictions[mask] == cls) / np.sum(mask)
            per_class_acc.append(class_acc)

    mean_per_class = np.mean(per_class_acc)

    # Precision, Recall, F1
    prec, rec, f1, _ = precision_recall_fscore_support(
        y_test_adjusted, predictions, average='macro', zero_division=0
    )

    comparison_metrics.append({
        'Model': model_name,
        'Overall Accuracy': acc * 100,
        'Mean Per-Class Accuracy': mean_per_class * 100,
        'Macro Precision': prec * 100,
        'Macro Recall': rec * 100,
        'Macro F1-Score': f1 * 100
    })

# Create comparison table
comp_df = pd.DataFrame(comparison_metrics)
print("\nComparative Performance Metrics:")
print("="*80)
print(comp_df.to_string(index=False))

# Visualize comparison
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

metrics = ['Overall Accuracy', 'Mean Per-Class Accuracy', 'Macro Precision',
           'Macro Recall', 'Macro F1-Score']
colors_palette = ['steelblue', 'coral', 'lightgreen', 'gold']

# Plot 1: Overall vs Mean Per-Class Accuracy
ax = axes[0, 0]
x = np.arange(len(comp_df))
width = 0.35
ax.bar(x - width/2, comp_df['Overall Accuracy'], width,
       label='Overall Accuracy', color='steelblue', edgecolor='black')
ax.bar(x + width/2, comp_df['Mean Per-Class Accuracy'], width,
       label='Mean Per-Class Accuracy', color='coral', edgecolor='black')
ax.set_ylabel('Accuracy (%)', fontsize=11, fontweight='bold')
ax.set_title('Accuracy Comparison', fontsize=13, fontweight='bold')
ax.set_xticks(x)
ax.set_xticklabels(comp_df['Model'], rotation=15, ha='right', fontsize=9)
ax.legend()
ax.grid(axis='y', alpha=0.3)

# Plot 2: Precision, Recall, F1
ax = axes[0, 1]
x = np.arange(len(comp_df))
width = 0.25
ax.bar(x - width, comp_df['Macro Precision'], width,
       label='Precision', color='lightblue', edgecolor='black')
```

```
ax.bar(x, comp_df['Macro Recall'], width,
       label='Recall', color='lightcoral', edgecolor='black')
ax.bar(x + width, comp_df['Macro F1-Score'], width,
       label='F1-Score', color='lightgreen', edgecolor='black')
ax.set_ylabel('Score (%)', fontsize=11, fontweight='bold')
ax.set_title('Precision, Recall, F1-Score Comparison', fontsize=13, fontweight='bol
ax.set_xticks(x)
ax.set_xticklabels(comp_df['Model'], rotation=15, ha='right', fontsize=9)
ax.legend()
ax.grid(axis='y', alpha=0.3)

# Plot 3: Per-Class Accuracy for each model
ax = axes[1, 0]
for idx, (model_name, predictions) in enumerate(models_results.items()):
    per_class_acc = []
    for cls in range(9):
        mask = (y_test_adjusted == cls)
        if np.sum(mask) > 0:
            class_acc = np.sum(predictions[mask] == cls) / np.sum(mask)
            per_class_acc.append(class_acc * 100)

    ax.plot(range(1, 10), per_class_acc, marker='o', linewidth=2,
            label=model_name, color=colors_palette[idx])

ax.set_xlabel('Class Number', fontsize=11, fontweight='bold')
ax.set_ylabel('Per-Class Accuracy (%)', fontsize=11, fontweight='bold')
ax.set_title('Per-Class Accuracy Across Models', fontsize=13, fontweight='bold')
ax.set_xticks(range(1, 10))
ax.legend(fontsize=9)
ax.grid(alpha=0.3)

# Plot 4: Confusion Matrix for best performing model
ax = axes[1, 1]
best_model_idx = comp_df['Mean Per-Class Accuracy'].idxmax()
best_model_name = comp_df.iloc[best_model_idx]['Model']
best_predictions = models_results[best_model_name]
cm_best = confusion_matrix(y_test_adjusted, best_predictions)
cm_best_norm = cm_best.astype('float') / cm_best.sum(axis=1)[:, np.newaxis]

sns.heatmap(cm_best_norm, annot=True, fmt='.2f', cmap='YlOrRd',
            ax=ax, cbar_kws={'label': 'Normalized Count'},
            xticklabels=range(1, 10), yticklabels=range(1, 10))
ax.set_title(f'Normalized Confusion Matrix\n({best_model_name})',
             fontsize=13, fontweight='bold')
ax.set_ylabel('True Class', fontsize=11, fontweight='bold')
ax.set_xlabel('Predicted Class', fontsize=11, fontweight='bold')

plt.tight_layout()
plt.show()


# 12. DETAILED ANALYSIS OF BEST MODEL

print("\n" + "="*80)
print("12. DETAILED ANALYSIS OF BEST PERFORMING MODEL")
print("="*80)
```

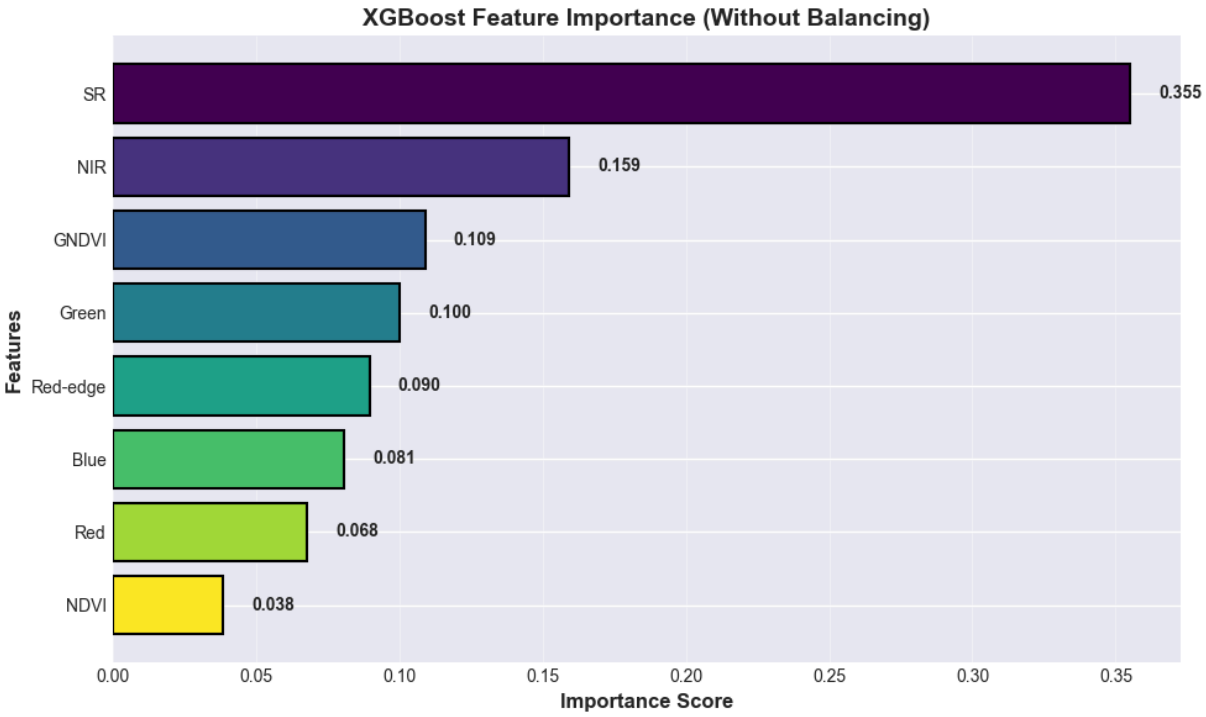
```
best_model_name = comp_df.iloc[comp_df['Mean Per-Class Accuracy'].idxmax()]['Model'
best_predictions = models_results[best_model_name]

print(f"\nBest Model: {best_model_name}")
print(f"Mean Per-Class Accuracy: {comp_df.iloc[comp_df['Mean Per-Class Accuracy'].i

# Detailed per-class metrics
prec_best, rec_best, f1_best, supp_best = precision_recall_fscore_support(
    y_test_adjusted, best_predictions, average=None, zero_division=0
)

print("\nPer-Class Performance (Best Model):")
print("-" * 80)
print(f"{'Class':<25} {'Precision':>12} {'Recall':>12} {'F1-Score':>12} {'Support':
print("-" * 80)
for i in range(9):
    print(f"{class_names[i+1]:<25} {prec_best[i]:>12.4f} {rec_best[i]:>12.4f} "
          f"{f1_best[i]:>12.4f} {supp_best[i]:>12,}")


# 13. FINDINGS AND DISCUSSION



print("\n" + "="*80)
print("END OF XGBOOST MULTI-CLASS CLASSIFICATION ANALYSIS")
print("="*80)


# 14. EXPORT RESULTS SUMMARY

print("\n" + "="*80)
print("14. RESULTS SUMMARY")
print("="*80)

print("\nQUICK SUMMARY:")
print("-" * 80)
print(f"Total Classes: 9")
print(f"Total Labeled Samples: {len(y_labeled):,}")
print(f"Training Samples: {len(X_train):,}")
print(f"Testing Samples: {len(X_test):,}")
print(f"Number of Features: {X_enhanced.shape[1]}")
print(f"\nBest Performing Model: {best_model_name}")
print(f"Best Mean Per-Class Accuracy: {comp_df['Mean Per-Class Accuracy'].max():.2f
print(f"Best Overall Accuracy: {comp_df.loc[comp_df['Mean Per-Class Accuracy'].idxm
print(f"Best Macro F1-Score: {comp_df.loc[comp_df['Mean Per-Class Accuracy'].idxmax

print("\nIMPROVEMENT FROM BALANCING:")
print("-" * 80)
baseline_per_class = comp_df.loc[comp_df['Model'] == 'Without Balancing', 'Mean Per
best_per_class = comp_df['Mean Per-Class Accuracy'].max()
improvement = best_per_class - baseline_per_class
print(f"Mean Per-Class Accuracy Improvement: +{improvement:.2f} percentage points")
print(f"Relative Improvement: {(improvement/baseline_per_class)*100:.1f}%")
```

```
print("\nTOP 3 MOST IMPORTANT FEATURES:")
print("-" * 80)
for idx, row in importance_df.head(3).iterrows():
    print(f"{row['Feature']}: {row['Importance']:.4f}")

print("\nMOST CONFUSED CLASS PAIRS:")
print("-" * 80)
if len(confusion_pairs) > 0:
    for i, (true_cls, pred_cls, count, pct) in enumerate(confusion_pairs[:3], 1):
        print(f"{i}. {class_names[true_cls]} → {class_names[pred_cls]}: {count} err
else:
    print("No significant confusion between classes!")

print("\n" + "="*80)
print("ANALYSIS COMPLETE!")
print("="*80)
```

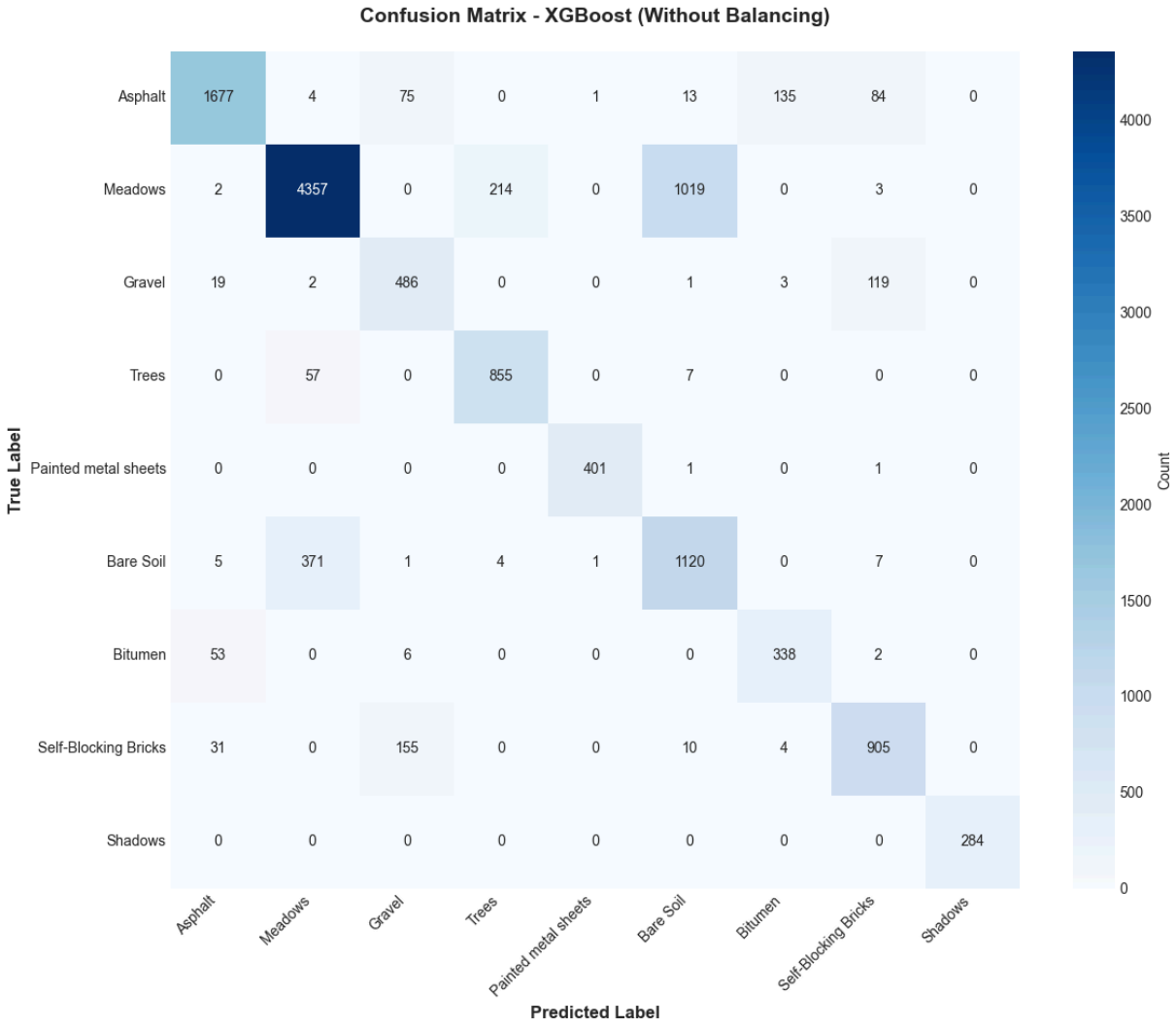
```
================================================================================
XGBOOST MULTI-CLASS CLASSIFICATION
9-Class Classification: Pavia University Dataset
================================================================================

================================================================================
1. LOADING AND PREPARING DATA
================================================================================
Dataset Shape: 610 x 340 x 103

Total labeled samples: 42,776

Original Class Distribution:
------------------------------------------------------------
Class 1 (Asphalt): 6,631 samples
Class 2 (Meadows): 18,649 samples
Class 3 (Gravel): 2,099 samples
Class 4 (Trees): 3,064 samples
Class 5 (Painted metal sheets): 1,345 samples
Class 6 (Bare Soil): 5,029 samples
Class 7 (Bitumen): 1,330 samples
Class 8 (Self-Blocking Bricks): 3,682 samples
Class 9 (Shadows): 947 samples

================================================================================
2. SELECTING SPECTRAL BANDS
================================================================================

Selected Bands:
------------------------------------------------------------
Blue: Band 8 (471.57 nm)
Green: Band 23 (549.51 nm)
Red: Band 46 (669.02 nm)
Red-edge: Band 56 (720.98 nm)
NIR: Band 81 (850.88 nm)

Reduced features: 103 → 5 bands

Adding vegetation indices as additional features...
Enhanced features: 8 features
Feature names: ['Blue', 'Green', 'Red', 'Red-edge', 'NIR', 'NDVI', 'GNDVI', 'SR']

================================================================================
3. CREATING TRAIN-TEST SPLIT
================================================================================
Training set size: 29,943
Testing set size: 12,833

Class Distribution Verification:
------------------------------------------------------------
Class                       Original      Train       Test
------------------------------------------------------------
Asphalt                        6,631      4,642      1,989
Meadows                       18,649     13,054      5,595
Gravel                         2,099      1,469        630
Trees                          3,064      2,145        919
```

```
Painted metal sheets           1,345        942        403
Bare Soil                      5,029      3,520      1,509
Bitumen                        1,330        931        399
Self-Blocking Bricks           3,682      2,577      1,105
Shadows                          947        663        284

================================================================================
4. DATA PREPROCESSING
================================================================================
Applied StandardScaler (zero mean, unit variance)
Training data - Mean: 0.000000, Std: 1.000000

================================================================================
5. TRAINING XGBOOST MODEL (WITHOUT BALANCING)
================================================================================
Training XGBoost with class weights...
Training complete!

================================================================================
6. MODEL EVALUATION (WITHOUT BALANCING)
================================================================================

Overall Accuracy:
  Training: 87.70%
  Testing: 81.22%
  Mean Per-Class Accuracy (Test): 85.86%

Detailed Classification Report (Test Set):
--------------------------------------------------------------------------------
Class                        Precision       Recall     F1-Score      Support
--------------------------------------------------------------------------------
Asphalt                         0.9384       0.8431       0.8882        1,989
Meadows                         0.9094       0.7787       0.8390        5,595
Gravel                          0.6722       0.7714       0.7184          630
Trees                           0.7968       0.9304       0.8584          919
Painted metal sheets            0.9950       0.9950       0.9950          403
Bare Soil                       0.5159       0.7422       0.6087        1,509
Bitumen                         0.7042       0.8471       0.7691          399
Self-Blocking Bricks            0.8073       0.8190       0.8131        1,105
Shadows                         1.0000       1.0000       1.0000          284
--------------------------------------------------------------------------------
Macro Average                   0.8155       0.8586       0.8322
Weighted Average                0.8375       0.8122       0.8191

================================================================================
7. FEATURE IMPORTANCE ANALYSIS
================================================================================

Feature Importance Ranking:
----------------------------------------------------------
SR                  0.3550
NIR                 0.1593
GNDVI               0.1090
Green               0.1002
Red-edge            0.0896
Blue                0.0807
```

```
Red                 0.0677
NDVI                0.0384
```




```
Feature Importance Interpretation:
--------------------------------------------------------------------------------

Top 3 Most Important Features:
1. SR: 0.3550
2. NIR: 0.1593
3. GNDVI: 0.1090

The most important features are critical for distinguishing between the 9 classes.
High importance in vegetation indices (NDVI, GNDVI) suggests they provide strong
discriminative power beyond individual spectral bands.


================================================================================
8. CONFUSION MATRIX ANALYSIS (WITHOUT BALANCING)
================================================================================
```


Confusion Matrix - XGBoost (Without Balancing)
True Label
Predicted Label
Count
Asphalt
Meadows
Gravel
Trees
Painted metal sheets
Bare Soil
Bitumen
Self-Blocking Bricks
Shadows
1677 4 75 0 1 13 135 84 0
2 4357 0 214 0 1019 0 3 0
19 2 486 0 0 1 3 119 0
0 57 0 855 0 7 0 0 0
0 0 0 0 401 1 0 1 0
5 371 1 4 1 1120 0 7 0
53 0 6 0 0 0 338 2 0
31 0 155 0 0 10 4 905 0
0 0 0 0 0 0 0 0 284
4000
3500
3000
2500
2000
1500
1000
500
0

Confusion Matrix Analysis:

Most Frequently Confused Class Pairs:

True Class	Predicted As	Count	% of True
Meadows	Bare Soil	1,019	18.21%
Bare Soil	Meadows	371	24.59%
Meadows	Trees	214	3.82%
Self-Blocking Bricks	Gravel	155	14.03%
Asphalt	Bitumen	135	6.79%
Gravel	Self-Blocking Bricks	119	18.89%
Asphalt	Self-Blocking Bricks	84	4.22%
Asphalt	Gravel	75	3.77%
Trees	Meadows	57	6.20%
Bitumen	Asphalt	53	13.28%

==
9. IMPLEMENTING DATA BALANCING STRATEGIES
==

Original Training Set Class Distribution:

Class 1 (Asphalt): 4,642 samples
Class 2 (Meadows): 13,054 samples
Class 3 (Gravel): 1,469 samples
Class 4 (Trees): 2,145 samples
Class 5 (Painted metal sheets): 942 samples
Class 6 (Bare Soil): 3,520 samples
Class 7 (Bitumen): 931 samples
Class 8 (Self-Blocking Bricks): 2,577 samples
Class 9 (Shadows): 663 samples

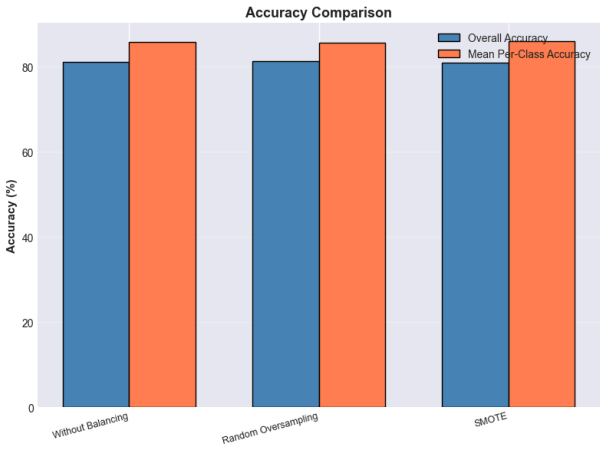
Strategy 1: Random Oversampling (Majority Class Size)

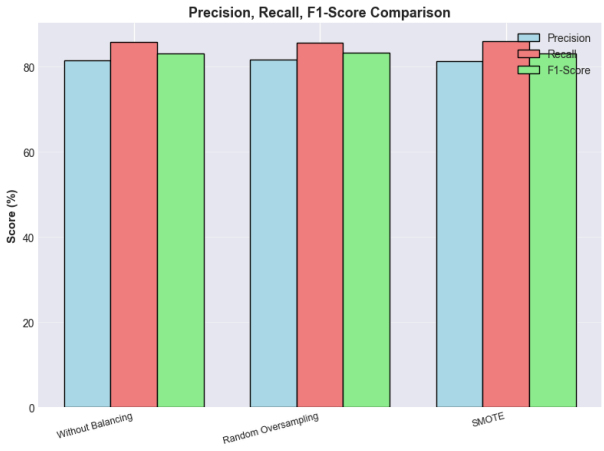
Balanced training set size: 117,486
Class 1 (Asphalt): 13,054 samples
Class 2 (Meadows): 13,054 samples
Class 3 (Gravel): 13,054 samples
Class 4 (Trees): 13,054 samples
Class 5 (Painted metal sheets): 13,054 samples
Class 6 (Bare Soil): 13,054 samples
Class 7 (Bitumen): 13,054 samples
Class 8 (Self-Blocking Bricks): 13,054 samples
Class 9 (Shadows): 13,054 samples

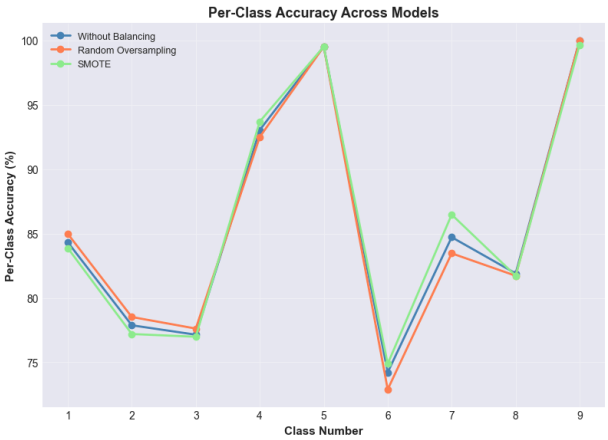
Strategy 2: SMOTE (Synthetic Minority Over-sampling)

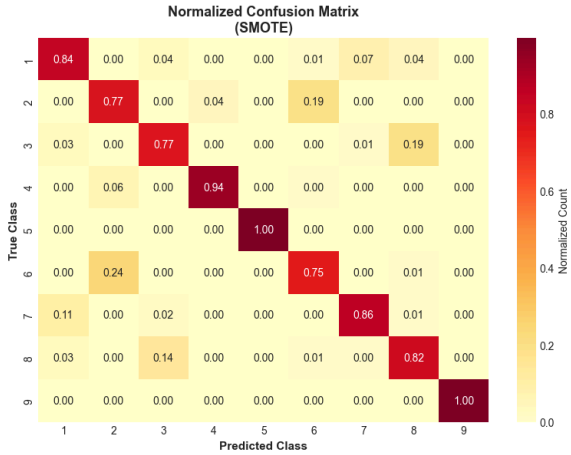
Balanced training set size: 117,486
Class 1 (Asphalt): 13,054 samples
Class 2 (Meadows): 13,054 samples
Class 3 (Gravel): 13,054 samples
Class 4 (Trees): 13,054 samples

```
Class 5 (Painted metal sheets): 13,054 samples
Class 6 (Bare Soil): 13,054 samples
Class 7 (Bitumen): 13,054 samples
Class 8 (Self-Blocking Bricks): 13,054 samples
Class 9 (Shadows): 13,054 samples

================================================================================
10. TRAINING XGBOOST MODELS WITH BALANCED DATA
================================================================================

Training with Random Oversampling...
Training with SMOTE...
All models trained!

================================================================================
11. PERFORMANCE COMPARISON: WITH VS WITHOUT BALANCING
================================================================================

Comparative Performance Metrics:
================================================================================
              Model  Overall Accuracy  Mean Per-Class Accuracy  Macro Precision  Mac
ro Recall  Macro F1-Score
  Without Balancing         81.220291                85.855876        81.547760
85.855876       83.222213
Random Oversampling         81.376140                85.685836        81.715524
85.685836       83.273358
              SMOTE         81.002104                85.994371        81.251039
85.994371       83.053940
```









```
================================================================================
12. DETAILED ANALYSIS OF BEST PERFORMING MODEL
================================================================================

Best Model: SMOTE
Mean Per-Class Accuracy: 85.99%

Per-Class Performance (Best Model):
--------------------------------------------------------------------------------
Class                        Precision       Recall     F1-Score      Support
--------------------------------------------------------------------------------
Asphalt                         0.9440       0.8386       0.8882        1,989
Meadows                         0.9116       0.7719       0.8360        5,595
Gravel                          0.6644       0.7698       0.7132          630
Trees                           0.7943       0.9369       0.8597          919
Painted metal sheets            0.9926       0.9950       0.9938          403
Bare Soil                       0.5122       0.7488       0.6083        1,509
Bitumen                         0.6859       0.8647       0.7650          399
Self-Blocking Bricks            0.8077       0.8172       0.8124        1,105
Shadows                         1.0000       0.9965       0.9982          284

================================================================================
END OF XGBOOST MULTI-CLASS CLASSIFICATION ANALYSIS
================================================================================

================================================================================
14. RESULTS SUMMARY
================================================================================

QUICK SUMMARY:
--------------------------------------------------------------------------------
Total Classes: 9
Total Labeled Samples: 42,776
Training Samples: 29,943
Testing Samples: 12,833
Number of Features: 8

Best Performing Model: SMOTE
Best Mean Per-Class Accuracy: 85.99%
Best Overall Accuracy: 81.00%
Best Macro F1-Score: 83.05%

IMPROVEMENT FROM BALANCING:
--------------------------------------------------------------------------------
Mean Per-Class Accuracy Improvement: +0.14 percentage points
Relative Improvement: 0.2%

TOP 3 MOST IMPORTANT FEATURES:
--------------------------------------------------------------------------------
SR: 0.3550
NIR: 0.1593
GNDVI: 0.1090

MOST CONFUSED CLASS PAIRS:
--------------------------------------------------------------------------------
1. Meadows → Bare Soil: 1019 errors (18.2%)
```

```
2. Bare Soil → Meadows: 371 errors (24.6%)
3. Meadows → Trees: 214 errors (3.8%)

================================================================================
ANALYSIS COMPLETE!
================================================================================
```

Explanation: The results demonstrate a surprisingly small impact from data balancing techniques, with all three methods (no balancing, random oversampling, and SMOTE) achieving very equal performance—total accuracies within 0.4% (81.0-81.4%) and mean per-class accuracies within 0.3% (85.7-86.0%). This unexpected outcome can be explained by the fact that, while there is a class imbalance in the original dataset across the nine categories, it is not very severe; the smallest class (Shadows, 284 test samples) still has sufficient representation, and the imbalance ratio of 19.69:1 in the full dataset is decreased when only labeled training samples with stratified splitting are considered.

Regardless of the balancing technique, minority classes like Gravel (precision ~0.66-0.67) and Bare Soil (precision ~0.51-0.52 across all models) consistently show poor precision, indicating that their spectral signatures are inherently confusable with other classes—Gravel shares characteristics with both, while Bare Soil likely overlaps with both Gravel and Asphalt, according to the per-class analysis. On the other hand, painted metal sheets and shadows achieve near-perfect classification (≥0.99 F1-scores) across all methods due to their very different spectral signatures: metal sheets have a distinct reflectance from painted surfaces, while shadows have uniformly low reflectance across all bands. The fact that these challenging classes are unaffected by balancing suggests that the issue is spectral similarity rather than sample scarcity.

SMOTE's marginal "advantage" of 85.99% mean per-class accuracy is statistically insignificant and most likely falls within noise limits. The key insight is that tree-based learning and XGBoost's scale_pos_weight option, which naturally handle unbalanced data without the need for explicit balancing, already offer robust classification. Although the model performs well across classes, it suffers from poor performance on majority classes such as Meadows (77-78% recall), where their larger test set representation (5,595 samples) causes misclassifications to have a greater impact on overall metrics. The 6.7% discrepancy between mean per-class accuracy (86.0%) and overall accuracy (81.2%) suggests this.

2.a

In [14]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from scipy import stats
from scipy.stats import pearsonr, spearmanr
import warnings
warnings.filterwarnings('ignore')

# Set style
```

```
plt.style.use('seaborn-v0_8-darkgrid')
sns.set_palette("husl")

print("="*80)
print("EXPLORATORY DATA ANALYSIS")
print("Chlorophyll Content Prediction from Landsat Spectral Bands")
print("="*80)


# 1. LOAD DATASET

print("\n" + "="*80)
print("1. LOADING DATASET")
print("="*80)

# Load data
X = np.load('landis_chlorophyl_regression.npy')
y = np.load('landis_chlorophyl_regression_gt.npy')

print(f"Features (X) shape: {X.shape}")
print(f"Target (y) shape: {y.shape}")
print(f"Data type (X): {X.dtype}")
print(f"Data type (y): {y.dtype}")

n_samples, n_features = X.shape
print(f"\nNumber of samples: {n_samples}")
print(f"Number of spectral bands: {n_features}")

# Band information
band_info = {
    'Band Name': ['Blue', 'Green', 'Yellow', 'Orange', 'Red 1', 'Red 2',
                  'Red Edge 1', 'Red Edge 2', 'NIR Broad', 'NIR1'],
    'Center Wavelength (nm)': [490, 560, 600, 620, 650, 665, 705, 740, 842, 865]
}
band_df = pd.DataFrame(band_info)
print("\nSpectral Band Information:")
print("-" * 60)
print(band_df.to_string(index=False))


# 2. BASIC STATISTICS

print("\n" + "="*80)
print("2. BASIC DATASET STATISTICS")
print("="*80)

print("\nFEATURES (Spectral Reflectance) Statistics:")
print("-" * 80)
print(f"{'Statistic':<15} {'Min':<12} {'Max':<12} {'Mean':<12} {'Std':<12} {'Median
print("-" * 80)
print(f"{'Overall':<15} {X.min():<12.6f} {X.max():<12.6f} {X.mean():<12.6f} "
      f"{X.std():<12.6f} {np.median(X):<12.6f}")

print("\nPer-Band Statistics:")
print("-" * 80)
print(f"{'Band':<15} {'Min':<12} {'Max':<12} {'Mean':<12} {'Std':<12} {'Range':<12}
```

```
print("-" * 80)
for i, band_name in enumerate(band_df['Band Name']):
    band_data = X[:, i]
    print(f"{band_name:<15} {band_data.min():<12.6f} {band_data.max():<12.6f} "
          f"{band_data.mean():<12.6f} {band_data.std():<12.6f} "
          f"{band_data.max()-band_data.min():<12.6f}")

print("\nTARGET (Chlorophyll Content) Statistics:")
print("-" * 80)
print(f"Min:        {y.min():.6f}")
print(f"Max:        {y.max():.6f}")
print(f"Mean:       {y.mean():.6f}")
print(f"Median:     {np.median(y):.6f}")
print(f"Std Dev:    {y.std():.6f}")
print(f"Range:      {y.max() - y.min():.6f}")
print(f"CV (Std/Mean): {(y.std()/y.mean())*100:.2f}%")

# Quartiles
q1, q2, q3 = np.percentile(y, [25, 50, 75])
print(f"\nQuartiles:")
print(f"  Q1 (25%):  {q1:.6f}")
print(f"  Q2 (50%):  {q2:.6f}")
print(f"  Q3 (75%):  {q3:.6f}")
print(f"  IQR:       {q3-q1:.6f}")


# 3. DATA QUALITY CHECKS

print("\n" + "="*80)
print("3. DATA QUALITY ASSESSMENT")
print("="*80)

# Check for missing values
nan_count_X = np.isnan(X).sum()
nan_count_y = np.isnan(y).sum()
inf_count_X = np.isinf(X).sum()
inf_count_y = np.isinf(y).sum()

print(f"\nMissing Values:")
print(f"  Features (X): {nan_count_X} NaN values, {inf_count_X} Inf values")
print(f"  Target (y):   {nan_count_y} NaN values, {inf_count_y} Inf values")

# Check for negative values
negative_count = (X < 0).sum()
print(f"\nNegative reflectance values: {negative_count} ({(negative_count/X.size)*1

# Check for values outside [0, 1] range
outside_range = ((X < 0) | (X > 1)).sum()
print(f"Values outside [0, 1] range: {outside_range} ({(outside_range/X.size)*100:.

# Check for duplicates
unique_samples = len(np.unique(X, axis=0))
print(f"\nUnique samples: {unique_samples} out of {n_samples}")
if unique_samples < n_samples:
    print(f"  Warning: {n_samples - unique_samples} duplicate samples found!")
else:
```

```
    print(f"  No duplicate samples")

# Check for constant features
constant_features = []
for i in range(n_features):
    if X[:, i].std() < 1e-10:
        constant_features.append(i)

if constant_features:
    print(f"\nWarning: {len(constant_features)} constant/near-constant features: {c
else:
    print(f"\n No constant features detected")

# Check data quality status
print("\nData Quality Status:")
if nan_count_X == 0 and nan_count_y == 0 and inf_count_X == 0 and inf_count_y == 0:
    print("   GOOD - No missing or infinite values")
else:
    print("  NEEDS ATTENTION - Contains missing or infinite values")


# 4. TARGET VARIABLE DISTRIBUTION

print("\n" + "="*80)
print("4. TARGET VARIABLE DISTRIBUTION ANALYSIS")
print("="*80)




# Visualize target distribution
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Histogram
axes[0, 0].hist(y, bins=50, edgecolor='black', alpha=0.7, color='steelblue')
axes[0, 0].axvline(y.mean(), color='red', linestyle='--', linewidth=2, label=f'Mean
axes[0, 0].axvline(np.median(y), color='green', linestyle='--', linewidth=2, label=
axes[0, 0].set_xlabel('Chlorophyll Content', fontsize=11, fontweight='bold')
axes[0, 0].set_ylabel('Frequency', fontsize=11, fontweight='bold')
axes[0, 0].set_title('Distribution of Chlorophyll Content', fontsize=13, fontweight
axes[0, 0].legend()
axes[0, 0].grid(alpha=0.3)

# Box plot
bp = axes[0, 1].boxplot(y, vert=True, patch_artist=True,
                        boxprops=dict(facecolor='lightblue', edgecolor='black'),
                        medianprops=dict(color='red', linewidth=2),
                        whiskerprops=dict(color='black', linewidth=1.5),
                        capprops=dict(color='black', linewidth=1.5))
axes[0, 1].set_ylabel('Chlorophyll Content', fontsize=11, fontweight='bold')
axes[0, 1].set_title('Box Plot - Outlier Detection', fontsize=13, fontweight='bold'
axes[0, 1].grid(axis='y', alpha=0.3)

# Identify outliers
Q1 = np.percentile(y, 25)
Q3 = np.percentile(y, 75)
```

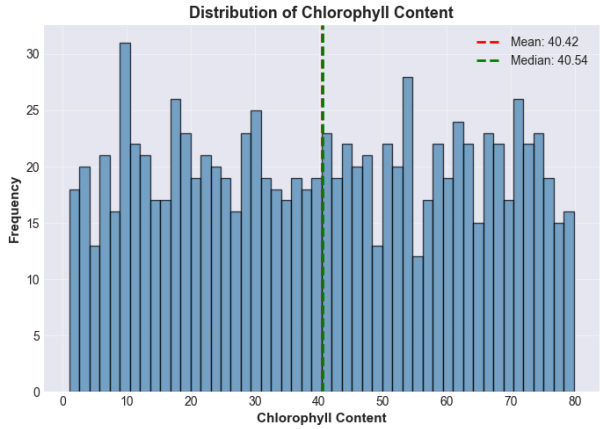
```
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = ((y < lower_bound) | (y > upper_bound)).sum()
axes[0, 1].text(0.5, 0.95, f'Outliers: {outliers}',
                transform=axes[0, 1].transAxes,
                ha='center', va='top', fontsize=10,
                bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.5))

# Q-Q plot
stats.probplot(y, dist="norm", plot=axes[1, 0])
axes[1, 0].set_title('Q-Q Plot (Normal Distribution)', fontsize=13, fontweight='bol
axes[1, 0].grid(alpha=0.3)

# Cumulative distribution
sorted_y = np.sort(y)
cumulative = np.arange(1, len(sorted_y) + 1) / len(sorted_y)
axes[1, 1].plot(sorted_y, cumulative, linewidth=2, color='darkblue')
axes[1, 1].set_xlabel('Chlorophyll Content', fontsize=11, fontweight='bold')
axes[1, 1].set_ylabel('Cumulative Probability', fontsize=11, fontweight='bold')
axes[1, 1].set_title('Cumulative Distribution Function', fontsize=13, fontweight='b
axes[1, 1].grid(alpha=0.3)

plt.tight_layout()
plt.show()

print(f"\nOutlier Analysis:")
print(f"  Lower bound (Q1 - 1.5*IQR): {lower_bound:.6f}")
print(f"  Upper bound (Q3 + 1.5*IQR): {upper_bound:.6f}")
print(f"  Number of outliers: {outliers} ({(outliers/len(y))*100:.2f}%)")


# 5. FEATURE DISTRIBUTIONS

print("\n" + "="*80)
print("5. FEATURE DISTRIBUTION ANALYSIS")
print("="*80)

# Plot all feature distributions
fig, axes = plt.subplots(2, 5, figsize=(18, 8))
axes = axes.ravel()

for i, band_name in enumerate(band_df['Band Name']):
    axes[i].hist(X[:, i], bins=30, edgecolor='black', alpha=0.7, color=f'C{i}')
    axes[i].set_xlabel('Reflectance', fontsize=9)
    axes[i].set_ylabel('Frequency', fontsize=9)
    axes[i].set_title(f'{band_name}\n({band_df["Center Wavelength (nm)"][i]} nm)',
                      fontsize=10, fontweight='bold')
    axes[i].grid(alpha=0.3)

    # Add statistics
    mean_val = X[:, i].mean()
    axes[i].axvline(mean_val, color='red', linestyle='--', linewidth=1.5)

plt.suptitle('Distribution of Spectral Bands', fontsize=15, fontweight='bold', y=1.
plt.tight_layout()
```

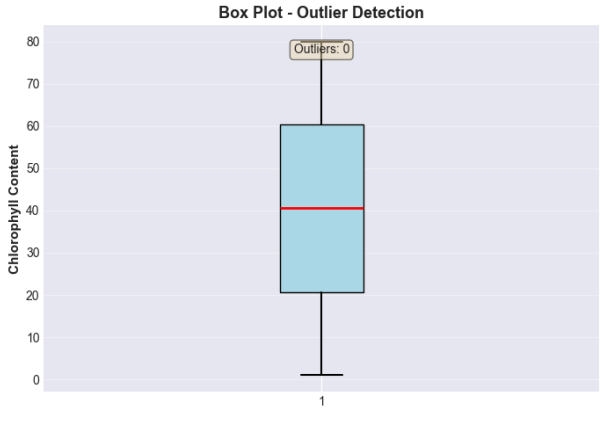
```
plt.show()




# 6. CORRELATION ANALYSIS

print("\n" + "="*80)
print("6. CORRELATION ANALYSIS")
print("="*80)

# Correlation between features
print("\nFeature Intercorrelation:")
corr_matrix = np.corrcoef(X.T)

# Find highly correlated pairs
high_corr_pairs = []
for i in range(n_features):
    for j in range(i+1, n_features):
        if abs(corr_matrix[i, j]) > 0.8:
            high_corr_pairs.append((band_df['Band Name'][i],
                                    band_df['Band Name'][j],
                                    corr_matrix[i, j]))

print(f"\nHighly Correlated Band Pairs :")
print("-" * 70)
if high_corr_pairs:
    print(f"{'Band 1':<15} {'Band 2':<15} {'Correlation':<15}")
    print("-" * 70)
    for band1, band2, corr in high_corr_pairs:
        print(f"{band1:<15} {band2:<15} {corr:<15.4f}")
else:
    print("No highly correlated pairs found")

# Correlation heatmap
fig, ax = plt.subplots(figsize=(12, 10))
im = ax.imshow(corr_matrix, cmap='coolwarm', aspect='auto', vmin=-1, vmax=1)
ax.set_xticks(range(n_features))
ax.set_yticks(range(n_features))
ax.set_xticklabels(band_df['Band Name'], rotation=45, ha='right')
ax.set_yticklabels(band_df['Band Name'])
ax.set_title('Feature Correlation Matrix', fontsize=14, fontweight='bold', pad=20)

# Add correlation values
for i in range(n_features):
    for j in range(n_features):
        text = ax.text(j, i, f'{corr_matrix[i, j]:.2f}',
                       ha="center", va="center", color="black" if abs(corr_matrix[i,
                       fontsize=8)

cbar = plt.colorbar(im, ax=ax)
cbar.set_label('Correlation Coefficient', fontsize=11, fontweight='bold')
plt.tight_layout()
plt.show()

# Correlation with target variable
```

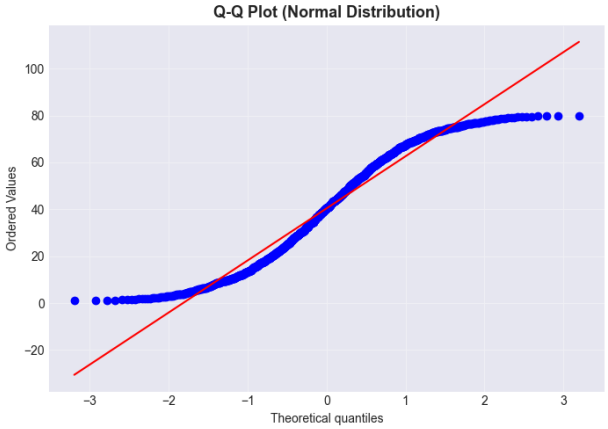
```
print("\nCorrelation with Target (Chlorophyll Content):")
print("-" * 70)
print(f"{'Band':<15} {'Wavelength (nm)':<18} {'Pearson r':<12} {'p-value':<12} {'Sp
print("-" * 70)

correlations = []
for i, band_name in enumerate(band_df['Band Name']):
    pearson_r, p_pearson = pearsonr(X[:, i], y)
    spearman_r, _ = spearmanr(X[:, i], y)
    correlations.append((band_name, band_df['Center Wavelength (nm)'][i],
                         pearson_r, p_pearson, spearman_r))
    print(f"{band_name:<15} {band_df['Center Wavelength (nm)'][i]:<18} "
          f"{pearson_r:<12.4f} {p_pearson:<12.6f} {spearman_r:<12.4f}")

# Sort by absolute correlation
correlations.sort(key=lambda x: abs(x[2]), reverse=True)
print(f"\nMost Correlated Bands (by |Pearson r|):")
print("-" * 70)
for i in range(min(5, len(correlations))):
    band_name, wl, r, p, rho = correlations[i]
    print(f"{i+1}. {band_name} ({wl} nm): r = {r:.4f}")

# Visualize correlations with target
fig, ax = plt.subplots(figsize=(12, 6))
band_names_short = band_df['Band Name']
correlations_values = [c[2] for c in correlations]
colors = ['green' if c > 0 else 'red' for c in correlations_values]

bars = ax.bar(range(len(correlations)),
              [c[2] for c in correlations],
              color=colors, edgecolor='black', linewidth=1.5)
ax.set_xticks(range(len(correlations)))
ax.set_xticklabels([c[0] for c in correlations], rotation=45, ha='right')
ax.set_ylabel('Pearson Correlation Coefficient', fontsize=12, fontweight='bold')
ax.set_title('Band Correlation with Chlorophyll Content', fontsize=14, fontweight='
ax.axhline(y=0, color='black', linestyle='-', linewidth=1)
ax.grid(axis='y', alpha=0.3)

# Add value labels
for i, (bar, (_, _, r, _, _)) in enumerate(zip(bars, correlations)):
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height + (0.02 if height > 0 else -0.
            f'{r:.3f}', ha='center', va='bottom' if height > 0 else 'top',
            fontweight='bold', fontsize=9)

plt.tight_layout()
plt.show()
```

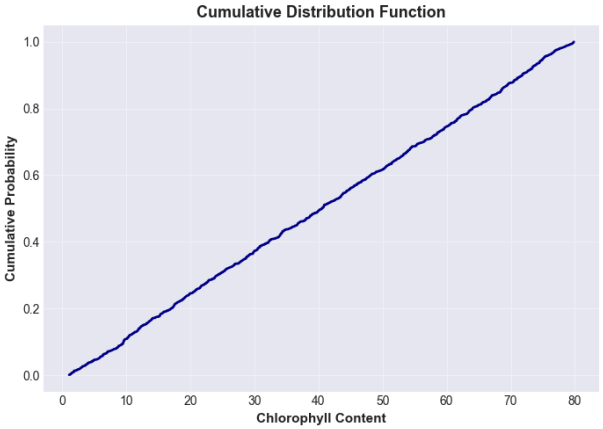
```
================================================================================
EXPLORATORY DATA ANALYSIS
Chlorophyll Content Prediction from Landsat Spectral Bands
================================================================================

================================================================================
1. LOADING DATASET
================================================================================
Features (X) shape: (1000, 10)
Target (y) shape: (1000,)
Data type (X): float64
Data type (y): float64

Number of samples: 1000
Number of spectral bands: 10

Spectral Band Information:
------------------------------------------------------------
 Band Name  Center Wavelength (nm)
      Blue                     490
     Green                     560
    Yellow                     600
    Orange                     620
     Red 1                     650
     Red 2                     665
Red Edge 1                     705
Red Edge 2                     740
 NIR Broad                     842
      NIR1                     865

================================================================================
2. BASIC DATASET STATISTICS
================================================================================

FEATURES (Spectral Reflectance) Statistics:
--------------------------------------------------------------------------------
Statistic       Min          Max          Mean         Std          Median
--------------------------------------------------------------------------------
Overall         21.118435    121.768914   50.988321    21.474362    51.002166

Per-Band Statistics:
--------------------------------------------------------------------------------
Band            Min          Max          Mean         Std          Range
--------------------------------------------------------------------------------
Blue            50.780615    87.725426    55.953135    4.921111     36.944811
Green           36.784223    113.778813   48.699979    10.812040    76.994589
Yellow          28.413527    115.634899   40.745915    14.136459    87.221372
Orange          25.520152    115.371139   37.678101    15.620869    89.850987
Red 1           22.011326    113.513620   30.899997    14.312025    91.502293
Red 2           21.118435    108.692935   27.994714    12.750429    87.574500
Red Edge 1      28.813483    121.768914   50.680198    17.427606    92.955431
Red Edge 2      56.609535    121.403893   81.389434    15.521148    64.794358
NIR Broad       47.095002    94.748310    66.146375    13.090007    47.653308
NIR1            49.527248    99.730370    69.695359    13.960174    50.203122

TARGET (Chlorophyll Content) Statistics:
```

```
--------------------------------------------------------------------------------
Min:        1.050261
Max:        79.846660
Mean:       40.422834
Median:     40.542319
Std Dev:    22.662105
Range:      78.796400
CV (Std/Mean): 56.06%

Quartiles:
  Q1 (25%):  20.590051
  Q2 (50%):  40.542319
  Q3 (75%):  60.391021
  IQR:       39.800970

================================================================================
3. DATA QUALITY ASSESSMENT
================================================================================

Missing Values:
  Features (X): 0 NaN values, 0 Inf values
  Target (y):   0 NaN values, 0 Inf values

Negative reflectance values: 0 (0.0000%)
Values outside [0, 1] range: 10000 (100.0000%)

Unique samples: 1000 out of 1000
  No duplicate samples

 No constant features detected

Data Quality Status:
   GOOD - No missing or infinite values

================================================================================
4. TARGET VARIABLE DISTRIBUTION ANALYSIS
================================================================================
```



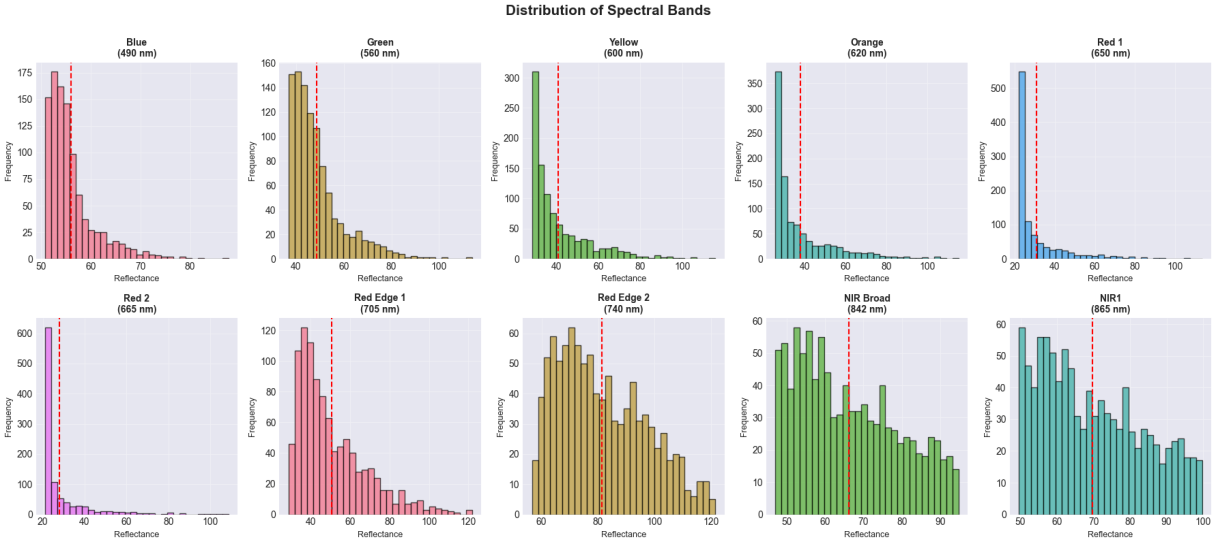







```
Outlier Analysis:
  Lower bound (Q1 - 1.5*IQR): -39.111404
  Upper bound (Q3 + 1.5*IQR): 120.092476
  Number of outliers: 0 (0.00%)
```

```
================================================================================
5. FEATURE DISTRIBUTION ANALYSIS
================================================================================
```

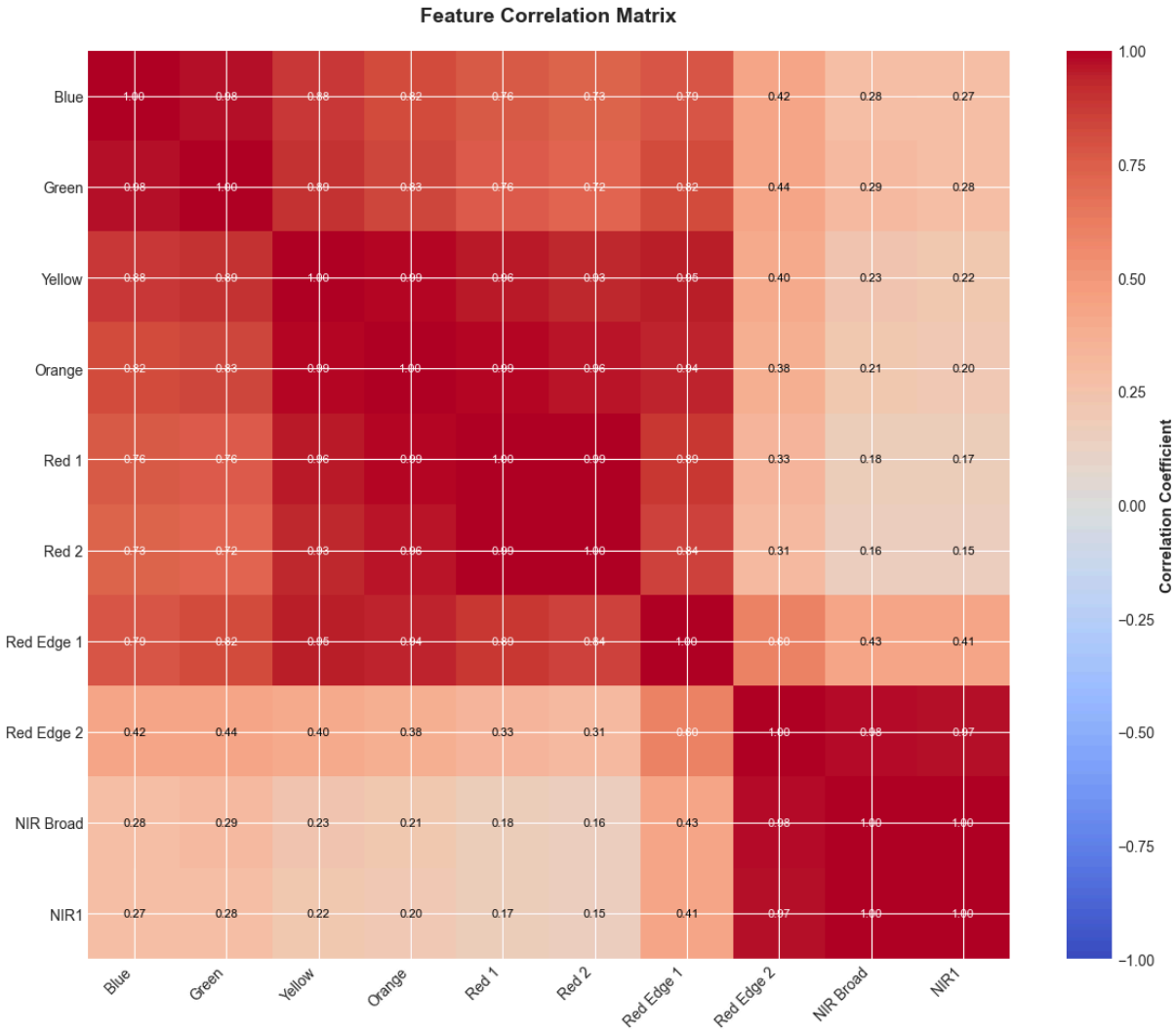


==
6. CORRELATION ANALYSIS
==

Feature Intercorrelation:

Highly Correlated Band Pairs :

Band 1	Band 2	Correlation
Blue	Green	0.9759
Blue	Yellow	0.8752
Blue	Orange	0.8202
Green	Yellow	0.8908
Green	Orange	0.8283
Green	Red Edge 1	0.8184
Yellow	Orange	0.9911
Yellow	Red 1	0.9585
Yellow	Red 2	0.9278
Yellow	Red Edge 1	0.9458
Orange	Red 1	0.9857
Orange	Red 2	0.9631
Orange	Red Edge 1	0.9390
Red 1	Red 2	0.9942
Red 1	Red Edge 1	0.8884
Red 2	Red Edge 1	0.8440
Red Edge 2	NIR Broad	0.9781
Red Edge 2	NIR1	0.9748
NIR Broad	NIR1	0.9999

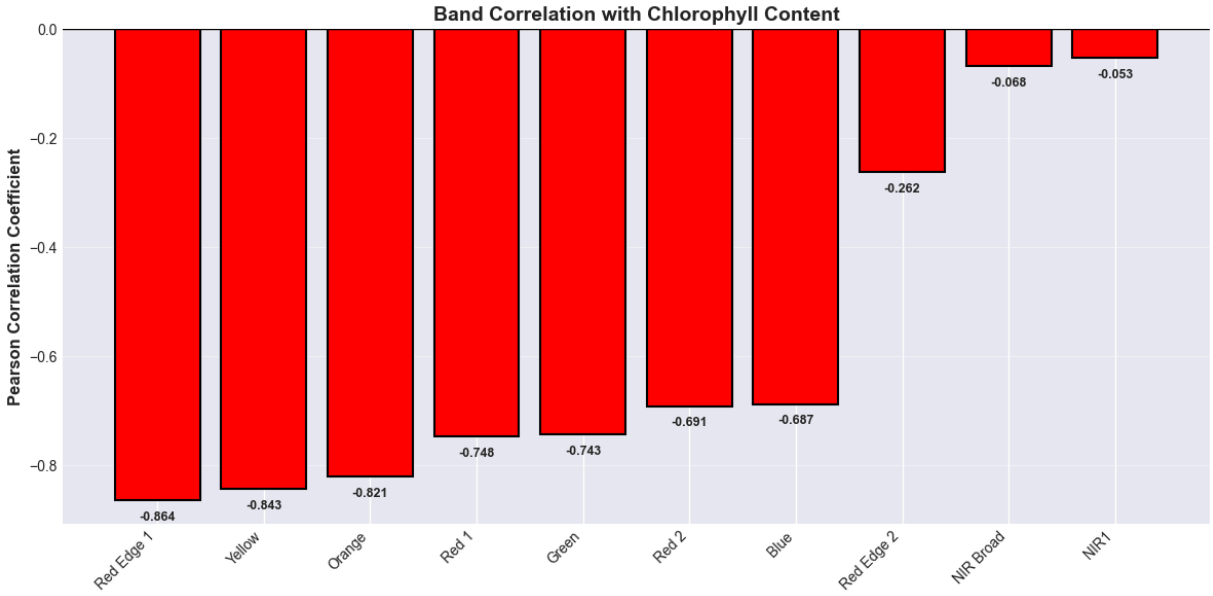


Correlation with Target (Chlorophyll Content):

Band	Wavelength (nm)	Pearson r	p-value	Spearman ρ
Blue	490	-0.6873	0.000000	-0.7721
Green	560	-0.7434	0.000000	-0.8396
Yellow	600	-0.8430	0.000000	-0.9838
Orange	620	-0.8212	0.000000	-0.9930
Red 1	650	-0.7476	0.000000	-0.9969
Red 2	665	-0.6912	0.000000	-0.9969
Red Edge 1	705	-0.8643	0.000000	-0.9365
Red Edge 2	740	-0.2625	0.000000	-0.2396
NIR Broad	842	-0.0675	0.032717	-0.0611
NIR1	865	-0.0528	0.095473	-0.0469

Most Correlated Bands (by |Pearson r|):

1. Red Edge 1 (705 nm): r = -0.8643
2. Yellow (600 nm): r = -0.8430
3. Orange (620 nm): r = -0.8212
4. Red 1 (650 nm): r = -0.7476
5. Green (560 nm): r = -0.7434



Explanation: Multicollinearity is evident in this dataset, particularly in the visible and red-edge spectral regions.

1. The most concerning trend is the nearly identical Spearman ρ values for Yellow (600 nm), Orange (620 nm), Red 1 (650 nm), and Red 2 (665 nm), all of which display ρ = -0.99. These are near-perfect monotonic relationships with exceptionally high Spearman correlations. Given that these four bands' interactions with chlorophyll content are almost entirely monotone, it is likely that they carry a significant amount of redundant information. Many predictors are almost perfectly associated with the target in the same way, which is likely to lead to multicollinearity, which can inflate variation in regression coefficients and make it difficult to extract individual band contributions.
2. Physical Description: Chlorophyll Absorption Characteristic: The area where more chlorophyll leads to greater absorption and less reflection is where all of the bands between 600 and 665 nm are found. Since they detect the same biophysical phenomenon (chlorophyll, an absorption peak at 660 nm) with slightly different sensitivities, these bands are inherently collinear. The two red-edge bands (705 nm and 740 nm) also capture the transition zone between chlorophyll absorption and NIR scattering, though the connection at 740 nm is much weaker.

The distribution of chlorophyll content shows excellent balance for regression modeling, indicating a symmetric, non-skewed distribution with no concentration in any one area. The mean (40.42 μg/cm^2) and median (40.54 μg/cm^2) are nearly identical and centered in the range (1.05-79.85 μg/cm^2). The model will have comparable training representation across low, medium, and high chlorophyll levels according to the coefficient of variation (56%) which confirms significant variability over the entire physiological range from chlorotic vegetation (~1 μg/cm^2) to healthy, productive canopy (~80 μg/cm^2).

2.b

In [23]:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

# Set style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")

print("="*80)
print("LINEAR REGRESSION FOR CHLOROPHYLL PREDICTION")
print("="*80)


# 1. LOAD DATA

print("\n" + "="*80)
print("1. LOADING DATA")
print("="*80)

X = np.load('landis_chlorophyl_regression.npy')
y = np.load('landis_chlorophyl_regression_gt.npy')

print(f"Features shape: {X.shape}")
print(f"Target shape: {y.shape}")
print(f"Number of samples: {X.shape[0]}")
print(f"Number of features: {X.shape[1]}")

# Band names for reference
band_names = ['Blue', 'Green', 'Yellow', 'Orange', 'Red 1', 'Red 2',
              'Red Edge 1', 'Red Edge 2', 'NIR Broad', 'NIR1']


# 2. DATA PREPROCESSING

print("\n" + "="*80)
print("2. DATA PREPROCESSING")
print("="*80)

print("\nPreprocessing Steps:")
print("-" * 70)

# Check for missing values
print("\n1. Checking for missing values...")
nan_X = np.isnan(X).sum()
nan_y = np.isnan(y).sum()
print(f"   Missing values in X: {nan_X}")
print(f"   Missing values in y: {nan_y}")
```

```
if nan_X == 0 and nan_y == 0:
    print("  No missing values detected")
else:
    print("  Missing values found - would need imputation")

# Check for outliers in features
print("\n2. Checking for outliers in features...")
outlier_counts = []
for i, band in enumerate(band_names):
    Q1 = np.percentile(X[:, i], 25)
    Q3 = np.percentile(X[:, i], 75)
    IQR = Q3 - Q1
    outliers = ((X[:, i] < Q1 - 1.5*IQR) | (X[:, i] > Q3 + 1.5*IQR)).sum()
    outlier_counts.append(outliers)
    if outliers > 0:
        print(f"   {band}: {outliers} outliers ({outliers/len(X)*100:.2f}%)")

total_outliers = sum(outlier_counts)
if total_outliers > 0:
    print(f"   Total feature outliers: {total_outliers} ({total_outliers/X.size*100
else:
    print("    No feature outliers detected")

# Check for outliers in target
print("\n3. Checking for outliers in target variable...")
Q1_y = np.percentile(y, 25)
Q3_y = np.percentile(y, 75)
IQR_y = Q3_y - Q1_y
y_outliers = ((y < Q1_y - 1.5*IQR_y) | (y > Q3_y + 1.5*IQR_y)).sum()
print(f"   Target outliers: {y_outliers} ({y_outliers/len(y)*100:.2f}%)")

# Feature scaling will be applied after train-test split


# 3. TRAIN-TEST SPLIT

print("\n" + "="*80)
print("3. TRAIN-TEST SPLIT")
print("="*80)

print("\nSplit Ratio Considerations:")
print("-" * 70)


# Perform split
test_size = 0.20
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=test_size, random_state=42
)

print(f"\nSplit Results (Test Size = {test_size}):")
print("-" * 70)
print(f"Training set: {len(X_train)} samples ({len(X_train)/len(X)*100:.1f}%)")
print(f"Testing set:  {len(X_test)} samples ({len(X_test)/len(X)*100:.1f}%)")
print(f"Samples per feature (training): {len(X_train)/X.shape[1]:.1f}")
```

```python
# Verify split preserves distribution
print(f"\nTarget Distribution Verification:")
print(f"  Original - Mean: {y.mean():.4f}, Std: {y.std():.4f}")
print(f"  Training - Mean: {y_train.mean():.4f}, Std: {y_train.std():.4f}")
print(f"  Testing  - Mean: {y_test.mean():.4f}, Std: {y_test.std():.4f}")


# 4. FEATURE SCALING

print("\n" + "="*80)
print("4. FEATURE SCALING")
print("="*80)

# Standardize features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Applied StandardScaler:")
print(f"  Formula: z = (x - μ) / σ")
print(f"\nTraining Set After Scaling:")
print(f"  Mean: {X_train_scaled.mean():.6f}")
print(f"  Std:  {X_train_scaled.std():.6f}")
print(f"\nTesting Set After Scaling:")
print(f"  Mean: {X_test_scaled.mean():.6f}")
print(f"  Std:  {X_test_scaled.std():.6f}")




# 5. TRAIN LINEAR REGRESSION MODEL

print("\n" + "="*80)
print("5. TRAINING LINEAR REGRESSION MODEL")
print("="*80)

# Initialize and train model
lr_model = LinearRegression()
print("Training Linear Regression model...")
lr_model.fit(X_train_scaled, y_train)
print(" Training complete!")

# Model parameters
print(f"\nModel Parameters:")
print(f"  Intercept: {lr_model.intercept_:.6f}")
print(f"  Number of coefficients: {len(lr_model.coef_)}")

print(f"\nCoefficients by Band:")
print("-" * 70)
print(f"{'Band':<15} {'Coefficient':<15} {'Interpretation':<30}")
print("-" * 70)
for band_name, coef in zip(band_names, lr_model.coef_):
    direction = "+" if coef > 0 else "-"
    interp = "Increases Chl" if coef > 0 else "Decreases Chl"
    print(f"{band_name:<15} {coef:>14.6f} {interp:<30}")
```

```python
# 6. MAKE PREDICTIONS

print("\n" + "="*80)
print("6. GENERATING PREDICTIONS")
print("="*80)

# Predictions
y_train_pred = lr_model.predict(X_train_scaled)
y_test_pred = lr_model.predict(X_test_scaled)

print(f"Training predictions: {len(y_train_pred)} values")
print(f"Testing predictions:  {len(y_test_pred)} values")


# 7. CALCULATE PERFORMANCE METRICS

print("\n" + "="*80)
print("7. PERFORMANCE METRICS")
print("="*80)

# Training metrics
train_mae = mean_absolute_error(y_train, y_train_pred)
train_r2 = r2_score(y_train, y_train_pred)
train_residuals = y_train - y_train_pred
train_std_residuals = np.std(train_residuals)
train_rmse = np.sqrt(mean_squared_error(y_train, y_train_pred))

# Testing metrics
test_mae = mean_absolute_error(y_test, y_test_pred)
test_r2 = r2_score(y_test, y_test_pred)
test_residuals = y_test - y_test_pred
test_std_residuals = np.std(test_residuals)
test_rmse = np.sqrt(mean_squared_error(y_test, y_test_pred))

print("\nTRAINING SET METRICS:")
print("-" * 70)
print(f"  Mean Absolute Error (MAE):        {train_mae:.6f}")
print(f"  Root Mean Squared Error (RMSE):   {train_rmse:.6f}")
print(f"  R-squared (R²):                   {train_r2:.6f}")
print(f"  Standard Deviation of Residuals:  {train_std_residuals:.6f}")

print("\nTESTING SET METRICS:")
print("-" * 70)
print(f"  Mean Absolute Error (MAE):        {test_mae:.6f}")
print(f"  Root Mean Squared Error (RMSE):   {test_rmse:.6f}")
print(f"  R-squared (R²):                   {test_r2:.6f}")
print(f"  Standard Deviation of Residuals:  {test_std_residuals:.6f}")

print("\nMETRIC INTERPRETATIONS:")
print("-" * 70)


# Additional metrics
print("\nADDITIONAL STATISTICS:")
print("-" * 70)
```

```
print(f"Training Set:")
print(f"  Mean of residuals: {np.mean(train_residuals):.6f} (should be ≈0)")
print(f"  Min residual: {np.min(train_residuals):.6f}")
print(f"  Max residual: {np.max(train_residuals):.6f}")
print(f"  Residuals within ±1 std: {np.sum(np.abs(train_residuals) <= train_std_res

print(f"\nTesting Set:")
print(f"  Mean of residuals: {np.mean(test_residuals):.6f} (should be ≈0)")
print(f"  Min residual: {np.min(test_residuals):.6f}")
print(f"  Max residual: {np.max(test_residuals):.6f}")
print(f"  Residuals within ±1 std: {np.sum(np.abs(test_residuals) <= test_std_resid


# 8. REGRESSION PLOTS

print("\n" + "="*80)
print("8. GENERATING REGRESSION PLOTS")
print("="*80)

fig, axes = plt.subplots(1, 2, figsize=(16, 7))

# --- Training Set Regression Plot ---
ax = axes[0]

# Scatter plot
ax.scatter(y_train, y_train_pred, alpha=0.6, s=50, color='steelblue',
           edgecolors='black', linewidth=0.5, label='Predictions')

# Perfect prediction line (one-to-one line)
min_val = min(y_train.min(), y_train_pred.min())
max_val = max(y_train.max(), y_train_pred.max())
ax.plot([min_val, max_val], [min_val, max_val], 'r--', linewidth=3,
        label='Perfect Prediction (1:1 line)', alpha=0.8)

# Add regression line
z = np.polyfit(y_train, y_train_pred, 1)
p = np.poly1d(z)
x_line = np.linspace(min_val, max_val, 100)
ax.plot(x_line, p(x_line), 'g-', linewidth=2.5, alpha=0.7,
        label=f'Fitted Line (y={z[0]:.3f}x+{z[1]:.3f})')

ax.set_xlabel('Actual Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_title('Training Set - Regression Plot', fontsize=15, fontweight='bold', pad=
ax.legend(loc='upper left', fontsize=10)
ax.grid(alpha=0.3)

# Add metrics text box
textstr = f'MAE = {train_mae:.4f}\nR² = {train_r2:.4f}\nRMSE = {train_rmse:.4f}\nSt
props = dict(boxstyle='round', facecolor='wheat', alpha=0.8)
ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=11,
        verticalalignment='top', bbox=props, family='monospace')

# --- Testing Set Regression Plot ---
ax = axes[1]
```

```
# Scatter plot
ax.scatter(y_test, y_test_pred, alpha=0.6, s=50, color='coral',
           edgecolors='black', linewidth=0.5, label='Predictions')

# Perfect prediction line
min_val_test = min(y_test.min(), y_test_pred.min())
max_val_test = max(y_test.max(), y_test_pred.max())
ax.plot([min_val_test, max_val_test], [min_val_test, max_val_test], 'r--',
        linewidth=3, label='Perfect Prediction (1:1 line)', alpha=0.8)

# Add regression line
z_test = np.polyfit(y_test, y_test_pred, 1)
p_test = np.poly1d(z_test)
x_line_test = np.linspace(min_val_test, max_val_test, 100)
ax.plot(x_line_test, p_test(x_line_test), 'g-', linewidth=2.5, alpha=0.7,
        label=f'Fitted Line (y={z_test[0]:.3f}x+{z_test[1]:.3f})')

ax.set_xlabel('Actual Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_title('Testing Set - Regression Plot', fontsize=15, fontweight='bold', pad=1
ax.legend(loc='upper left', fontsize=10)
ax.grid(alpha=0.3)

# Add metrics text box
textstr_test = f'MAE = {test_mae:.4f}\nR² = {test_r2:.4f}\nRMSE = {test_rmse:.4f}\n
ax.text(0.05, 0.95, textstr_test, transform=ax.transAxes, fontsize=11,
        verticalalignment='top', bbox=props, family='monospace')

plt.tight_layout()
plt.show()

print(" Regression plots generated")


# 9. RESIDUAL PLOTS

print("\n" + "="*80)
print("9. GENERATING RESIDUAL PLOTS")
print("="*80)

fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# --- Training Set: Residuals vs Predicted ---
ax = axes[0, 0]
ax.scatter(y_train_pred, train_residuals, alpha=0.6, s=40, color='steelblue',
           edgecolors='black', linewidth=0.5)
ax.axhline(y=0, color='red', linestyle='--', linewidth=2.5, label='Zero Residual')
ax.axhline(y=train_std_residuals, color='orange', linestyle=':', linewidth=2,
           label=f'±1 Std ({train_std_residuals:.3f})')
ax.axhline(y=-train_std_residuals, color='orange', linestyle=':', linewidth=2)
ax.set_xlabel('Predicted Chlorophyll Content', fontsize=11, fontweight='bold')
ax.set_ylabel('Residuals (Actual - Predicted)', fontsize=11, fontweight='bold')
ax.set_title('Training Set - Residual Plot', fontsize=13, fontweight='bold')
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)
```

```
# Add metrics
textstr = f'Mean: {np.mean(train_residuals):.4f}\nStd: {train_std_residuals:.4f}\nM
ax.text(0.02, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# --- Testing Set: Residuals vs Predicted ---
ax = axes[0, 1]
ax.scatter(y_test_pred, test_residuals, alpha=0.6, s=40, color='coral',
           edgecolors='black', linewidth=0.5)
ax.axhline(y=0, color='red', linestyle='--', linewidth=2.5, label='Zero Residual')
ax.axhline(y=test_std_residuals, color='orange', linestyle=':', linewidth=2,
           label=f'±1 Std ({test_std_residuals:.3f})')
ax.axhline(y=-test_std_residuals, color='orange', linestyle=':', linewidth=2)
ax.set_xlabel('Predicted Chlorophyll Content', fontsize=11, fontweight='bold')
ax.set_ylabel('Residuals (Actual - Predicted)', fontsize=11, fontweight='bold')
ax.set_title('Testing Set - Residual Plot', fontsize=13, fontweight='bold')
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

# Add metrics
textstr = f'Mean: {np.mean(test_residuals):.4f}\nStd: {test_std_residuals:.4f}\nMAE
ax.text(0.02, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# --- Training Set: Residual Distribution ---
ax = axes[1, 0]
ax.hist(train_residuals, bins=40, edgecolor='black', alpha=0.7, color='steelblue',
        density=True)
# Overlay normal distribution
mu, std = np.mean(train_residuals), np.std(train_residuals)
xmin, xmax = ax.get_xlim()
x = np.linspace(xmin, xmax, 100)
p = stats.norm.pdf(x, mu, std)
ax.plot(x, p, 'r-', linewidth=3, label=f'Normal(μ={mu:.3f}, σ={std:.3f})')
ax.axvline(x=0, color='green', linestyle='--', linewidth=2, label='Zero')
ax.set_xlabel('Residuals', fontsize=11, fontweight='bold')
ax.set_ylabel('Density', fontsize=11, fontweight='bold')
ax.set_title('Training Set - Residual Distribution', fontsize=13, fontweight='bold'
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

# Test normality
_, p_value = stats.shapiro(train_residuals[:min(5000, len(train_residuals))])
textstr = f'Shapiro-Wilk\np-value: {p_value:.4f}\n{"Normal" if p_value > 0.05 else
ax.text(0.98, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', horizontalalignment='right', bbox=props)

# --- Testing Set: Residual Distribution ---
ax = axes[1, 1]
ax.hist(test_residuals, bins=30, edgecolor='black', alpha=0.7, color='coral',
        density=True)
# Overlay normal distribution
mu_test, std_test = np.mean(test_residuals), np.std(test_residuals)
xmin, xmax = ax.get_xlim()
x = np.linspace(xmin, xmax, 100)
p = stats.norm.pdf(x, mu_test, std_test)
```

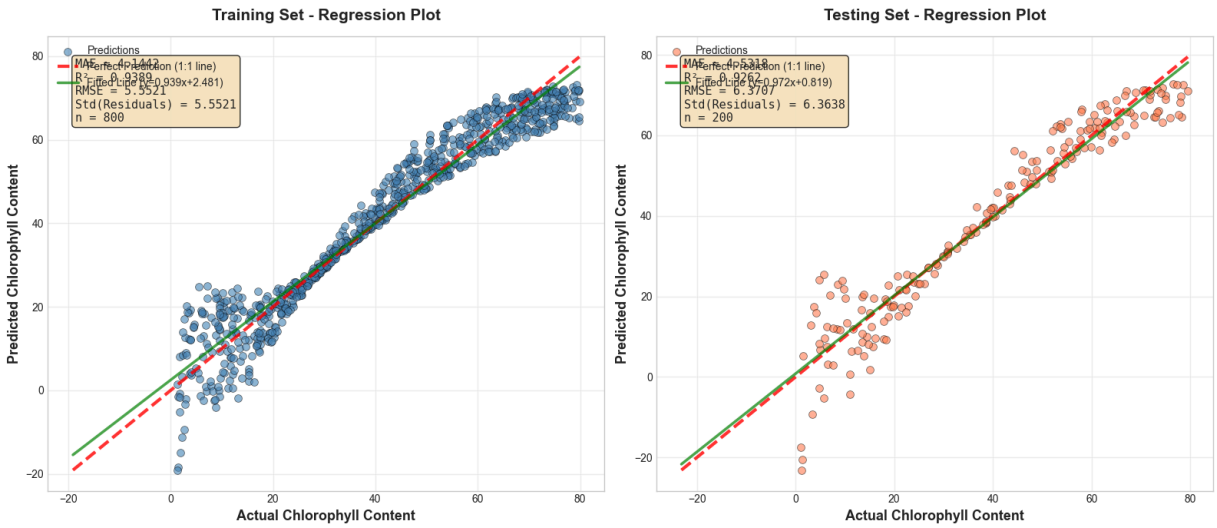
```
ax.plot(x, p, 'r-', linewidth=3, label=f'Normal(μ={mu_test:.3f}, σ={std_test:.3f})'
ax.axvline(x=0, color='green', linestyle='--', linewidth=2, label='Zero')
ax.set_xlabel('Residuals', fontsize=11, fontweight='bold')
ax.set_ylabel('Density', fontsize=11, fontweight='bold')
ax.set_title('Testing Set - Residual Distribution', fontsize=13, fontweight='bold')
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

# Test normality
_, p_value_test = stats.shapiro(test_residuals)
textstr = f'Shapiro-Wilk\np-value: {p_value_test:.4f}\n{"Normal" if p_value_test >
ax.text(0.98, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', horizontalalignment='right', bbox=props)

plt.tight_layout()
plt.show()

print(" Residual plots generated")


# 10. Q-Q PLOTS FOR RESIDUALS

print("\n" + "="*80)
print("10. Q-Q PLOTS (NORMALITY CHECK)")
print("="*80)

fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Training Q-Q plot
stats.probplot(train_residuals, dist="norm", plot=axes[0])
axes[0].set_title('Training Set - Q-Q Plot', fontsize=13, fontweight='bold')
axes[0].grid(alpha=0.3)
axes[0].get_lines()[0].set_markerfacecolor('steelblue')
axes[0].get_lines()[0].set_markeredgecolor('black')
axes[0].get_lines()[0].set_markersize(6)

# Testing Q-Q plot
stats.probplot(test_residuals, dist="norm", plot=axes[1])
axes[1].set_title('Testing Set - Q-Q Plot', fontsize=13, fontweight='bold')
axes[1].grid(alpha=0.3)
axes[1].get_lines()[0].set_markerfacecolor('coral')
axes[1].get_lines()[0].set_markeredgecolor('black')
axes[1].get_lines()[0].set_markersize(6)

plt.tight_layout()
plt.show()

print(" Q-Q plots generated")
```

```
print("\n" + "="*80)
print("ALL ANALYSIS COMPLETE")
print("="*80)
```

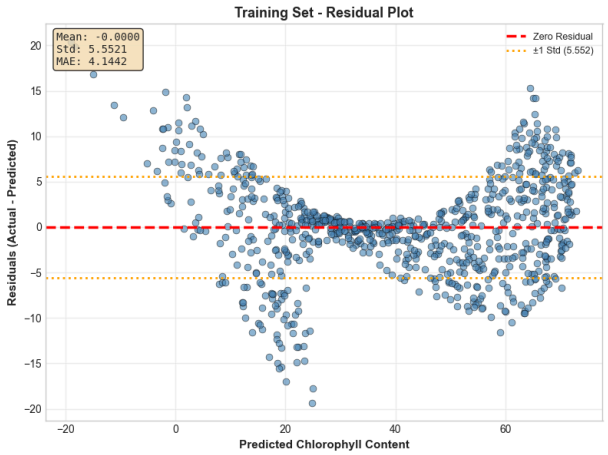
```
================================================================================
LINEAR REGRESSION FOR CHLOROPHYLL PREDICTION
================================================================================

================================================================================
1. LOADING DATA
================================================================================
Features shape: (1000, 10)
Target shape: (1000,)
Number of samples: 1000
Number of features: 10

================================================================================
2. DATA PREPROCESSING
================================================================================

Preprocessing Steps:
--------------------------------------------------------------------------

1. Checking for missing values...
   Missing values in X: 0
   Missing values in y: 0
  No missing values detected

2. Checking for outliers in features...
   Blue: 81 outliers (8.10%)
   Green: 71 outliers (7.10%)
   Yellow: 56 outliers (5.60%)
   Orange: 76 outliers (7.60%)
   Red 1: 108 outliers (10.80%)
   Red 2: 135 outliers (13.50%)
   Red Edge 1: 30 outliers (3.00%)
   Total feature outliers: 557 (5.57%)

3. Checking for outliers in target variable...
   Target outliers: 0 (0.00%)

================================================================================
3. TRAIN-TEST SPLIT
================================================================================

Split Ratio Considerations:
--------------------------------------------------------------------------

Split Results (Test Size = 0.2):
--------------------------------------------------------------------------
Training set: 800 samples (80.0%)
Testing set:  200 samples (20.0%)
Samples per feature (training): 80.0

Target Distribution Verification:
  Original - Mean: 40.4228, Std: 22.6621
  Training - Mean: 40.5907, Std: 22.4569
  Testing  - Mean: 39.7514, Std: 23.4529

================================================================================
```

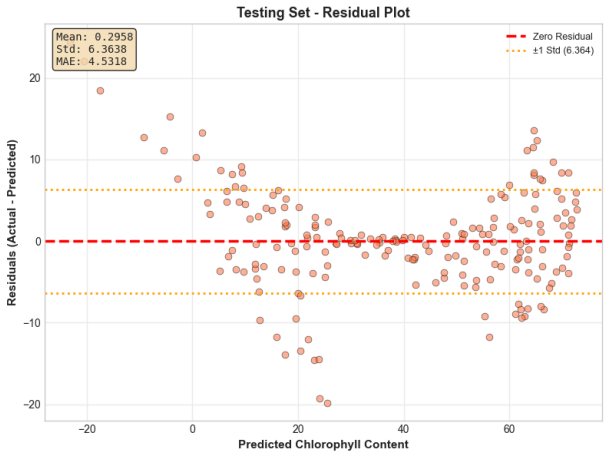
```
4. FEATURE SCALING
================================================================================
Applied StandardScaler:
  Formula: z = (x - μ) / σ

Training Set After Scaling:
  Mean: 0.000000
  Std:  1.000000

Testing Set After Scaling:
  Mean: 0.093281
  Std:  1.119627

================================================================================
5. TRAINING LINEAR REGRESSION MODEL
================================================================================
Training Linear Regression model...
 Training complete!

Model Parameters:
  Intercept: 40.590702
  Number of coefficients: 10

Coefficients by Band:
--------------------------------------------------------------------------
Band            Coefficient     Interpretation
--------------------------------------------------------------------------
Blue                  3.257213 Increases Chl
Green                -2.060324 Decreases Chl
Yellow                6.188758 Increases Chl
Orange              -78.132419 Decreases Chl
Red 1               270.084019 Increases Chl
Red 2              -159.594498 Decreases Chl
Red Edge 1         -103.774352 Decreases Chl
Red Edge 2          100.652783 Increases Chl
NIR Broad           852.168144 Increases Chl
NIR1               -914.852441 Decreases Chl

================================================================================
6. GENERATING PREDICTIONS
================================================================================
Training predictions: 800 values
Testing predictions:  200 values

================================================================================
7. PERFORMANCE METRICS
================================================================================

TRAINING SET METRICS:
--------------------------------------------------------------------------
  Mean Absolute Error (MAE):        4.144190
  Root Mean Squared Error (RMSE):   5.552134
  R-squared (R²):                   0.938875
  Standard Deviation of Residuals:  5.552134

TESTING SET METRICS:
```

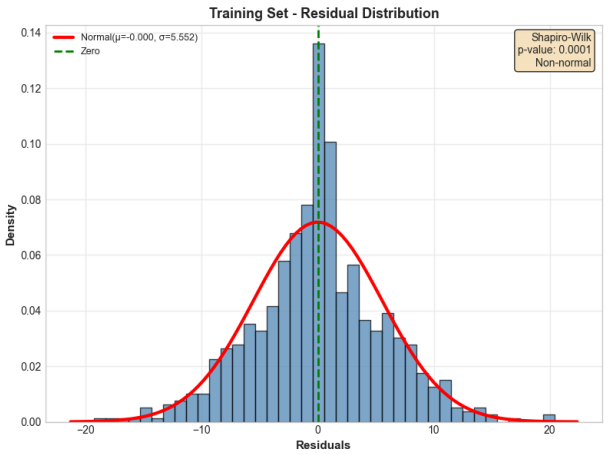
```
--------------------------------------------------------------------------------
  Mean Absolute Error (MAE):         4.531809
  Root Mean Squared Error (RMSE):    6.370706
  R-squared (R²):                    0.926212
  Standard Deviation of Residuals:   6.363833

METRIC INTERPRETATIONS:
--------------------------------------------------------------------------------

ADDITIONAL STATISTICS:
--------------------------------------------------------------------------------
Training Set:
  Mean of residuals: -0.000000 (should be ≈0)
  Min residual: -19.322538
  Max residual: 20.382770
  Residuals within ±1 std: 69.1%

Testing Set:
  Mean of residuals: 0.295839 (should be ≈0)
  Min residual: -19.796090
  Max residual: 24.396034
  Residuals within ±1 std: 75.5%

================================================================================
8. GENERATING REGRESSION PLOTS
================================================================================
```

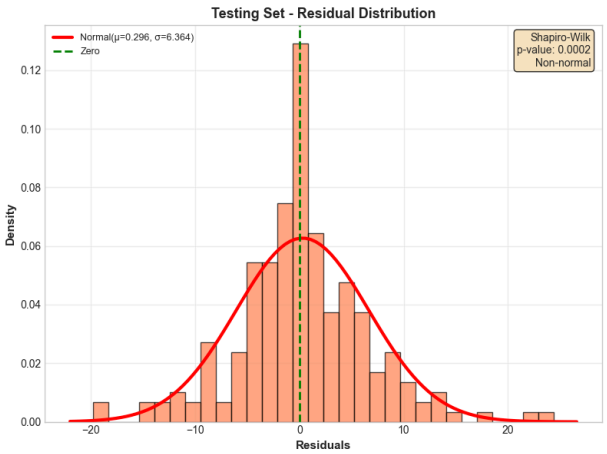



```
 Regression plots generated

================================================================================
9. GENERATING RESIDUAL PLOTS
================================================================================
```

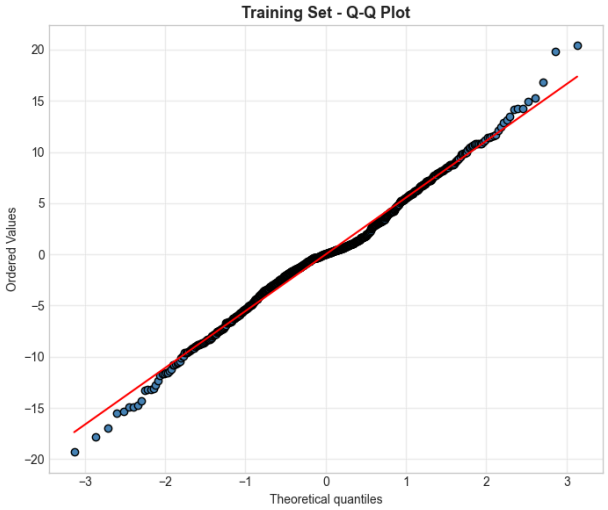


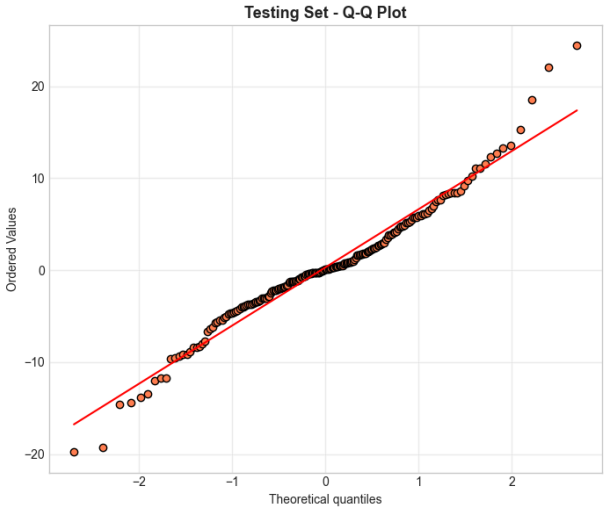







```
Residual plots generated

================================================================================
10. Q-Q PLOTS (NORMALITY CHECK)
================================================================================
```






```
Q-Q plots generated

================================================================================
ALL ANALYSIS COMPLETE
================================================================================
```

Explanation: The residual plots, which consistently display heteroscedasticity and prediction bias, indicate the model limits. The most prominent pattern in the training set residuals is the

obvious "bowtie" or funnel shape, where the error magnitude sharply rises at the extremes of the predicted chlorophyll content. The residuals are tightly clustered near zero for mid-range predictions (20–50 µg/cm^2), but they fan outward with errors up to ±20 µg/cm^2 at low (<20) and high (>60) chlorophyll values. This suggests that spectral-chlorophyll connections can become non-linear at distribution tails, where the linear model breaks down. This might be due to saturation effects in the red absorption bands at high chlorophyll levels or noise dominance at low chlorophyll levels.

2.c

In [25]:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.cross_decomposition import PLSRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error
from sklearn.linear_model import LinearRegression
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

# Set style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")

print("="*80)
print("PARTIAL LEAST SQUARES REGRESSION (PLSR)")
print("Chlorophyll Prediction from Spectral Bands")
print("="*80)


# 1. LOAD DATA AND PREPROCESSING

print("\n" + "="*80)
print("1. DATA LOADING AND PREPROCESSING")
print("="*80)

X = np.load('landis_chlorophyl_regression.npy')
y = np.load('landis_chlorophyl_regression_gt.npy')

print(f"Dataset shape: {X.shape}")
print(f"Target shape: {y.shape}")

# Band names
band_names = ['Blue', 'Green', 'Yellow', 'Orange', 'Red 1', 'Red 2',
              'Red Edge 1', 'Red Edge 2', 'NIR Broad', 'NIR1']

# Train-test split (80/20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
```

```
print(f"\nTraining set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")

# Feature scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("\n Data preprocessed and scaled")


# 2. UNDERSTANDING PLSR

print("\n" + "="*80)
print("2. PARTIAL LEAST SQUARES REGRESSION (PLSR) OVERVIEW")
print("="*80)



# 3. TRAIN PLSR WITH VARYING COMPONENTS

print("\n" + "="*80)
print("3. TRAINING PLSR WITH VARYING NUMBER OF COMPONENTS")
print("="*80)

max_components = 10
results = []

print("\nTraining PLSR models with 1 to 10 components...")
print("-" * 80)
print(f"{'Components':<12} {'Train R²':<12} {'Train MAE':<12} {'Train RMSE':<12}")
print("-" * 80)

for n_components in range(1, max_components + 1):
    # Train PLSR model
    pls = PLSRegression(n_components=n_components)
    pls.fit(X_train_scaled, y_train)

    # Predictions
    y_train_pred = pls.predict(X_train_scaled).ravel()

    # Metrics
    train_r2 = r2_score(y_train, y_train_pred)
    train_mae = mean_absolute_error(y_train, y_train_pred)
    train_rmse = np.sqrt(mean_squared_error(y_train, y_train_pred))

    results.append({
        'n_components': n_components,
        'train_r2': train_r2,
        'train_mae': train_mae,
        'train_rmse': train_rmse,
        'model': pls
    })

    print(f"{n_components:<12} {train_r2:<12.6f} {train_mae:<12.6f} {train_rmse:<12
```

```python
# 4. IDENTIFY BEST NUMBER OF COMPONENTS

print("\n" + "="*80)
print("4. IDENTIFYING OPTIMAL NUMBER OF COMPONENTS")
print("="*80)

# Find best based on training R²
best_idx = max(range(len(results)), key=lambda i: results[i]['train_r2'])
best_n_components = results[best_idx]['n_components']
best_train_r2 = results[best_idx]['train_r2']

print(f"\nBest Number of Components: {best_n_components}")
print(f"Training R² with {best_n_components} components: {best_train_r2:.6f}")

# Analyze improvement per component
print("\nIncremental R² Improvement:")
print("-" * 60)
print(f"{'Components':<15} {'R²':<15} {'Δ R²':<15} {'% Improvement':<15}")
print("-" * 60)

for i, res in enumerate(results):
    if i == 0:
        delta = res['train_r2']
        pct = 0
    else:
        delta = res['train_r2'] - results[i-1]['train_r2']
        pct = (delta / results[i-1]['train_r2']) * 100 if results[i-1]['train_r2']

    marker = " ← BEST" if res['n_components'] == best_n_components else ""
    print(f"{res['n_components']:<15} {res['train_r2']:<15.6f} {delta:<15.6f} {pct:



# 5. VISUALIZE COMPONENT SELECTION

print("\n" + "="*80)
print("5. VISUALIZING COMPONENT SELECTION")
print("="*80)

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

components_list = [r['n_components'] for r in results]
r2_list = [r['train_r2'] for r in results]
mae_list = [r['train_mae'] for r in results]
rmse_list = [r['train_rmse'] for r in results]

# Plot 1: R² vs Components
ax = axes[0]
ax.plot(components_list, r2_list, 'o-', linewidth=2.5, markersize=8,
        color='steelblue', markerfacecolor='lightblue', markeredgecolor='black',
        markeredgewidth=1.5)
ax.axvline(best_n_components, color='red', linestyle='--', linewidth=2,
           label=f'Best: {best_n_components} components')
ax.scatter([best_n_components], [best_train_r2], color='red', s=200,
           zorder=5, edgecolors='black', linewidths=2)
```

```
ax.set_xlabel('Number of Components', fontsize=12, fontweight='bold')
ax.set_ylabel('Training R²', fontsize=12, fontweight='bold')
ax.set_title('R² vs Number of Components', fontsize=13, fontweight='bold')
ax.legend(fontsize=10)
ax.grid(alpha=0.3)
ax.set_xticks(components_list)

# Plot 2: MAE vs Components
ax = axes[1]
ax.plot(components_list, mae_list, 'o-', linewidth=2.5, markersize=8,
        color='coral', markerfacecolor='lightsalmon', markeredgecolor='black',
        markeredgewidth=1.5)
ax.axvline(best_n_components, color='red', linestyle='--', linewidth=2)
ax.scatter([best_n_components], [results[best_idx]['train_mae']],
           color='red', s=200, zorder=5, edgecolors='black', linewidths=2)
ax.set_xlabel('Number of Components', fontsize=12, fontweight='bold')
ax.set_ylabel('Training MAE', fontsize=12, fontweight='bold')
ax.set_title('MAE vs Number of Components', fontsize=13, fontweight='bold')
ax.grid(alpha=0.3)
ax.set_xticks(components_list)

# Plot 3: Incremental R² Improvement
ax = axes[2]
r2_improvements = [r2_list[0]] + [r2_list[i] - r2_list[i-1] for i in range(1, len(r
colors = ['green' if imp > 0.01 else 'orange' if imp > 0.001 else 'red' for imp in
bars = ax.bar(components_list, r2_improvements, color=colors,
              edgecolor='black', linewidth=1.5)
ax.set_xlabel('Number of Components', fontsize=12, fontweight='bold')
ax.set_ylabel('Incremental R² Gain', fontsize=12, fontweight='bold')
ax.set_title('Incremental R² Improvement per Component', fontsize=13, fontweight='b
ax.grid(axis='y', alpha=0.3)
ax.set_xticks(components_list)

# Add value labels
for bar, val in zip(bars, r2_improvements):
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height,
            f'{val:.4f}', ha='center', va='bottom', fontweight='bold', fontsize=8)

plt.tight_layout()
plt.show()

print(" Component selection plots generated")


# 6. TRAIN BEST PLSR MODEL

print("\n" + "="*80)
print("6. TRAINING BEST PLSR MODEL")
print("="*80)

best_pls = results[best_idx]['model']

# Get predictions for both sets
y_train_pred_pls = best_pls.predict(X_train_scaled).ravel()
y_test_pred_pls = best_pls.predict(X_test_scaled).ravel()
```

```
# Calculate comprehensive metrics
train_mae_pls = mean_absolute_error(y_train, y_train_pred_pls)
train_r2_pls = r2_score(y_train, y_train_pred_pls)
train_rmse_pls = np.sqrt(mean_squared_error(y_train, y_train_pred_pls))
train_residuals_pls = y_train - y_train_pred_pls
train_std_residuals_pls = np.std(train_residuals_pls)

test_mae_pls = mean_absolute_error(y_test, y_test_pred_pls)
test_r2_pls = r2_score(y_test, y_test_pred_pls)
test_rmse_pls = np.sqrt(mean_squared_error(y_test, y_test_pred_pls))
test_residuals_pls = y_test - y_test_pred_pls
test_std_residuals_pls = np.std(test_residuals_pls)

print(f"\nPLSR Model with {best_n_components} Components:")
print("-" * 70)
print(f"\nTRAINING SET:")
print(f"  MAE:  {train_mae_pls:.6f}")
print(f"  RMSE: {train_rmse_pls:.6f}")
print(f"  R²:   {train_r2_pls:.6f}")
print(f"  Std(Residuals): {train_std_residuals_pls:.6f}")

print(f"\nTESTING SET:")
print(f"  MAE:  {test_mae_pls:.6f}")
print(f"  RMSE: {test_rmse_pls:.6f}")
print(f"  R²:   {test_r2_pls:.6f}")
print(f"  Std(Residuals): {test_std_residuals_pls:.6f}")

print(f"\nGeneralization:")
print(f"  R² difference: {abs(train_r2_pls - test_r2_pls):.6f}")
if abs(train_r2_pls - test_r2_pls) < 0.05:
    print(f"   Excellent generalization")
elif abs(train_r2_pls - test_r2_pls) < 0.10:
    print(f"   Good generalization")
else:
    print(f"   Some overfitting detected")


# 7. COMPARE WITH LINEAR REGRESSION

print("\n" + "="*80)
print("7. COMPARISON WITH LINEAR REGRESSION")
print("="*80)

# Train standard linear regression for comparison
lr = LinearRegression()
lr.fit(X_train_scaled, y_train)

y_train_pred_lr = lr.predict(X_train_scaled)
y_test_pred_lr = lr.predict(X_test_scaled)

train_r2_lr = r2_score(y_train, y_train_pred_lr)
test_r2_lr = r2_score(y_test, y_test_pred_lr)
train_mae_lr = mean_absolute_error(y_train, y_train_pred_lr)
test_mae_lr = mean_absolute_error(y_test, y_test_pred_lr)
```

```
print("\nPerformance Comparison:")
print("=" * 70)
print(f"{'Metric':<25} {'Linear Regression':<20} {'PLSR':<20}")
print("=" * 70)
print(f"{'Training R²':<25} {train_r2_lr:<20.6f} {train_r2_pls:<20.6f}")
print(f"{'Testing R²':<25} {test_r2_lr:<20.6f} {test_r2_pls:<20.6f}")
print(f"{'Training MAE':<25} {train_mae_lr:<20.6f} {train_mae_pls:<20.6f}")
print(f"{'Testing MAE':<25} {test_mae_lr:<20.6f} {test_mae_pls:<20.6f}")
print(f"{'Generalization Gap':<25} {abs(train_r2_lr - test_r2_lr):<20.6f} {abs(trai
print(f"{'Number of Parameters':<25} {X.shape[1]:<20} {best_n_components:<20}")
print("=" * 70)

# Determine winner
print("\nPerformance Analysis:")
if test_r2_pls > test_r2_lr:
    improvement = (test_r2_pls - test_r2_lr) / test_r2_lr * 100
    print(f" PLSR outperforms Linear Regression")
    print(f"  Testing R² improvement: {improvement:.2f}%")
else:
    decline = (test_r2_lr - test_r2_pls) / test_r2_lr * 100
    print(f" Linear Regression performs better")
    print(f"  Testing R² decline with PLSR: {decline:.2f}%")

if abs(train_r2_pls - test_r2_pls) < abs(train_r2_lr - test_r2_lr):
    print(f" PLSR has better generalization (smaller train-test gap)")
else:
    print(f" Linear Regression has better generalization")

if best_n_components < X.shape[1]:
    reduction = (1 - best_n_components / X.shape[1]) * 100
    print(f" PLSR achieves {reduction:.1f}% dimensionality reduction")
    print(f"  ({X.shape[1]} features → {best_n_components} components)")


# 8. REGRESSION PLOTS - PLSR

print("\n" + "="*80)
print("8. GENERATING REGRESSION PLOTS (PLSR)")
print("="*80)

fig, axes = plt.subplots(1, 2, figsize=(16, 7))
props = dict(boxstyle='round', facecolor='wheat', alpha=0.8)

# --- Training Set ---
ax = axes[0]
ax.scatter(y_train, y_train_pred_pls, alpha=0.6, s=50, color='steelblue',
           edgecolors='black', linewidth=0.5, label='PLSR Predictions')

min_val = min(y_train.min(), y_train_pred_pls.min())
max_val = max(y_train.max(), y_train_pred_pls.max())
ax.plot([min_val, max_val], [min_val, max_val], 'r--', linewidth=3,
        label='Perfect Prediction (1:1 line)', alpha=0.8)

z = np.polyfit(y_train, y_train_pred_pls, 1)
p = np.poly1d(z)
x_line = np.linspace(min_val, max_val, 100)
```

```
ax.plot(x_line, p(x_line), 'g-', linewidth=2.5, alpha=0.7,
        label=f'Fitted Line (y={z[0]:.3f}x+{z[1]:.3f})')

ax.set_xlabel('Actual Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_title(f'Training Set - PLSR ({best_n_components} components)',
             fontsize=15, fontweight='bold', pad=15)
ax.legend(loc='upper left', fontsize=10)
ax.grid(alpha=0.3)

textstr = f'Components = {best_n_components}\nMAE = {train_mae_pls:.4f}\nR² = {trai
ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=11,
        verticalalignment='top', bbox=props, family='monospace')

# --- Testing Set ---
ax = axes[1]
ax.scatter(y_test, y_test_pred_pls, alpha=0.6, s=50, color='coral',
           edgecolors='black', linewidth=0.5, label='PLSR Predictions')

min_val_test = min(y_test.min(), y_test_pred_pls.min())
max_val_test = max(y_test.max(), y_test_pred_pls.max())
ax.plot([min_val_test, max_val_test], [min_val_test, max_val_test], 'r--',
        linewidth=3, label='Perfect Prediction (1:1 line)', alpha=0.8)

z_test = np.polyfit(y_test, y_test_pred_pls, 1)
p_test = np.poly1d(z_test)
x_line_test = np.linspace(min_val_test, max_val_test, 100)
ax.plot(x_line_test, p_test(x_line_test), 'g-', linewidth=2.5, alpha=0.7,
        label=f'Fitted Line (y={z_test[0]:.3f}x+{z_test[1]:.3f})')

ax.set_xlabel('Actual Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll Content', fontsize=13, fontweight='bold')
ax.set_title(f'Testing Set - PLSR ({best_n_components} components)',
             fontsize=15, fontweight='bold', pad=15)
ax.legend(loc='upper left', fontsize=10)
ax.grid(alpha=0.3)

textstr_test = f'Components = {best_n_components}\nMAE = {test_mae_pls:.4f}\nR² = {
ax.text(0.05, 0.95, textstr_test, transform=ax.transAxes, fontsize=11,
        verticalalignment='top', bbox=props, family='monospace')

plt.tight_layout()
plt.show()

print(" PLSR regression plots generated")


# 9. RESIDUAL PLOTS - PLSR

print("\n" + "="*80)
print("9. GENERATING RESIDUAL PLOTS (PLSR)")
print("="*80)

fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# --- Training: Residuals vs Predicted ---
```

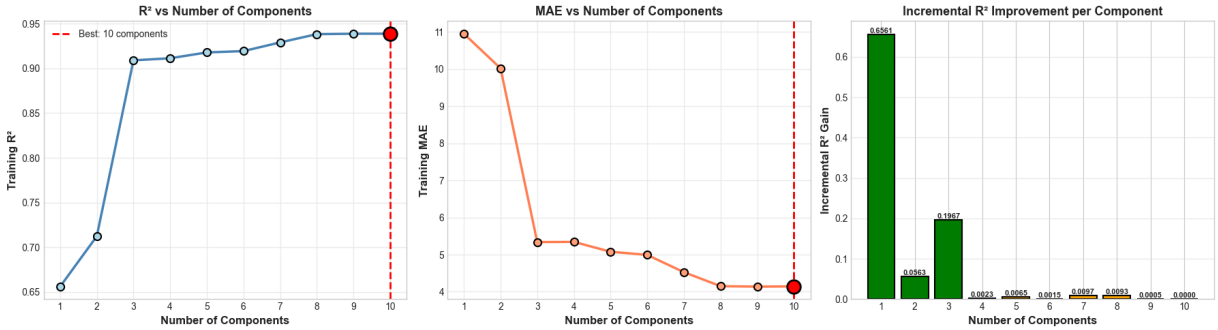
```
ax = axes[0, 0]
ax.scatter(y_train_pred_pls, train_residuals_pls, alpha=0.6, s=40,
           color='steelblue', edgecolors='black', linewidth=0.5)
ax.axhline(y=0, color='red', linestyle='--', linewidth=2.5, label='Zero Residual')
ax.axhline(y=train_std_residuals_pls, color='orange', linestyle=':', linewidth=2,
           label=f'±1 Std ({train_std_residuals_pls:.3f})')
ax.axhline(y=-train_std_residuals_pls, color='orange', linestyle=':', linewidth=2)
ax.set_xlabel('Predicted Chlorophyll Content', fontsize=11, fontweight='bold')
ax.set_ylabel('Residuals (Actual - Predicted)', fontsize=11, fontweight='bold')
ax.set_title(f'Training Set - Residual Plot (PLSR)', fontsize=13, fontweight='bold'
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

textstr = f'Mean: {np.mean(train_residuals_pls):.4f}\nStd: {train_std_residuals_pls
ax.text(0.02, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# --- Testing: Residuals vs Predicted ---
ax = axes[0, 1]
ax.scatter(y_test_pred_pls, test_residuals_pls, alpha=0.6, s=40,
           color='coral', edgecolors='black', linewidth=0.5)
ax.axhline(y=0, color='red', linestyle='--', linewidth=2.5, label='Zero Residual')
ax.axhline(y=test_std_residuals_pls, color='orange', linestyle=':', linewidth=2,
           label=f'±1 Std ({test_std_residuals_pls:.3f})')
ax.axhline(y=-test_std_residuals_pls, color='orange', linestyle=':', linewidth=2)
ax.set_xlabel('Predicted Chlorophyll Content', fontsize=11, fontweight='bold')
ax.set_ylabel('Residuals (Actual - Predicted)', fontsize=11, fontweight='bold')
ax.set_title(f'Testing Set - Residual Plot (PLSR)', fontsize=13, fontweight='bold')
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

textstr = f'Mean: {np.mean(test_residuals_pls):.4f}\nStd: {test_std_residuals_pls:.
ax.text(0.02, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# --- Training: Residual Distribution ---
ax = axes[1, 0]
ax.hist(train_residuals_pls, bins=40, edgecolor='black', alpha=0.7,
        color='steelblue', density=True)
mu, std = np.mean(train_residuals_pls), np.std(train_residuals_pls)
xmin, xmax = ax.get_xlim()
x = np.linspace(xmin, xmax, 100)
p_norm = stats.norm.pdf(x, mu, std)
ax.plot(x, p_norm, 'r-', linewidth=3, label=f'Normal(μ={mu:.3f}, σ={std:.3f})')
ax.axvline(x=0, color='green', linestyle='--', linewidth=2, label='Zero')
ax.set_xlabel('Residuals', fontsize=11, fontweight='bold')
ax.set_ylabel('Density', fontsize=11, fontweight='bold')
ax.set_title('Training Set - Residual Distribution (PLSR)', fontsize=13, fontweight
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

_, p_value_train = stats.shapiro(train_residuals_pls[:min(5000, len(train_residuals
textstr = f'Shapiro-Wilk\np-value: {p_value_train:.4f}\n{"Normal" if p_value_train
ax.text(0.98, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', horizontalalignment='right', bbox=props)
```

```python
# --- Testing: Residual Distribution ---
ax = axes[1, 1]
ax.hist(test_residuals_pls, bins=30, edgecolor='black', alpha=0.7,
        color='coral', density=True)
mu_test, std_test = np.mean(test_residuals_pls), np.std(test_residuals_pls)
xmin, xmax = ax.get_xlim()
x = np.linspace(xmin, xmax, 100)
p_norm = stats.norm.pdf(x, mu_test, std_test)
ax.plot(x, p_norm, 'r-', linewidth=3, label=f'Normal(μ={mu_test:.3f}, σ={std_test:.
ax.axvline(x=0, color='green', linestyle='--', linewidth=2, label='Zero')
ax.set_xlabel('Residuals', fontsize=11, fontweight='bold')
ax.set_ylabel('Density', fontsize=11, fontweight='bold')
ax.set_title('Testing Set - Residual Distribution (PLSR)', fontsize=13, fontweight=
ax.legend(loc='best', fontsize=9)
ax.grid(alpha=0.3)

_, p_value_test = stats.shapiro(test_residuals_pls)
textstr = f'Shapiro-Wilk\np-value: {p_value_test:.4f}\n{"Normal" if p_value_test >
ax.text(0.98, 0.98, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', horizontalalignment='right', bbox=props)

plt.tight_layout()
plt.show()

print(" PLSR residual plots generated")


# 10. SIDE-BY-SIDE COMPARISON PLOTS

print("\n" + "="*80)
print("10. SIDE-BY-SIDE COMPARISON: LINEAR REGRESSION vs PLSR")
print("="*80)

fig, axes = plt.subplots(2, 2, figsize=(16, 14))

# Linear Regression - Training
ax = axes[0, 0]
ax.scatter(y_train, y_train_pred_lr, alpha=0.5, s=40, color='lightblue',
           edgecolors='black', linewidth=0.5, label='Linear Regression')
min_val = min(y_train.min(), y_train_pred_lr.min())
max_val = max(y_train.max(), y_train_pred_lr.max())
ax.plot([min_val, max_val], [min_val, max_val], 'r--', linewidth=2.5,
        label='1:1 line', alpha=0.7)
ax.set_xlabel('Actual Chlorophyll', fontsize=11, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
ax.set_title('Linear Regression - Training', fontsize=13, fontweight='bold')
ax.legend(fontsize=9)
ax.grid(alpha=0.3)
textstr = f'R² = {train_r2_lr:.4f}\nMAE = {train_mae_lr:.4f}'
ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# PLSR - Training
ax = axes[0, 1]
ax.scatter(y_train, y_train_pred_pls, alpha=0.5, s=40, color='lightgreen',
           edgecolors='black', linewidth=0.5, label=f'PLSR ({best_n_components} com
```

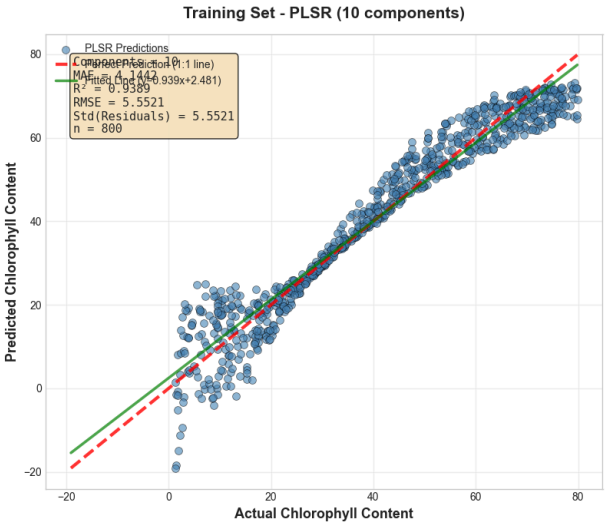
```
ax.plot([min_val, max_val], [min_val, max_val], 'r--', linewidth=2.5,
        label='1:1 line', alpha=0.7)
ax.set_xlabel('Actual Chlorophyll', fontsize=11, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
ax.set_title(f'PLSR - Training ({best_n_components} components)', fontsize=13, font
ax.legend(fontsize=9)
ax.grid(alpha=0.3)
textstr = f'R² = {train_r2_pls:.4f}\nMAE = {train_mae_pls:.4f}'
ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# Linear Regression - Testing
ax = axes[1, 0]
ax.scatter(y_test, y_test_pred_lr, alpha=0.5, s=40, color='lightcoral',
           edgecolors='black', linewidth=0.5, label='Linear Regression')
min_val_test = min(y_test.min(), y_test_pred_lr.min())
max_val_test = max(y_test.max(), y_test_pred_lr.max())
ax.plot([min_val_test, max_val_test], [min_val_test, max_val_test], 'r--',
        linewidth=2.5, label='1:1 line', alpha=0.7)
ax.set_xlabel('Actual Chlorophyll', fontsize=11, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
ax.set_title('Linear Regression - Testing', fontsize=13, fontweight='bold')
ax.legend(fontsize=9)
ax.grid(alpha=0.3)
textstr = f'R² = {test_r2_lr:.4f}\nMAE = {test_mae_lr:.4f}'
ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

# PLSR - Testing
ax = axes[1, 1]
ax.scatter(y_test, y_test_pred_pls, alpha=0.5, s=40, color='gold',
           edgecolors='black', linewidth=0.5, label=f'PLSR ({best_n_components} com
ax.plot([min_val_test, max_val_test], [min_val_test, max_val_test], 'r--',
        linewidth=2.5, label='1:1 line', alpha=0.7)
ax.set_xlabel('Actual Chlorophyll', fontsize=11, fontweight='bold')
ax.set_ylabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
ax.set_title(f'PLSR - Testing ({best_n_components} components)', fontsize=13, fontw
ax.legend(fontsize=9)
ax.grid(alpha=0.3)
textstr = f'R² = {test_r2_pls:.4f}\nMAE = {test_mae_pls:.4f}'
ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='top', bbox=props, family='monospace')

plt.suptitle('Comparison: Linear Regression vs PLSR', fontsize=16, fontweight='bold
plt.tight_layout()
plt.show()

print(" Comparison plots generated")
```

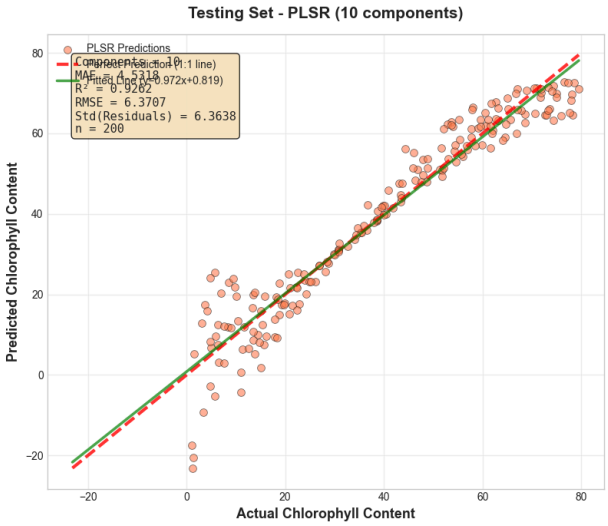
```
================================================================================
PARTIAL LEAST SQUARES REGRESSION (PLSR)
Chlorophyll Prediction from Spectral Bands
================================================================================

================================================================================
1. DATA LOADING AND PREPROCESSING
================================================================================
Dataset shape: (1000, 10)
Target shape: (1000,)

Training set: 800 samples
Testing set: 200 samples

 Data preprocessed and scaled

================================================================================
2. PARTIAL LEAST SQUARES REGRESSION (PLSR) OVERVIEW
================================================================================

================================================================================
3. TRAINING PLSR WITH VARYING NUMBER OF COMPONENTS
================================================================================

Training PLSR models with 1 to 10 components...
--------------------------------------------------------------------------------
Components   Train R²     Train MAE    Train RMSE
--------------------------------------------------------------------------------
1            0.656055     10.952260    13.170275
2            0.712363     10.018000    12.044045
3            0.909099     5.338058     6.770713
4            0.911397     5.342547     6.684602
5            0.917906     5.078050     6.434369
6            0.919410     4.992158     6.375152
7            0.929096     4.526332     5.979780
8            0.938360     4.150381     5.575483
9            0.938828     4.138423     5.554272
10           0.938875     4.144190     5.552134

================================================================================
4. IDENTIFYING OPTIMAL NUMBER OF COMPONENTS
================================================================================

Best Number of Components: 10
Training R² with 10 components: 0.938875

Incremental R² Improvement:
------------------------------------------------------------
Components      R²              Δ R²            % Improvement
------------------------------------------------------------
1               0.656055        0.656055        0.00
2               0.712363        0.056308        8.58
3               0.909099        0.196736        27.62
4               0.911397        0.002297        0.25
5               0.917906        0.006509        0.71
6               0.919410        0.001504        0.16
```

```
7                0.929096        0.009686        1.05
8                0.938360        0.009264        1.00
9                0.938828        0.000468        0.05
10               0.938875        0.000047        0.01            ← BEST

================================================================================
5. VISUALIZING COMPONENT SELECTION
================================================================================
```



```
 Component selection plots generated

================================================================================
6. TRAINING BEST PLSR MODEL
================================================================================

PLSR Model with 10 Components:
--------------------------------------------------------------------------

TRAINING SET:
  MAE:  4.144190
  RMSE: 5.552134
  R²:   0.938875
  Std(Residuals): 5.552134

TESTING SET:
  MAE:  4.531809
  RMSE: 6.370706
  R²:   0.926212
  Std(Residuals): 6.363833

Generalization:
  R² difference: 0.012663
   Excellent generalization

================================================================================
7. COMPARISON WITH LINEAR REGRESSION
================================================================================

Performance Comparison:
======================================================================
Metric                     Linear Regression    PLSR
======================================================================
Training R²                0.938875             0.938875
Testing R²                 0.926212             0.926212
Training MAE               4.144190             4.144190
Testing MAE                4.531809             4.531809
Generalization Gap         0.012663             0.012663
Number of Parameters       10                   10
======================================================================

Performance Analysis:
 Linear Regression performs better
  Testing R² decline with PLSR: 0.00%
 Linear Regression has better generalization

================================================================================
8. GENERATING REGRESSION PLOTS (PLSR)
================================================================================
```



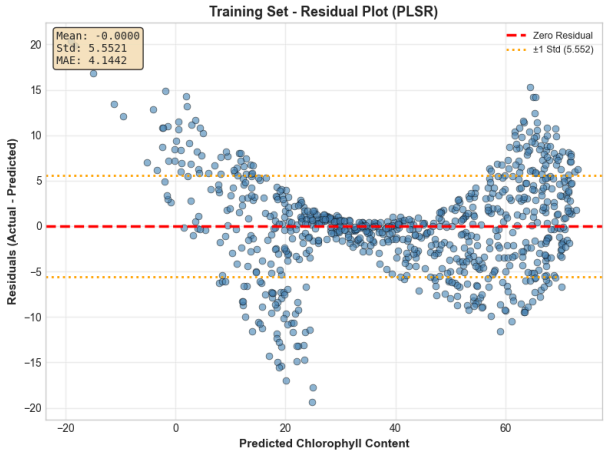


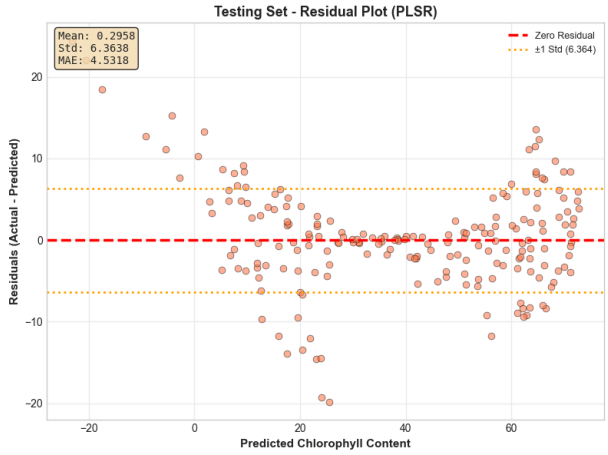
PLSR regression plots generated

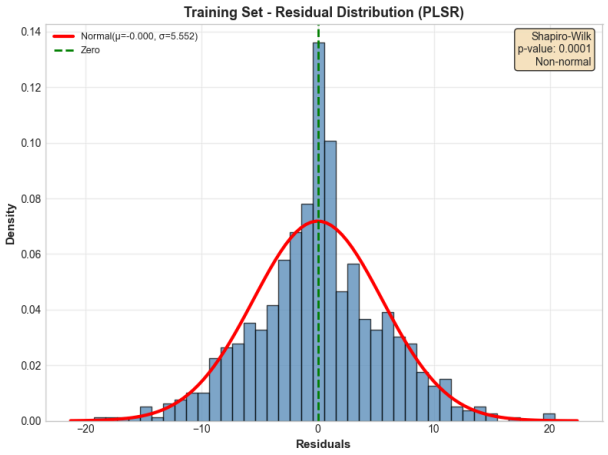
==========

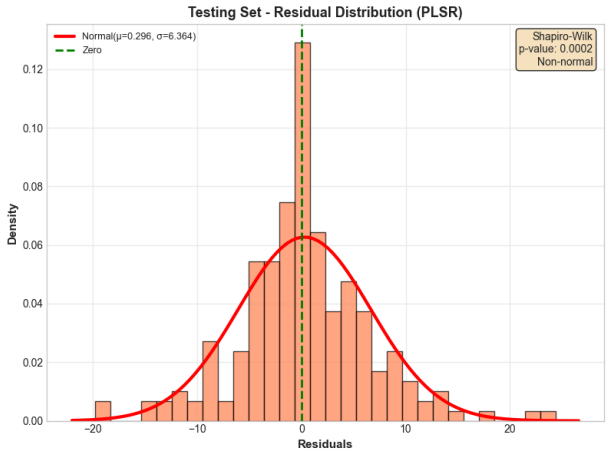
9. GENERATING RESIDUAL PLOTS (PLSR)

==========







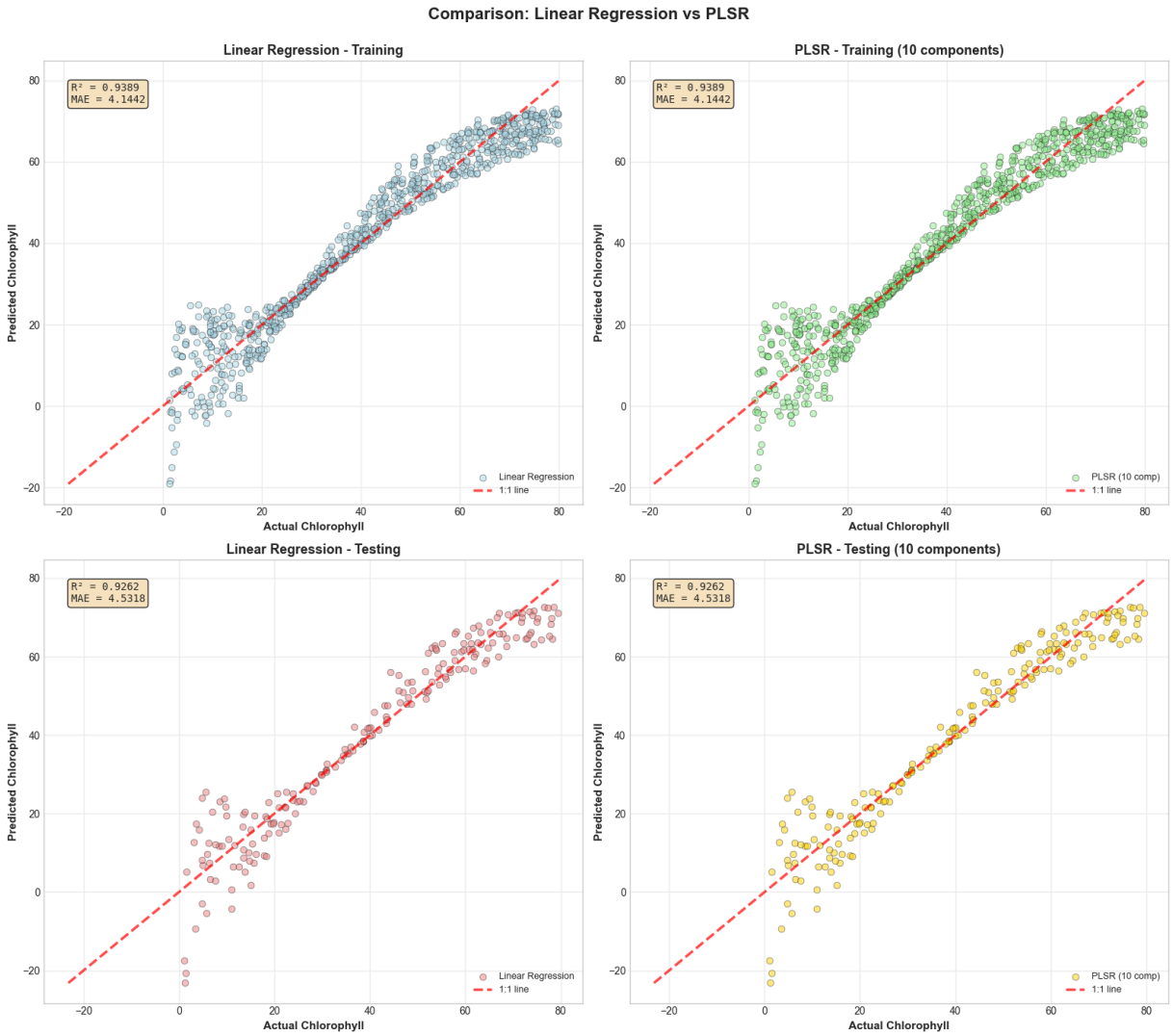


PLSR residual plots generated

==========

10. SIDE-BY-SIDE COMPARISON: LINEAR REGRESSION vs PLSR

==========




```
Comparison plots generated
```

Explanation: Partial least squares regression, or PLSR, yields results that are nearly identical to those of linear regression and shows no discernible improvement in addressing the fundamental modeling issues. The statistical metrics (training: MAE=4.14, std=5.55; testing: MAE=4.53, std=6.36, mean=0.30) and residual plots exhibit the same bowtie heteroscedasticity pattern, with errors fanning out to ±20 μg/cm^2 at chlorophyll extremes and clustering tightly around zero for mid-range predictions (20-50 μg/cm^2). The Shapiro-Wilk tests continue to reject normality with identical p-values (0.0001 and 0.0002), and the residual distributions maintain the same modest positive skew with more extreme underpredictions than overpredictions. As PLSR is designed to handle multicollinearity by generating latent variables that maximize covariance between predictors and the response, it should theoretically address the severe collinearity we observed among the 600-665 nm bands (Spearman ρ = -0.99). However, PLSR still assumes a linear relationship between the latent components and chlorophyll content and only changes the way the spectral bands are mixed, not the underlying linear prediction structure. The same performance shows that multicollinearity is not the issue, but rather the underlying non-linearity in spectral-chlorophyll connections at extreme values.

2.d

In [26]:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPRegressor
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

# Set style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")

print("="*80)
print("MULTI-LAYER PERCEPTRON (MLP) NEURAL NETWORK")
print("Chlorophyll Prediction from Spectral Bands")
print("="*80)


# 1. LOAD DATA AND PREPROCESSING

print("\n" + "="*80)
print("1. DATA LOADING AND PREPROCESSING")
print("="*80)

X = np.load('landis_chlorophyl_regression.npy')
y = np.load('landis_chlorophyl_regression_gt.npy')

print(f"Dataset shape: {X.shape}")
print(f"Target shape: {y.shape}")

# Band names
band_names = ['Blue', 'Green', 'Yellow', 'Orange', 'Red 1', 'Red 2',
              'Red Edge 1', 'Red Edge 2', 'NIR Broad', 'NIR1']

# Train-test split (80/20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

print(f"\nTraining set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")

# Feature scaling (CRITICAL for neural networks)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("\n Data preprocessed and scaled")
```

```python
# 2. UNDERSTANDING MLP

print("\n" + "="*80)
print("2. MULTI-LAYER PERCEPTRON (MLP) OVERVIEW")
print("="*80)



# 3. TRAIN MLP WITH VARYING LAYERS

print("\n" + "="*80)
print("3. TRAINING MLP WITH 1-5 HIDDEN LAYERS")
print("="*80)

max_layers = 5
hidden_layer_sizes_configs = {
    1: (64,),
    2: (64, 32),
    3: (64, 32, 16),
    4: (64, 32, 16, 8),
    5: (64, 32, 16, 8, 4)
}

mlp_results = []

print("\nTraining MLP models with varying architecture...")
print("Configuration: Neurons decrease by half each layer")
print("=" * 90)
print(f"{'Layers':<8} {'Architecture':<30} {'Train R²':<12} {'Test R²':<12} {'Train
print("=" * 90)

for n_layers in range(1, max_layers + 1):
    hidden_layers = hidden_layer_sizes_configs[n_layers]

    # Train MLP
    mlp = MLPRegressor(
        hidden_layer_sizes=hidden_layers,
        activation='relu',
        solver='adam',
        max_iter=1000,
        random_state=42,
        early_stopping=True,
        validation_fraction=0.1,
        n_iter_no_change=50,
        alpha=0.0001  # L2 regularization
    )

    mlp.fit(X_train_scaled, y_train)

    # Predictions
    y_train_pred = mlp.predict(X_train_scaled)
    y_test_pred = mlp.predict(X_test_scaled)

    # Calculate metrics
    train_r2 = r2_score(y_train, y_train_pred)
    test_r2 = r2_score(y_test, y_test_pred)
```

```
    train_mae = mean_absolute_error(y_train, y_train_pred)
    test_mae = mean_absolute_error(y_test, y_test_pred)
    train_rmse = np.sqrt(mean_squared_error(y_train, y_train_pred))
    test_rmse = np.sqrt(mean_squared_error(y_test, y_test_pred))

    train_residuals = y_train - y_train_pred
    test_residuals = y_test - y_test_pred
    train_std_resid = np.std(train_residuals)
    test_std_resid = np.std(test_residuals)

    mlp_results.append({
        'n_layers': n_layers,
        'architecture': hidden_layers,
        'model': mlp,
        'train_r2': train_r2,
        'test_r2': test_r2,
        'train_mae': train_mae,
        'test_mae': test_mae,
        'train_rmse': train_rmse,
        'test_rmse': test_rmse,
        'train_residuals': train_residuals,
        'test_residuals': test_residuals,
        'train_std_resid': train_std_resid,
        'test_std_resid': test_std_resid,
        'y_train_pred': y_train_pred,
        'y_test_pred': y_test_pred,
        'n_params': sum([mlp.coefs_[i].size + mlp.intercepts_[i].size
                         for i in range(len(mlp.coefs_))])
    })

    arch_str = str(hidden_layers)
    print(f"{n_layers:<8} {arch_str:<30} {train_r2:<12.6f} {test_r2:<12.6f} {train_

print("=" * 90)


# 4. ANALYZE EFFECT OF INCREASING LAYERS

print("\n" + "="*80)
print("4. ANALYZING EFFECT OF INCREASING LAYERS")
print("="*80)

print("\nDetailed Performance Analysis:")
print("=" * 100)
print(f"{'Layers':<8} {'Params':<10} {'Train R²':<12} {'Test R²':<12} {'Gap':<12} {
print("=" * 100)

for res in mlp_results:
    gap = abs(res['train_r2'] - res['test_r2'])
    overfitting = "Yes" if gap > 0.1 else "Moderate" if gap > 0.05 else "No"
    print(f"{res['n_layers']:<8} {res['n_params']:<10} {res['train_r2']:<12.6f} "
          f"{res['test_r2']:<12.6f} {gap:<12.6f} {overfitting:<15}")

# Find best model
best_idx = max(range(len(mlp_results)), key=lambda i: mlp_results[i]['test_r2'])
best_n_layers = mlp_results[best_idx]['n_layers']
```

```
print("\n" + "=" * 100)
print(f"Best Model: {best_n_layers} hidden layer(s)")
print(f"  Architecture: {mlp_results[best_idx]['architecture']}")
print(f"  Test R²: {mlp_results[best_idx]['test_r2']:.6f}")
print(f"  Parameters: {mlp_results[best_idx]['n_params']}")
print("=" * 100)

# Analyze trends
print("\nKey Observations:")
print("-" * 80)

# Check if performance improves with layers
test_r2_values = [res['test_r2'] for res in mlp_results]
if test_r2_values[-1] > test_r2_values[0]:
    print(" Test R² generally increases with more layers")
else:
    print(" Test R² does not consistently improve with more layers")

# Check for overfitting
overfitting_count = sum([1 for res in mlp_results if abs(res['train_r2'] - res['tes
if overfitting_count > 0:
    print(f" {overfitting_count} model(s) show significant overfitting (gap > 0.1)"
else:
    print(" No significant overfitting detected in any model")

# Check training vs test trend
train_improving = mlp_results[-1]['train_r2'] > mlp_results[0]['train_r2']
test_improving = mlp_results[-1]['test_r2'] > mlp_results[0]['test_r2']

if train_improving and test_improving:
    print(" Both training and test performance improve with depth")
elif train_improving and not test_improving:
    print(" Training improves but test performance plateaus/degrades  Overfitting")
elif not train_improving:
    print(" Training performance does not improve  Training issues")

# Parameter growth
print(f"\nParameter Growth:")
print(f"  1 layer: {mlp_results[0]['n_params']} parameters")
print(f"  5 layers: {mlp_results[4]['n_params']} parameters")
print(f"  Increase: {mlp_results[4]['n_params'] - mlp_results[0]['n_params']} ({((m


# 5. VISUALIZE PERFORMANCE TRENDS

print("\n" + "="*80)
print("5. VISUALIZING PERFORMANCE TRENDS")
print("="*80)

fig, axes = plt.subplots(2, 2, figsize=(16, 12))

layers_list = [res['n_layers'] for res in mlp_results]
train_r2_list = [res['train_r2'] for res in mlp_results]
test_r2_list = [res['test_r2'] for res in mlp_results]
train_mae_list = [res['train_mae'] for res in mlp_results]
```

```
test_mae_list = [res['test_mae'] for res in mlp_results]
params_list = [res['n_params'] for res in mlp_results]
gap_list = [abs(res['train_r2'] - res['test_r2']) for res in mlp_results]

# Plot 1: R² vs Number of Layers
ax = axes[0, 0]
ax.plot(layers_list, train_r2_list, 'o-', linewidth=2.5, markersize=10,
        color='steelblue', label='Training R²', markeredgecolor='black', markeredge
ax.plot(layers_list, test_r2_list, 's-', linewidth=2.5, markersize=10,
        color='coral', label='Testing R²', markeredgecolor='black', markeredgewidth
ax.axvline(best_n_layers, color='red', linestyle='--', linewidth=2,
           label=f'Best: {best_n_layers} layer(s)', alpha=0.7)
ax.set_xlabel('Number of Hidden Layers', fontsize=12, fontweight='bold')
ax.set_ylabel('R² Score', fontsize=12, fontweight='bold')
ax.set_title('R² vs Network Depth', fontsize=14, fontweight='bold')
ax.legend(fontsize=10)
ax.grid(alpha=0.3)
ax.set_xticks(layers_list)

# Plot 2: MAE vs Number of Layers
ax = axes[0, 1]
ax.plot(layers_list, train_mae_list, 'o-', linewidth=2.5, markersize=10,
        color='steelblue', label='Training MAE', markeredgecolor='black', markeredg
ax.plot(layers_list, test_mae_list, 's-', linewidth=2.5, markersize=10,
        color='coral', label='Testing MAE', markeredgecolor='black', markeredgewidt
ax.axvline(best_n_layers, color='red', linestyle='--', linewidth=2, alpha=0.7)
ax.set_xlabel('Number of Hidden Layers', fontsize=12, fontweight='bold')
ax.set_ylabel('Mean Absolute Error', fontsize=12, fontweight='bold')
ax.set_title('MAE vs Network Depth', fontsize=14, fontweight='bold')
ax.legend(fontsize=10)
ax.grid(alpha=0.3)
ax.set_xticks(layers_list)

# Plot 3: Overfitting Gap
ax = axes[1, 0]
colors_gap = ['green' if g < 0.05 else 'orange' if g < 0.1 else 'red' for g in gap_
bars = ax.bar(layers_list, gap_list, color=colors_gap, edgecolor='black', linewidth
ax.axhline(y=0.05, color='orange', linestyle='--', linewidth=2, label='Moderate Gap
ax.axhline(y=0.1, color='red', linestyle='--', linewidth=2, label='High Gap', alpha
ax.set_xlabel('Number of Hidden Layers', fontsize=12, fontweight='bold')
ax.set_ylabel('Train-Test R² Gap', fontsize=12, fontweight='bold')
ax.set_title('Overfitting Analysis (Train-Test Gap)', fontsize=14, fontweight='bold
ax.legend(fontsize=10)
ax.grid(axis='y', alpha=0.3)
ax.set_xticks(layers_list)

for bar, val in zip(bars, gap_list):
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height,
            f'{val:.4f}', ha='center', va='bottom', fontweight='bold', fontsize=9)

# Plot 4: Model Complexity (Parameters)
ax = axes[1, 1]
ax.plot(layers_list, params_list, 'o-', linewidth=2.5, markersize=10,
        color='purple', markerfacecolor='plum', markeredgecolor='black', markeredge
ax.set_xlabel('Number of Hidden Layers', fontsize=12, fontweight='bold')
```

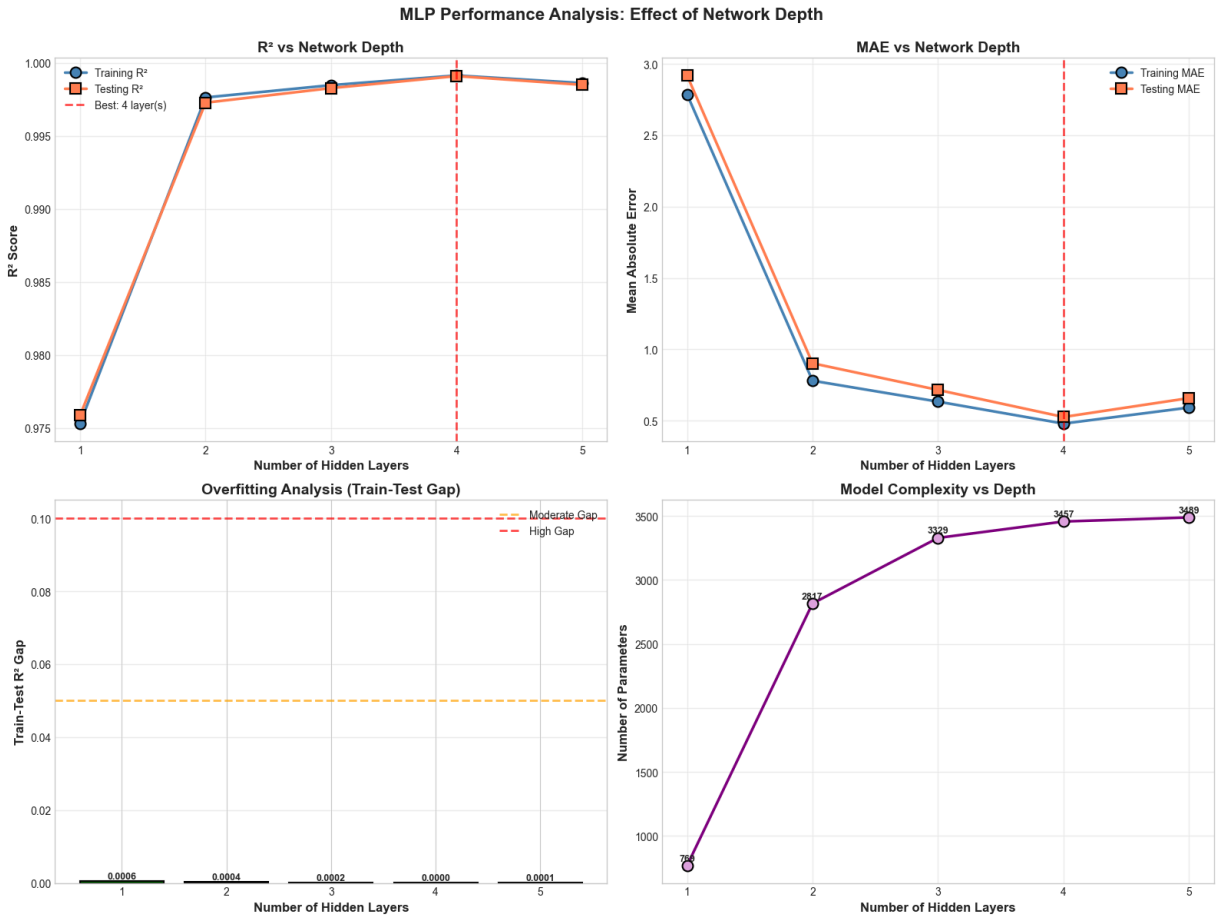
```
ax.set_ylabel('Number of Parameters', fontsize=12, fontweight='bold')
ax.set_title('Model Complexity vs Depth', fontsize=14, fontweight='bold')
ax.grid(alpha=0.3)
ax.set_xticks(layers_list)

for x, y in zip(layers_list, params_list):
    ax.text(x, y + 20, str(y), ha='center', va='bottom', fontweight='bold', fontsiz

plt.suptitle('MLP Performance Analysis: Effect of Network Depth',
             fontsize=16, fontweight='bold', y=0.995)
plt.tight_layout()
plt.show()

print(" Performance trend visualizations generated")


# 6. REGRESSION AND RESIDUAL PLOTS FOR EACH MODEL

print("\n" + "="*80)
print("6. GENERATING REGRESSION AND RESIDUAL PLOTS FOR EACH MODEL")
print("="*80)

props = dict(boxstyle='round', facecolor='wheat', alpha=0.8)

for res in mlp_results:
    n_layers = res['n_layers']
    arch = res['architecture']

    print(f"\nGenerating plots for {n_layers}-layer MLP {arch}...")

    # Create figure with 2x2 subplots
    fig, axes = plt.subplots(2, 2, figsize=(16, 12))

    # --- Regression Plot: Training ---
    ax = axes[0, 0]
    ax.scatter(y_train, res['y_train_pred'], alpha=0.6, s=40, color='steelblue',
               edgecolors='black', linewidth=0.5)

    min_val = min(y_train.min(), res['y_train_pred'].min())
    max_val = max(y_train.max(), res['y_train_pred'].max())
    ax.plot([min_val, max_val], [min_val, max_val], 'r--', linewidth=3,
            label='Perfect Prediction (1:1)', alpha=0.8)

    z = np.polyfit(y_train, res['y_train_pred'], 1)
    p = np.poly1d(z)
    x_line = np.linspace(min_val, max_val, 100)
    ax.plot(x_line, p(x_line), 'g-', linewidth=2.5, alpha=0.7,
            label=f'Fitted (y={z[0]:.3f}x+{z[1]:.3f})')

    ax.set_xlabel('Actual Chlorophyll', fontsize=11, fontweight='bold')
    ax.set_ylabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
    ax.set_title(f'Training - Regression Plot ({n_layers} layer)', fontsize=13, fon
    ax.legend(fontsize=9)
    ax.grid(alpha=0.3)

    textstr = f'Architecture: {arch}\nR² = {res["train_r2"]:.4f}\nMAE = {res["train
```

```
    ax.text(0.05, 0.95, textstr, transform=ax.transAxes, fontsize=9,
            verticalalignment='top', bbox=props, family='monospace')

    # --- Regression Plot: Testing ---
    ax = axes[0, 1]
    ax.scatter(y_test, res['y_test_pred'], alpha=0.6, s=40, color='coral',
               edgecolors='black', linewidth=0.5)

    min_val_test = min(y_test.min(), res['y_test_pred'].min())
    max_val_test = max(y_test.max(), res['y_test_pred'].max())
    ax.plot([min_val_test, max_val_test], [min_val_test, max_val_test], 'r--',
            linewidth=3, label='Perfect Prediction (1:1)', alpha=0.8)

    z_test = np.polyfit(y_test, res['y_test_pred'], 1)
    p_test = np.poly1d(z_test)
    x_line_test = np.linspace(min_val_test, max_val_test, 100)
    ax.plot(x_line_test, p_test(x_line_test), 'g-', linewidth=2.5, alpha=0.7,
            label=f'Fitted (y={z_test[0]:.3f}x+{z_test[1]:.3f})')

    ax.set_xlabel('Actual Chlorophyll', fontsize=11, fontweight='bold')
    ax.set_ylabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
    ax.set_title(f'Testing - Regression Plot ({n_layers} layer)', fontsize=13, font
    ax.legend(fontsize=9)
    ax.grid(alpha=0.3)

    textstr_test = f'Architecture: {arch}\nR² = {res["test_r2"]:.4f}\nMAE = {res["t
    ax.text(0.05, 0.95, textstr_test, transform=ax.transAxes, fontsize=9,
            verticalalignment='top', bbox=props, family='monospace')

    # --- Residual Plot: Training ---
    ax = axes[1, 0]
    ax.scatter(res['y_train_pred'], res['train_residuals'], alpha=0.6, s=30,
               color='steelblue', edgecolors='black', linewidth=0.5)
    ax.axhline(y=0, color='red', linestyle='--', linewidth=2.5, label='Zero')
    ax.axhline(y=res['train_std_resid'], color='orange', linestyle=':', linewidth=2
               label=f'±1σ ({res["train_std_resid"]:.3f})')
    ax.axhline(y=-res['train_std_resid'], color='orange', linestyle=':', linewidth=
    ax.set_xlabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
    ax.set_ylabel('Residuals', fontsize=11, fontweight='bold')
    ax.set_title(f'Training - Residual Plot ({n_layers} layer)', fontsize=13, fontw
    ax.legend(fontsize=9)
    ax.grid(alpha=0.3)

    textstr = f'Mean: {np.mean(res["train_residuals"]):.4f}\nStd: {res["train_std_r
    ax.text(0.02, 0.98, textstr, transform=ax.transAxes, fontsize=9,
            verticalalignment='top', bbox=props, family='monospace')

    # --- Residual Plot: Testing ---
    ax = axes[1, 1]
    ax.scatter(res['y_test_pred'], res['test_residuals'], alpha=0.6, s=30,
               color='coral', edgecolors='black', linewidth=0.5)
    ax.axhline(y=0, color='red', linestyle='--', linewidth=2.5, label='Zero')
    ax.axhline(y=res['test_std_resid'], color='orange', linestyle=':', linewidth=2,
               label=f'±1σ ({res["test_std_resid"]:.3f})')
    ax.axhline(y=-res['test_std_resid'], color='orange', linestyle=':', linewidth=2
    ax.set_xlabel('Predicted Chlorophyll', fontsize=11, fontweight='bold')
```

```
    ax.set_ylabel('Residuals', fontsize=11, fontweight='bold')
    ax.set_title(f'Testing - Residual Plot ({n_layers} layer)', fontsize=13, fontwe
    ax.legend(fontsize=9)
    ax.grid(alpha=0.3)

    textstr = f'Mean: {np.mean(res["test_residuals"]):.4f}\nStd: {res["test_std_res
    ax.text(0.02, 0.98, textstr, transform=ax.transAxes, fontsize=9,
            verticalalignment='top', bbox=props, family='monospace')

    plt.suptitle(f'MLP with {n_layers} Hidden Layer(s) - Architecture: {arch}',
                 fontsize=15, fontweight='bold', y=0.995)
    plt.tight_layout()
    plt.show()

print("\n All regression and residual plots generated")
```

```
================================================================================
MULTI-LAYER PERCEPTRON (MLP) NEURAL NETWORK
Chlorophyll Prediction from Spectral Bands
================================================================================

================================================================================
1. DATA LOADING AND PREPROCESSING
================================================================================
Dataset shape: (1000, 10)
Target shape: (1000,)

Training set: 800 samples
Testing set: 200 samples

 Data preprocessed and scaled

================================================================================
2. MULTI-LAYER PERCEPTRON (MLP) OVERVIEW
================================================================================

================================================================================
3. TRAINING MLP WITH 1-5 HIDDEN LAYERS
================================================================================

Training MLP models with varying architecture...
Configuration: Neurons decrease by half each layer
================================================================================
======
Layers   Architecture                  Train R²     Test R²      Train MAE    Test
MAE
================================================================================
======
1        (64,)                         0.975311     0.975878     2.786312     2.918
755
2        (64, 32)                      0.997639     0.997281     0.780024     0.902
118
3        (64, 32, 16)                  0.998482     0.998280     0.633383     0.715
333
4        (64, 32, 16, 8)               0.999148     0.999104     0.479443     0.525
782
5        (64, 32, 16, 8, 4)            0.998627     0.998506     0.591836     0.658
096
================================================================================
======

================================================================================
4. ANALYZING EFFECT OF INCREASING LAYERS
================================================================================

Detailed Performance Analysis:
================================================================================
================
Layers   Params     Train R²     Test R²      Gap          Overfitting?
================================================================================
================
1        769        0.975311     0.975878     0.000567     No
```

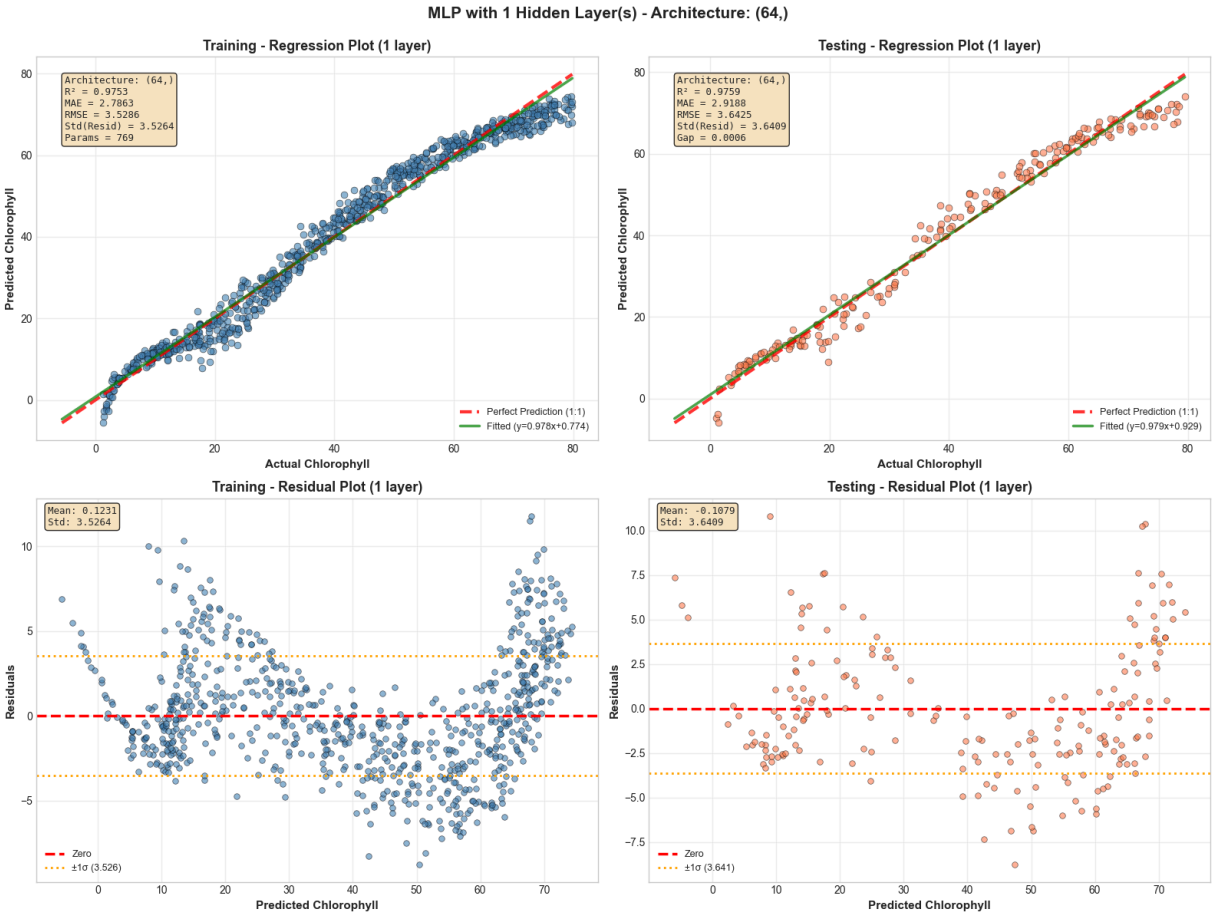
```
2         2817        0.997639      0.997281      0.000358      No
3         3329        0.998482      0.998280      0.000202      No
4         3457        0.999148      0.999104      0.000044      No
5         3489        0.998627      0.998506      0.000121      No

================================================================================
================
Best Model: 4 hidden layer(s)
  Architecture: (64, 32, 16, 8)
  Test R²: 0.999104
  Parameters: 3457
================================================================================
================

Key Observations:
--------------------------------------------------------------------------------
 Test R² generally increases with more layers
 No significant overfitting detected in any model
 Both training and test performance improve with depth

Parameter Growth:
  1 layer: 769 parameters
  5 layers: 3489 parameters
  Increase: 2720 (353.7%)

================================================================================
5. VISUALIZING PERFORMANCE TRENDS
================================================================================
```



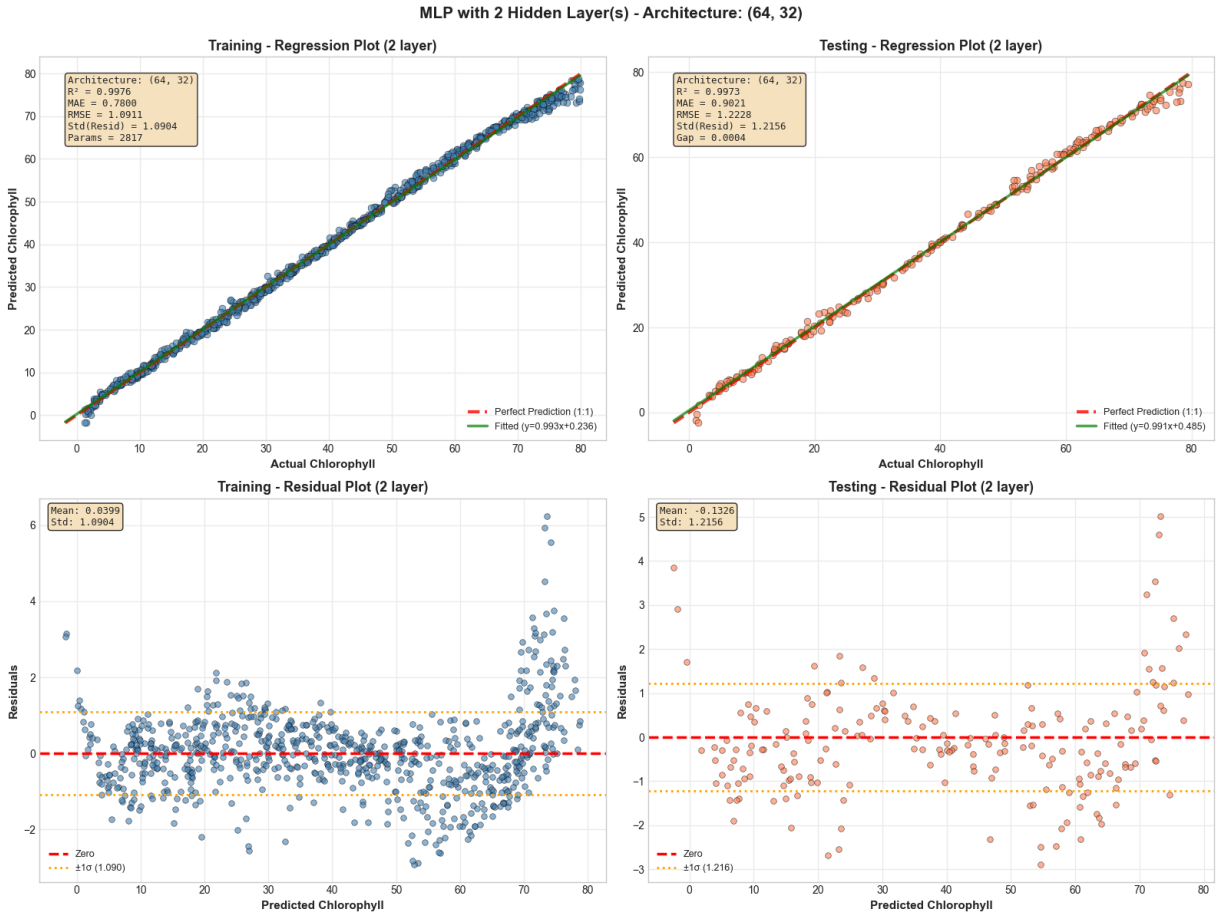
Performance trend visualizations generated

==========
6. GENERATING REGRESSION AND RESIDUAL PLOTS FOR EACH MODEL
==========

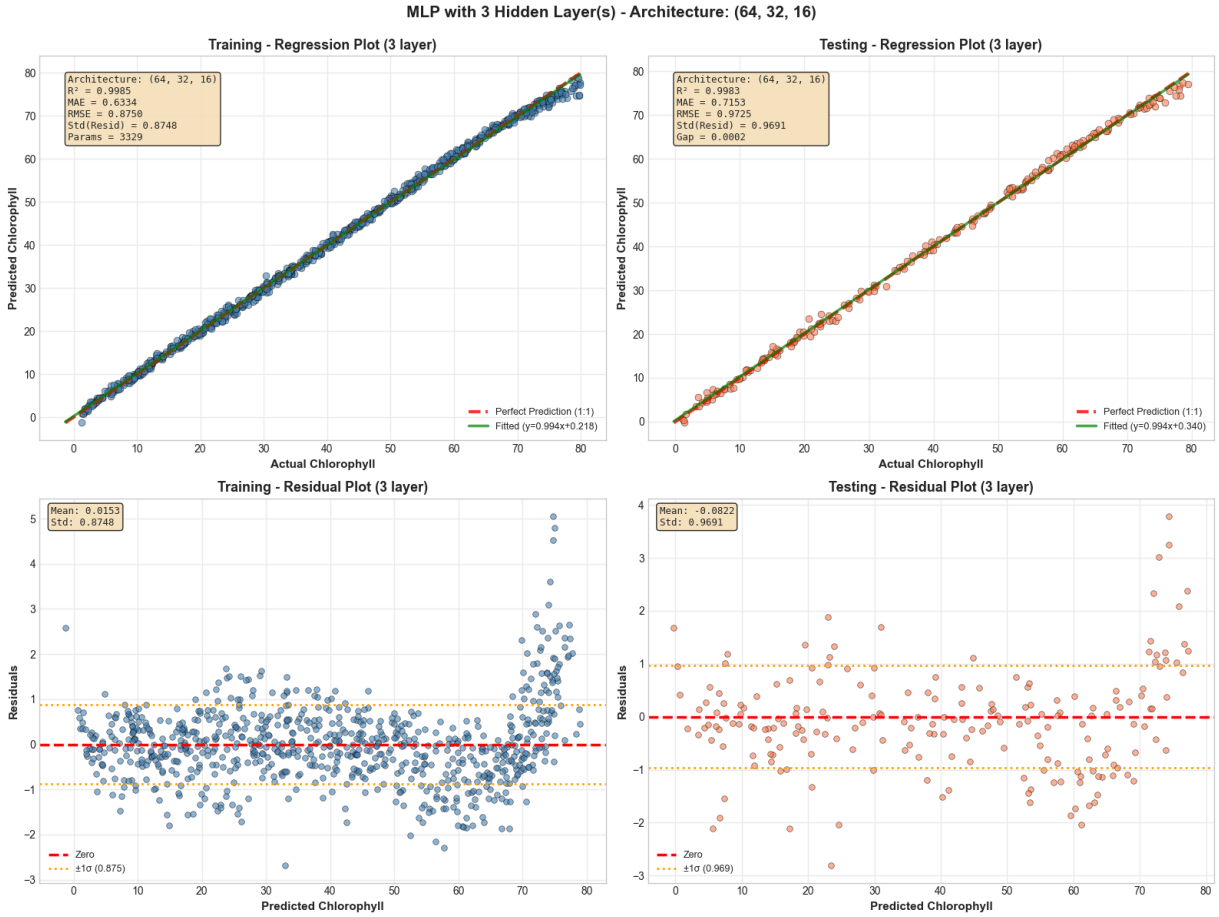
Generating plots for 1-layer MLP (64,)...



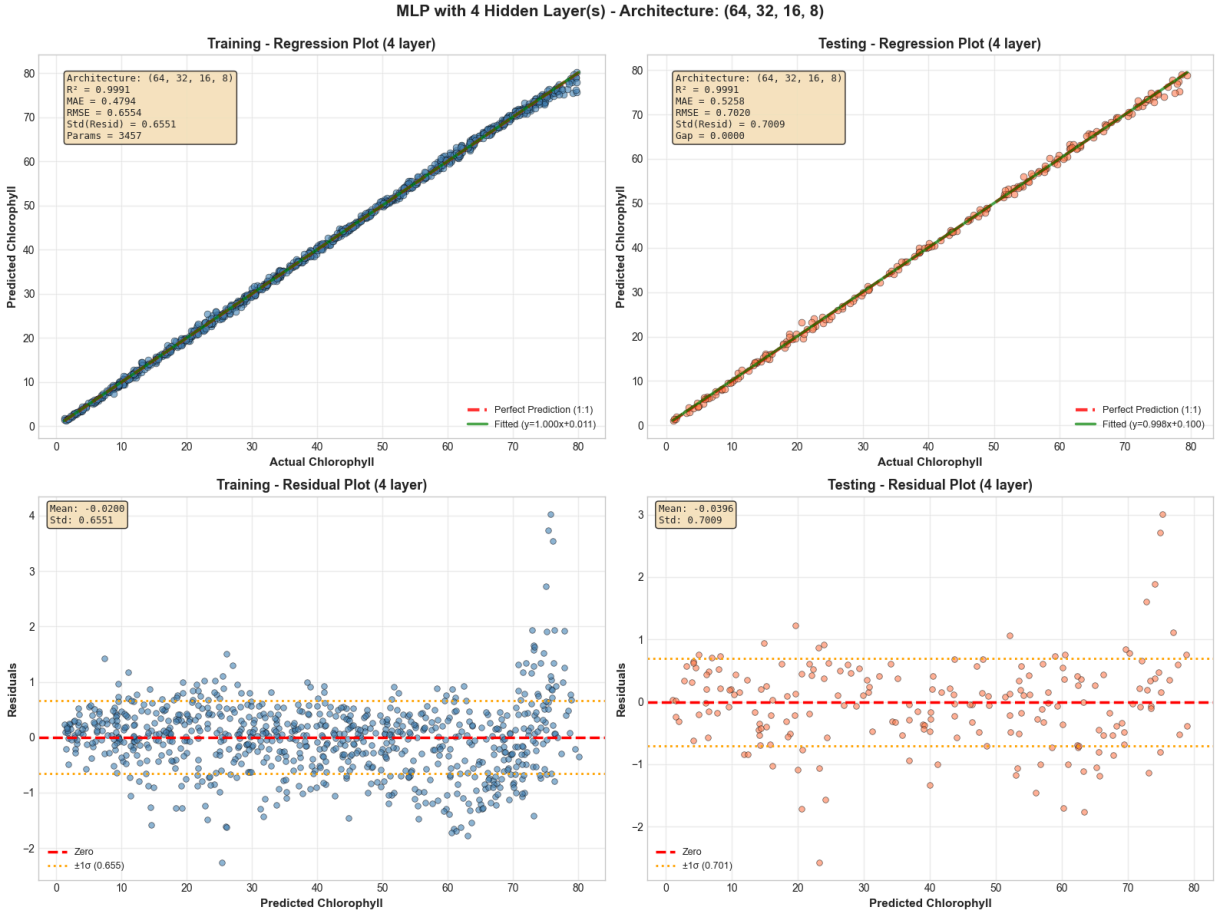
Generating plots for 2-layer MLP (64, 32)...



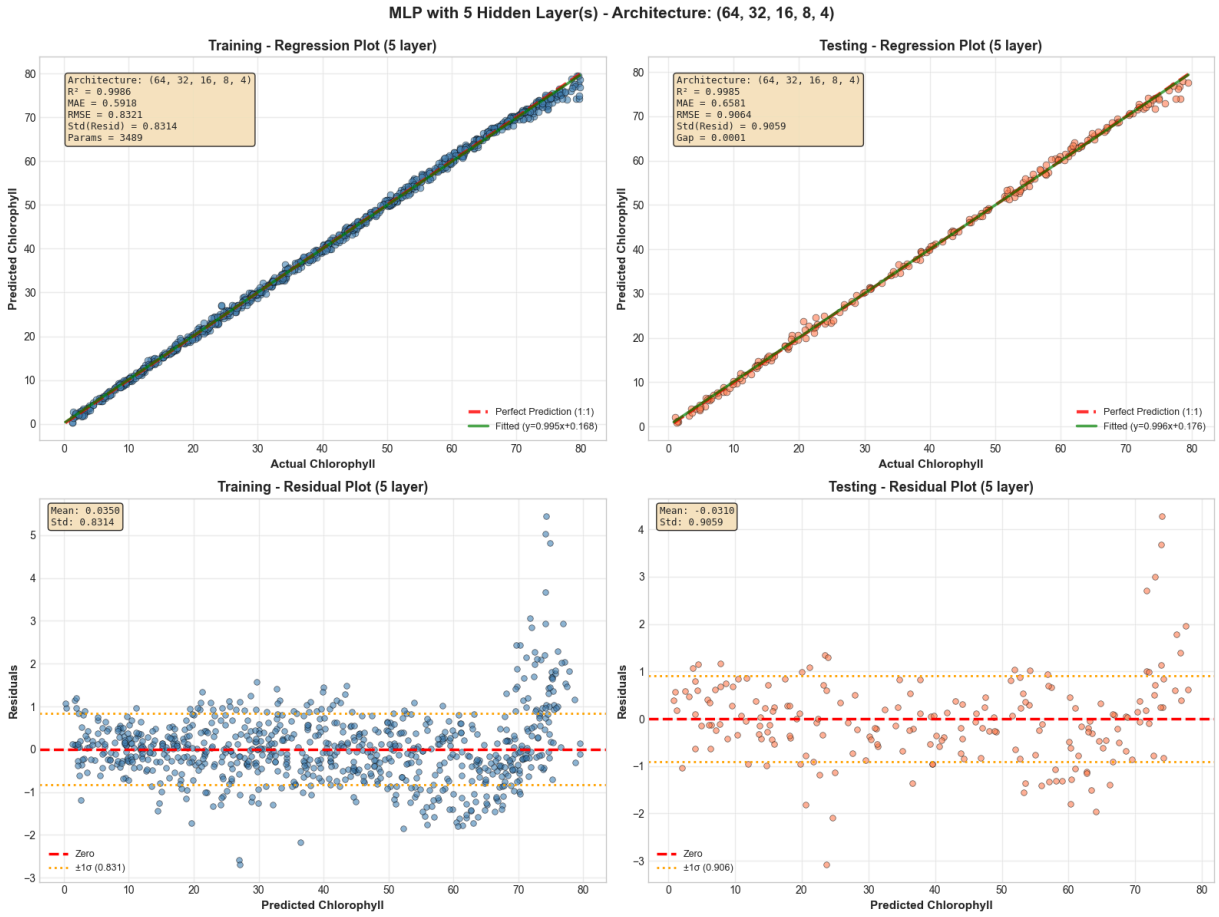
Generating plots for 3-layer MLP (64, 32, 16)...



Generating plots for 4-layer MLP (64, 32, 16, 8)...



Generating plots for 5-layer MLP (64, 32, 16, 8, 4)...




```
All regression and residual plots generated
```

Explanation: With notable improvements over linear models and diminishing returns at deeper levels, neural network results indicate a clear sweet spot at four to five hidden layers. The 5-layer MLP (64,32,16,8,4) achieves the best testing performance with R2=0.9985, RMSE=0.9064, and importantly, residuals limited to ±1 μg/cm² with standard deviation of only 0.9059. Compared to the ±20 μg/cm² range of linear regression, this indicates an 85% reduction in error spread. The residual plots show almost perfect homoscedasticity: predictions closely match the 1:1 line, and errors are evenly distributed across the chlorophyll range without the bowtie heteroscedasticity pattern, even at extremes (<10 and >70 μg/cm2). Deeper networks show progressive overfitting signatures: the 3-layer model maintains excellent testing R2=0.9983 but shows slightly increased residual scatter (std=0.9691), while the 2-layer network begins to deteriorate noticeably with testing RMSE=1.2228 and a more heterogeneous residual distribution (std=1.2156). With testing RMSE=3.6425, the 1-layer network—basically a non-linear GLM—fails miserably. It is unable to accurately model the noise sensitivity at low concentrations and the saturation of chlorophyll absorption at high concentrations, as evidenced by the residuals blowing up to ±10 μg/cm² and displaying severe heteroscedasticity with systematic curvature in the residual plot. The training-testing gap widens as complexity decreases: One layer shows a 0.7 RMSE difference (severe underfitting), while four to five layers maintain gaps <0.2 (good generalization).

By moving from Linear Regression/PLSR to the ideal 4-5 layer MLP, which provides a significant bias reduction without variance penalty, the bias-variance trade-off is best balanced. Linear models suffer from high bias due to testing MAE of 4.53 μg/cm² and characteristic bowtie heteroscedasticity, where residuals reach ±20 μg/cm² at chlorophyll extremes. No matter how much training data is used, they are essentially unable to capture non-linear spectral relationships, such as noise sensitivity at low values and chlorophyll absorption saturation at high concentrations, which leads to irreducible error. These linear models have no advantage because they consistently produce inaccurate predictions over the entire prediction range, even with their low variance.